

New Inferential Methods for Astrophysical Discoveries

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It still feels unreal that my PhD journey has reached its end. Even now, the memories of 2020, when we navigated the uncertainties and challenges of our first year entirely online due to the COVID pandemic, remain vividly close, as if they had occurred only moments ago. As I reflect on those early days and all that followed, I am profoundly grateful for the kindness, support, and guidance I have received along the way. Indeed, this journey would not have been possible without the remarkable individuals who believed in me, challenged me to grow, and stood by me through every difficulty and obstacle. For this reason, I would like to express my deepest gratitude to all those who have supported, encouraged, and walked alongside me throughout this extraordinary chapter. I will carry their lessons forward and strive to repay this generosity in the future.

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Abstract

The development of modern astronomical instruments has generated massive datasets for astrophysical research, raising fundamental questions about the origins of observed and unobserved phenomena and their consistency with existing theories. Answering these questions requires identifying meaningful signals amid noise and background sources, and comparing observations against theoretical models—both of which demand robust statistical tools capable of handling complex theoretical and practical settings. This dissertation addresses this need by advancing statistical methodological developments toward a unified, efficient, and accurate inferential framework that integrates model validation, background calibration, and signal detection. These methods are demonstrated through applications to specific astrophysical problems. In particular, we introduce an inferential framework that integrates smooth tests with likelihood ratio tests for signal discovery in the presence of uncertain background and demonstrate its usefulness in the study of the symbiotic star RT Cru using Chandra/LETGS data. Furthermore, we extend the theory of smooth tests by developing a new family of asymptotically distribution-free smooth test statistics based on the Khmaladze-2 (K2) transform introduced by [Khmaladze \[2013, 2016\]](#). This transform enables accurate and efficient implementation of smooth tests to multiple complex astrophysical models, even with moderately large samples, in a matter of seconds. The K2 transform is not limited to the context of smooth tests but applies to most goodness-of-fit settings, maintaining both distribution-free properties and computational efficiency. We therefore employ it to validate angular power spectrum models of the stochastic gravitational-wave background using Advanced LIGO–Virgo O3 data. Collectively, these contributions establish a unified inferential framework that is statistically robust and directly applicable to complex astrophysical problems.

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Chapter 1

Introduction

The search for new signals under high background is a ubiquitous challenge in physical explorations. Here, the term *background* refers to the sources of noise that can contribute to masking the signal of interest. The primary challenge arising in such searches affected by a high background is that misspecifying the background distribution can significantly affect the sensitivity of the analysis. When the background is incorrectly modeled, it can lead to false detections or missed signals, thereby compromising the accuracy and reliability of statistical inference. To solve this problem, Chapter 2 develops a novel statistical approach to identify emission features or set upper limits in high-resolution spectra in the presence of a high background. The method relies on detecting differences from the background using smooth tests and using classical likelihood ratio tests to characterize known shapes like emission lines. The method is illustrated through application to a Chandra LETGS+HRC-S observation of the symbiotic star RT Cru, successfully detecting previously known features like the Fe line emission in the 6-7 keV range and the Iridium-edge due to the mirror coating on *Chandra*. A search for thermal emission lines from Ne X, Fe XVII, O VIII, and O VII yields no detections, with upper limits on their intensities placed consistent with an approximately 1 keV plasma. Serendipitously, a line at 16.93 Å (0.732 keV) is detected and attributed to photoionization or a reflection component.

The study conducted in Chapter 3 aims to address some limitations of classical smooth tests relevant in astrophysical applications. In particular, while the asymptotic null distributions of smooth test statistics are known, in practical applications, their implementation requires very large sample sizes to reach these asymptotic limits. Although the parametric bootstrap can be employed to derive the null distribution of smooth test statistics, such a numerical approach becomes impractical when sampling from complex distributions or when parameter estimation is computationally expensive. Moreover, the computational cost increases substantially when testing different models that define competing astrophysical theories, as simulations must be performed separately for each candidate model. To overcome these limitations, a fundamental shift in perspective is introduced. Rather than selecting the orthonormal basis for smooth tests somewhat arbitrarily, we construct such a basis to ensure that any test statistic based on its elements is asymptotically distribution-free by design. This construction is achieved via the *Khmaladze-2 (K2) transform* introduced by [Khmaladze \[2016\]](#). Analogous to the probability integral transform in classical nonparametric goodness-of-fit testing, the K2 transform maps a wide range of projected empirical processes into standard projections with the same limiting null distribution. Simulation and real data analysis demonstrate the accuracy and efficiency of smooth tests under this construction even when parameters are estimated, model selection is performed, and sample sizes are only moderately large. Chapter 3 also introduces an alternative to the classical parametric bootstrap that further enhances computational efficiency.

The K2 transform naturally applies to most contexts in which goodness-of-fit (GOF) tests are required, including regression. In this setting, GOF via the K2 transform is especially valuable in that it requires no distributional assumptions on the errors to perform distribution-free inference. This property proves crucial for analyzing the angular power spectrum of the stochastic gravitational-wave background (SGWB). Specifically, in Chapter 4, we show that standard estimators of the SGWB angular power spectrum lack closed-form likelihood expressions and typically cannot be accurately approximated by Gaussian

likelihoods. Our K2 transform-based framework circumvents these limitations by enabling efficient and robust statistical inference without specifying the distribution of the estimators of the angular power spectrum. Moreover, we derive a new, consistent estimator for the angular power spectrum's covariance matrix and apply the proposed methodology to estimate and test angular power spectral models for the SGWB. The approach is demonstrated on an analysis of data from the third observing run (O3) of Advanced LIGO and Advanced Virgo.

Chapter 2

A novel approach to detect line emission under high background in high-resolution X-ray spectra

2.1 Overview

Recently, several new X-ray missions and mission concepts have been developed with the goal of obtaining high-resolution spectra (e.g., XRISM [[XRISM Science Team 2022](#), [Ishisaki 2022](#)], Athena [[Barret 2022](#)], Arcus [[Smith 2020](#)], LEM [[Kraft 2022](#)]). With such spectra expected to become a ubiquitous feature of high-energy investigations, it is necessary to consider the challenge of detecting weak features when the spectrum is also contaminated by high levels of background of both instrumental and cosmic origins. Traditional methods employed for spectral analysis, like globally fitting simplified spectral models, or carrying out isolated analyses of individual spectral lines, will fail to produce robust results; the former when the data show more structure than is encoded in the models, and the latter when statistical fluctuations in the background confound the detection of weak features.

Here, we present a newly developed statistical approach to evaluate the significance

of deviations that may be present in high-resolution spectra even in the presence of high background [Algeri, 2020a, 2021]. The method works by comparing a postulated model for the background and a source-free background dataset, to assess the validity of the former and, if needed, provide a data-driven correction for it based on the information carried by the latter. Once the background model has been “trained” on the signal-free sample, our strategy is to employ *smooth tests*, originally introduced by Neyman [1937], to assess whether the source spectrum of interest shows any *shape* differences from the re-calibrated background model. Whenever significant deviations cannot be found, we then search for the presence of the most prominent spectral lines at well-defined locations by means of likelihood ratio tests and place strict upper limits on their fluxes following the framework of Kashyap [2010]. The statistical properties of the procedure proposed, i.e., power and probability of type I error, are also investigated.

Our method is applicable to high-resolution spectra, i.e., one where the resolution of the detector is sufficient to separate individual emission lines. In the case of low-resolution spectra, the presence or absence of individual lines must be inferred by modeling [see, e.g., Park et al., 2008]. In this setting, Bayesian solutions are often implemented in order to overcome the difficulties associated with low bin counts and the lack of model identifiability within the digitization limits of the spectra. In contrast, here we implement an *unbinned* analysis on data obtained as lists of photons. The goal is to detect differences between the likelihoods under the background-only and the background+source hypotheses.

Estimation and inference can then be facilitated by conducting a separate analysis over different regions of the spectrum following a frequentist statistical paradigm. The inferential results are ultimately combined by implementing adequate corrections for multiple hypothesis testing.

To illustrate the necessity of the method proposed here in the analysis of high-resolution spectra, we apply it to study the symbiotic stellar system RT Cru through the analysis of the *Chandra* LETGS+HRC-S dataset. We emphasize that the method itself has been

developed to have general applicability to high-energy high-resolution datasets where weak spectral features are expected over strong background contamination. We describe the RT Cru system and the *Chandra* data below in Section 2.2.1, and demonstrate the need for a principled statistical method to obtain robust inferences. We then provide a detailed description of *smooth models* and *smooth tests* in Section 2.3. In Section 2.4, we describe the analysis that we carry out. Specifically, in Section 2.4.1 we focus on the implementation of data-driven background correction, in Section 2.4.2 we implement the proposed testing procedure to detect spectral features, and in Section 2.4.3 we construct upper limits for specified line locations where no signal is detected. The applicability of the method and the results of the analysis are discussed in Section 2.5. We summarise our work in Section 2.5.

2.2 The relevance of RT Cru

In order to demonstrate the suitability of smooth tests in detecting new signals when the data is contaminated by high background, we apply it to answer questions that arose during the analysis of *Chandra* spectra of RT Cru, and which could not be dealt with using existing methods. Below, we describe the astronomical object, and the dataset, and illustrate how an improved statistical analysis is necessary.

RT Cru is a symbiotic system (distance 2.5 kpc; GAIA EDR3) where a high-mass white dwarf (WD; $M_{WD} > 1.3 M_{\odot}$) accretes from the wind of an M5 III red giant companion [Cieslinski, 1994]. Symbiotic systems have been invoked as potential progenitors of a fraction of SN type Ia - key cosmological indicators. Symbiotic systems, especially those containing a high-mass ($> 1.3 M_{\odot}$) white dwarf component are important for understanding the possibility of a single degenerate path toward SN Ia.

Symbiotic systems produce soft X-ray spectra from the accretion disk surrounding the WD, through quasi-steady burning of the accreting material on the WD surface [Muerset, 1997], and from a jet.

A handful of systems have also been detected in hard X-rays, including RT Cru (INTEGRAL, [Bird 2007](#); *Swift*, [Kennea 2009](#); *Suzaku*, [Ducci 2016](#), *Chandra*, [Luna and Sokoloski 2007](#), [Danehkar 2021a](#); *NuSTAR+Suzaku+Swift*, [Luna 2018](#)). The nature of the hard X-ray emission is not well-established, though it is thought to be attributable to the presence of an accretion disk: the hard X-rays have been variously modeled as isobaric cooling flows [[Luna and Sokoloski, 2007](#)]; or partially covered [[Kennea, 2009](#)] or clumpy absorption of thermal emission [[Danehkar, 2021a](#)]; or post-shock regions above the polar caps of a magnetized white dwarf [[Ducci, 2016](#)], and are able to explain the features of the spectrum at high energies ($E \gtrsim 3$ keV).

RT Cru also exhibits several variability features like aperiodic flickering at timescales of a few ks, and a strong correlation of spectral hardness with overall brightness, with higher intensities corresponding to softer spectra [see [Danehkar, 2021a](#)]. Such features are characteristic of emission driven by changes near the inner boundary of the accretion disk. The question that arises then is what the origin of this variability could be. As noted above, the observed X-ray spectrum has been variously modeled as an intrinsic change in the soft thermal emission component as well as changes in a continuum component due to intervening absorption. The presence of spectral lines during increases of soft flux, especially if they are the dominant contributors to the soft emission, would support the former scenario, while the lack of such lines would favor the latter scenario. The *Chandra*/LETGS+HRC-S data we focus on below (Section [2.2.1](#)) was originally obtained to settle this question, but the analysis was limited because of the relatively high background encountered. While an adequate spectral fit that included partial covering of a thermal component was obtained by [Danehkar \[2021a\]](#), the fit was driven by the higher energy signal. Thus, the nature of the soft emission remained unsettled.

Passbands of interest (W_r)	Passband range [Å] [keV]		Dominant lines and Features [Å]		N_r	n_r
W_1	1.65-2.05	6.05-7.5	Fe K	1.785/1.853/1.944	6879	368
W_2	5.5-10.2	1.2-2.25	Ir-M edge	≈ 6.5	75699	3311
W_3	11.5-13.0	0.95-1.08	Ne X	12.131	22809	730
W_4	14.6-15.15	0.82-0.85	Fe XVII	15.014	8354	247
W_5	16.2-17.4	0.71-0.765	Fe XVII	17.051	17331	471
W_6	18.5-19.5	0.65-0.67	O VIII	18.967	14060	390
W_7	21.3-21.75	0.57-0.582	O VII(r)	21.602	6192	179
W_8	21.65-22.0	0.564-0.573	O VII(i)	21.804	4927	145
W_9	21.9-23.0	0.539-0.566	O VII(f)	22.101	15255	390

Table 2.1: Summary of wavelength bands of interest. The second and third columns of the table report the wavelength range (in both Å and keV) of each of the regions, W_r , $r = 1, \dots, 9$, considered. The expected spectral lines and their expected positions (in Å) on each of the regions are listed in the fourth and fifth columns. The size of the source-free sample (N_r) and the physics sample (n_r) for each of the region r are given in the last two columns.

2.2.1 Data collection: the *Chandra*/LETGS+HRC-S observation

RT Cru has been observed extensively with both *Swift*/*XRT* and the *Chandra* gratings. Here we focus specifically on the *Chandra*/LETGS+HRC-S observations that were carried out in Nov 2015 for a total of 78.9 ks (PI: M. Karovska; ObsIDs 16688 and 17810) and whose analysis was described by [Daneħkar \[2021a\]](#). [Daneħkar](#) modeled the spectrum as a combination of a power-law component (with index $\Gamma \approx 1.7$), a thermal soft excess (with temperature ≈ 1.3 keV), and an ad hoc triplet of emission lines from Fe K α , Fe XXV, and Fe XXVI in the 6-7 keV region. The observed spectrum is shown in the top panel of Figure 2.1 in the $\lambda=[1.56, 30]$ Å wavelength range (corresponding to photon energies $E = [0.41, 8]$ keV) as the black histogram, along with the best-fit astrophysical model spectrum (together with the estimated background contribution) as the red curve. The hard emission in the 6-7 keV range likely originates close to the accretor, e.g., from a region of the boundary layer of the accretion disk or a bright spot. It could also be associated with collisional excitation in the vicinity of the WD accretor, including in the inner-jet regions (e.g., [Kennea 2009](#), [Karovska 2010](#), [Eze 2014](#)).

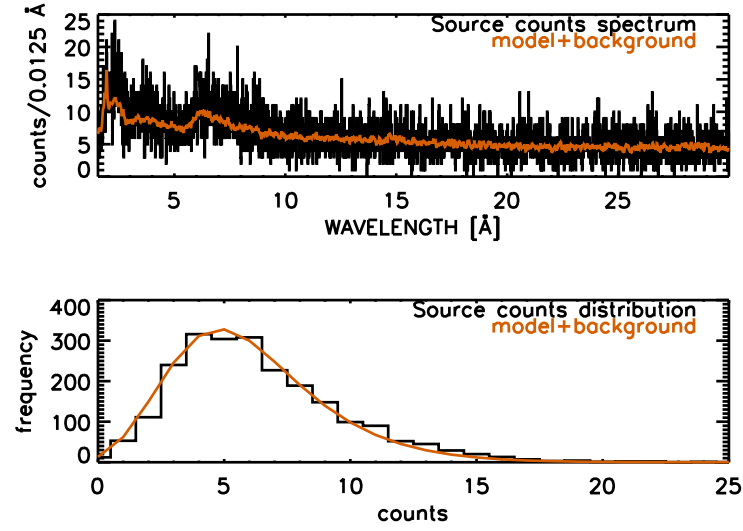


Figure 2.1: Comparing the spectrum of RT Cru with the model from [Danehkar \[2021a\]](#). *Top*: The *Chandra*/LETGS+HRC-S counts spectrum is shown as the black histogram, and the sum of the estimated background and the predicted model are shown in red. *Bottom*: The distribution of counts in each bin in the observed spectrum is shown as the black histogram, and the expected Poisson distribution based on the model intensity and the estimated background is shown as the red curve.

However, the soft component of this model is not well constrained, partly due to the lack of sensitivity of Swift and *Chandra*/ACIS detectors at $E \lesssim 2$ keV and the high instrument background present in *Chandra*/LETGS+HRC-S. The advantage of high spectral resolution observations is that emission lines, if they exist, can be located, and their presence and identification can provide useful diagnostic information about the emitting plasma. The high background in the LETGS+HRC-S spectrum makes this a difficult problem. The upper panel of Figure 2.1 shows several spikes that could be tagged as spectral lines, but the question arises as to how significant each of those identifications would be. Methods that are typically used in these situations compare the heights of the spikes to the baseline intensity levels, and flag them as significant if they exceed a certain threshold. But the large number of bins present in high-resolution spectra implies that the chances of false positives being flagged as a line are high. This is demonstrated in the bottom panel of Figure 2.1, which compares the observed distribution of counts in the bins (black stepped histogram) with the expected Poisson distribution derived from the model and background (smooth red curve). Typically, a threshold is set based on the expected distribution such that a tolerable number of false detections are accepted; this tolerance can be set as accepting one false detection over the sample [as is done in, e.g., `wavdetect`; Freeman, 2002]. There are ≈ 2280 bins in the observed spectrum shown in the upper panel, yielding $p < 0.000044$ as the required threshold value (note that the usual “ 3σ ” detection criterion often used in astronomy corresponds to $p < 0.0027$, and adopting it would result in as many as 6 false claims of detection; this issue will worsen when higher resolution spectra covering larger wavelength ranges are obtained), which translates to the requirement that a fluctuation exceeds 20 counts before it can be considered significant.

The goal of this observation was to obtain a spectrum at wavelengths $\lambda \gtrsim 6$ Å (energies $E \lesssim 2$ keV), where the sensitivity of the instrument is higher on average than previous observations, in particular at the locations dominated by Ne X $\lambda 12.13$ and the two Fe XVII lines at $\lambda \lambda 15.01, 17.05$. In addition, we also consider other wavelength ranges like the

Fe K region (where spectral lines have indeed been detected and modeled) and wavelength regions where O VIII $\lambda 18.96$ and the O VII He-like triplets at $\lambda\lambda 21.6, 21.8, 22.1$ are to be found. These wavelength bands of interest are denoted with W_r , $r = 1, \dots, 9$ and are listed in Table 2.1; the Table shows the wavelength range, the dominant structure expected in these bands, and the number of events within the passbands in both the source region (n_r) as well as the source-free background region (N_r , collected in an area $39.4\times$ greater than the source region). The observed spectra (black histograms) and model predicted spectra (green curves) are shown, along with the expected background (red curves), zoomed in to these wavelength regions in Figure 2.2; the expected locations of spectral lines of interest are marked with yellow shaded regions centered around vertical green lines.

Inspection of the spectral regions of interest in Figure 2.2 shows hints that some emission structure may be present in some cases, e.g., in passbands W_3 (Ne X), W_5 (Fe XVII), W_6 (O VIII), W_7 (O VII(r)), and W_8 (O VII(i)). We carry out a naive estimation of the line intensities by counting the number of photons within $\pm 0.1\text{\AA}$ (corresponding to the width of the line response) at the putative locations of the lines, collected within both the source and background datasets, and computing the Bayesian posterior density of the line intensities [viz. [Primini and Kashyap, 2014](#)]. Table 2.2 lists the mode and the 68% highest posterior density (HPD) intervals for the resulting posterior distributions for each line or band of interest. Several line intensities are estimated formally at significance $> 1\sigma$. An analysis that ignores the problem of multiple tests would claim detections of these lines, but it is possible that such claims would turn out to be false positives. A more careful and principled method must be used to evaluate their reality. We, therefore, employ the method proposed in [[Algeri, 2020a, 2021](#)] to adequately model background shapes on each of the regions considered. Moreover, since our analysis requires us to test for the presence of one or more lines on each of the above-mentioned regions, adequate corrections for multiple hypotheses testing are also implemented. Such step is necessary to ensure that the overall probability of a false discovery across any of the regions considered does not exceed the

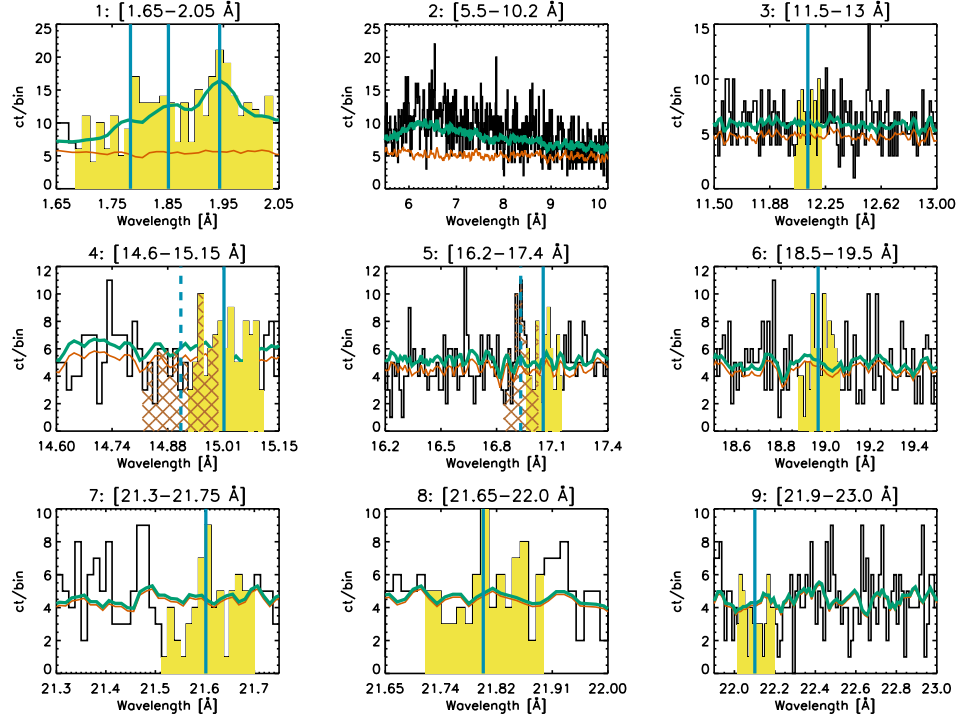


Figure 2.2: The counts spectrum of RT Cru (Figure 2.1), without background subtraction, zoomed in to the various bands of interest (see Table 2.1). The counts spectrum is shown as the black histogram, with the best-fit model spectrum from Danehkar [2021a] shown as the green curve. The estimated smoothed background is shown as the red curve. The blue vertical lines indicate the lines of interest in each band (except for W_2 , which covers an instrument feature). The shaded yellow regions cover the approximate expected widths of the line response for each of the lines studied here. The brown cross-hatched regions in passbands W_4 and W_5 mark serendipitous testing locations (see Discussion).

Wavelength Band	Model ^a Expected	Estimate ^b Observed
W_1	175.7	186 $\begin{smallmatrix} <205 \\ >167 \end{smallmatrix}$
W_2	1021.0	1370 $\begin{smallmatrix} <1430 \\ >1310 \end{smallmatrix}$
$12.131 \pm 0.1 \text{ \AA} \subset W_3$	16.4	20.0 $\begin{smallmatrix} <30.6 \\ >10.9 \end{smallmatrix}$
$15.014 \pm 0.1 \text{ \AA} \subset W_4$	12.7	20.6 $\begin{smallmatrix} <31.2 \\ >11.4 \end{smallmatrix}$
$17.051 \pm 0.1 \text{ \AA} \subset W_5$	9.15	5.6 $\begin{smallmatrix} <15.2 \\ >1.4 \end{smallmatrix}$
$18.967 \pm 0.1 \text{ \AA} \subset W_6$	6.78	16.1 $\begin{smallmatrix} <26.3 \\ >7.7 \end{smallmatrix}$
$21.602 \pm 0.1 \text{ \AA} \subset W_7$	2.8	0 $\begin{smallmatrix} <6.7 \\ \geq 0 \end{smallmatrix}$
$21.804 \pm 0.1 \text{ \AA} \subset W_8$	2.6	9.8 $\begin{smallmatrix} <19.5 \\ >3.5 \end{smallmatrix}$
$22.101 \pm 0.1 \text{ \AA} \subset W_9$	2.5	0 $\begin{smallmatrix} <4.8 \\ \geq 0 \end{smallmatrix}$

a: Expected model counts based on [Danezhkar \[2021a\]](#)

b: Mode of the posterior density distribution of source intensity [[Primini and Kashyap, 2014](#)] and the 68% HPD (highest posterior density) uncertainty bounds

Table 2.2: Model predicted and estimated counts at line locations of interest.

desired significance level. Finally, we also test whether the feature present at $16.93 \text{ \AA}$ in W_5 , shortward of the Fe XVII $\lambda 17.04$ line by $\approx 2100 \text{ km s}^{-1}$ is real, in the sense of being detectable.

We find that none of the hypothesised thermal spectral lines listed in Table 2.2 are detectable (see Section 2.4.2.2), and thus we place upper limits on the strengths of the lines that are well above reasonable variations in the fitted spectral model (cf. expected model counts listed in Table 2.2). In contrast, we find that the serendipitously identified line at $16.93 \text{ \AA}$ is detectable with a corrected p -value ≈ 0.01 . We discuss the possible origins of this line in Section 2.5.

2.3 Statistical methods

As outlined in Sections 2.1-2.2, the presence of high background makes it necessary to adequately characterise its shape and discern its characteristic features from random fluctuations. We show here that this can be done by means of *smooth models* and *smooth tests* originally introduced in statistical literature by [Neyman \[1937\]](#).

In the context of the analysis of high-resolution spectra, smooth tests are particularly

useful in that they naturally integrate estimation and inference to adequately handle uncertain background shapes. Moreover, when aiming to detect new signals, they provide a trade-off between tests of hypotheses and goodness-of-fit tests. Specifically, while tests of hypotheses, such as the Likelihood Ratio Test (LRT) [e.g., [Wilks, 1938](#)], are known to have high power towards a well-specified alternative model, their validity is severely undermined if the models under comparison are misspecified. In our case, this would occur, for instance, when searching for a few strong spectral line at the wrong position, or when the deviations from the background are due to the signal of a continuous, widely spread source. A classical approach to obviate this problem is that of relying on classical goodness-of-fit tests, such as Kolmogorov-Smirnov [[Kolmogorov, 1933](#), [Smirnov, 1939](#)], or Pearson’s χ^2 [[Pearson, 1900](#)], and which are expected to have some power against all possible deviations from the background model. However, since different tests have higher power towards different alternatives, they may not be sensible in detecting deviations towards the desired direction. Smooth tests, on the other hand, enjoy high power only towards a finite number of possible directions. Interestingly, the user can rely on suitable data-driven procedures to select the most “significant” directions towards which deviations from the background occur. Therefore, they effectively let the data decide “where to look”.

From a methodological perspective, the scope of our analysis is three folds. Our first goal is to assess the validity of our postulated background model and, when needed, provide a data-driven correction for it. Second, we aim to determine whether the Chandra LETGS+HRC-S Data provides significant evidence to conclude that spectral lines are present in addition to the background. Third, in regions where no significant deviations from the background model are detected, we proceed by setting upper limits on the intensity of the expected signals. In order to achieve these goals, we rely on suitably constructed *smooth tests* and likelihood-ratio tests.

2.3.1 Smooth models and smooth tests

Smooth tests are an inferential procedure for model assessment originally introduced by [Neyman \[1937\]](#). In the context of astronomical spectra, they are particularly advantageous because, in addition to validating the postulated astrophysical models, they also provide valuable insights on how misspecified models can be improved by means of smooth models. The latter consist of an “updated” version of the initial proposed model on the basis of the data observed. Sections [2.3.1-2.3.3](#) outline the main steps characterizing these procedures.

2.3.1.1 Data-driven Model Calibration via Smooth Models

Let X be a continuous random variable with an unknown cumulative distribution function (CDF) F , and an unknown probability density function (PDF) f . F corresponds to the true distribution from which our data were generated. Despite F being unknown, we aim to assess if it can be reasonably approximated by a hypothesised distribution G , with PDF g . The goal is to test the hypotheses

$$H_0 : F = G; \quad vs \quad H_1 : F \neq G \quad (2.2.3.1)$$

To perform the test in [\(2.2.3.1\)](#), we begin by rewriting the true density, f , as a function of the postulated model g , i.e.,

$$f(x) = g(x)d(G(x); G, F) \quad (2.2.3.2)$$

where $d(G(x); G, F) = \frac{f(x)}{g(x)}$; that is, $d(G(x); G, F)$ corresponds to the density ratio between f and g and can be expressed in the quantile domain as

$$d(u; G, F) = \frac{f(G^{-1}(u))}{g(G^{-1}(u))} \quad \text{with } u = G(x) \in [0, 1]$$

Table 2.3: Summary of the notation used throughout the manuscript.

Symbol	Definition
Section 2.3.1:	
X	Random variable with unknown distribution F
x	The value of X observed on the data
F	Unknown cumulative distribution function (cdf) of X
f	Unknown probability density function (pdf) of X
G	Hypothesised cdf of X
g	Hypothesised pdf of X
$d(u; G, F)$	ratio between the densities f and g evaluated at $u = G(x)$
$\{h_j(u)\}_{j \geq 0}$	A series of orthonormal basis functions on $[0, 1]$
θ_j	j^{th} coefficient of the orthonormal expansion for $d(u; G, F)$
$\hat{\theta}_j$	Estimate of θ_j
m	Number of terms used in the expansion for $d(u; G, F)$
$\hat{d}(u; G, F)$	Estimate of $d(u; G, F)$
$\hat{f}$	Smooth estimator of f
D_m	Deviance statistic
d_m	The value of D_m observed on the data
K_m	K-statistic
k_m	The value of K_m observed on the data
α	A pre-specified significance level
Section 2.3.2:	
M	Upper bound for the selection of m
$\hat{\theta}_{(j)}$	The j^{th} largest (in magnitude) among the coefficients θ_j
Section 2.3.3:	
$D_{(m)}$	Post-selection Deviance statistic
$d_{(m)}$	The value of $D_{(m)}$ observed on the data
$K_{(m)}$	Post-selection K-statistic
$k_{(m)}$	The value of $K_{(m)}$ observed on the data
Section 2.4.1:	
W_r	r th wavelength band of interest, $r = 1, \dots, 9$
B_{W_r}	True unknown background distribution for region W_r
b_{W_r}	True unknown background density for region W_r
g_{W_r}	Postulated uniform background density for region W_r
$\hat{B}_{W_r}$	Corrected background distribution for region W_r
$\hat{b}_{W_r}$	Corrected background density for region W_r
C_w	w th calibration region, $w = 1, \dots, 5$
$\hat{b}_{C_w}$	Corrected background density for combined region C_w
N_r	The size of the source-free sample for the region r
n_r	The size of the physics sample for the region r
Section 2.4.2:	
F_r	Unknown distribution for physics sample for region W_r
f_r	Unknown density for physics sample for region W_r
Section 2.4.3:	
s_r	Density of spectral line for region W_r , $r = 3, \dots, 9$
η_r	Relative intensity of the expected signal for region W_r
q_r	Density of physics sample for region W_r

where $G^{-1}(u)$ is the quantile function of X under model G . In statistics literature, $d(u; G, F)$ is also referred to as "comparison density" [e.g. [Parzen, 2004](#)]; such nomenclature is used to emphasise that $d(u; G, F)$ is the PDF of the random variable $U = G(X)$ [see proposition 3.1 in [Algeri, 2021](#)].

Under the assumption that the density $d(u; G, F)$ is square integrable on the unit interval, we can represent it via a series of orthonormal basis functions $\{h_j(u)\}_{j \geq 0}$, i.e.,

$$d(u; G, F) = 1 + \sum_{j=1}^{\infty} \theta_j h_j(u) \quad (2.2.3.3)$$

with $h_0(u) = 1$ and $\theta_j = \int_0^1 h_j(u) d(u; G, F) du$. The choice of the basis functions h_j in (2.2.3.3) is arbitrary. For example, in virtue of their simple implementation and generalisability to the discrete setting, a popular choice are the normalised shifted Legendre polynomials [e.g., [Neyman, 1937](#), [Ledwina, 1994](#)].

A *smooth model* can then be specified by truncating the series in (2.2.3.3) at a point m . Moreover, once a set of independent and identically distributed observations, $x_1, \dots, x_n$, has been collected, the coefficients θ_j are estimated via

$$\hat{\theta}_j = \frac{1}{n} \sum_{i=1}^n h_j(G(x_i)) = \frac{1}{n} \sum_{i=1}^n h_j(u_i). \quad (2.2.3.4)$$

It follows that an estimator of $d(G(x); G, F)$ in (2.2.3.3) is

$$\hat{d}(G(x); G, F) = 1 + \sum_{j=1}^m \hat{\theta}_j h_j(G(x)). \quad (2.2.3.5)$$

Finally, a *smooth estimator* of the true density f is

$$\hat{f}(x) = g(x) \hat{d}(G(x); G, F) = g(x) \left[1 + \sum_{j=1}^m \hat{\theta}_j h_j(G(x)) \right]. \quad (2.2.3.6)$$

As described in Section 2.3.1.2, this modeling strategy naturally leads to the inferential

framework needed to test the hypotheses in (2.2.3.1).

2.3.1.2 Smooth Tests

We begin by considering the simplified scenario where the point of truncation, m , in (2.2.3.5) is fixed and all the first m coefficients are included in our estimators in (2.2.3.5) and (2.2.3.6). This simplifying assumption allows us to introduce the main asymptotic results at the core of our inferential strategy. A detailed discussion on model selection and adequate inferential adjustments is postponed to Sections 2.3.2-2.3.3.

The decomposition in (2.2.3.2) allows us to rewrite the hypotheses in (2.2.3.1) as

$$\begin{aligned} H_0 : d(u; G, F) &= 1 \quad \text{for all } u \in [0, 1], \quad \text{versus} \\ H_1 : d(u; G, F) &\neq 1 \quad \text{for some } u \in [0, 1] \end{aligned} \tag{2.2.3.7}$$

Furthermore, by exploiting our estimator in (2.2.3.5), we can reformulate (2.2.3.7) as

$$\begin{aligned} H_0 : \theta_1 &= \dots = \theta_m = 0 \quad \text{versus} \\ H_1 : \theta_j &\neq 0 \text{ for at least one } j = 1, \dots, m \end{aligned} \tag{2.2.3.8}$$

Notice that H_0 in (2.2.3.7) implies H_0 in (2.2.3.8) but H_1 in (2.2.3.7) does not imply H_1 in (2.2.3.8). Whereas, H_1 in (2.2.3.8) implies H_1 in (2.2.3.7) and thus by testing H_0 and H_1 in (2.2.3.8) we can determine if the true distribution F deviates significantly from our hypothesised distribution G .

To test the hypotheses in (2.2.3.8), we rely on two different test statistics. A common choice in the context of smooth tests [e.g. [Ledwina, 1994](#), [Kallenberg and Ledwina, 1997a](#)] is the deviance statistic, i.e.,

$$D_m = n \sum_{j=1}^m \hat{\theta}_j^2; \tag{2.2.3.9}$$

whereas, in this manuscript, we propose, in addition to (2.2.3.9), the use of the *K-statistic*

which specifies as

$$K_m = \max_{j=1,\dots,m} n\widehat{\theta}_j^2.$$

It is easy to show [e.g. [Algeri, 2020a](#)] that, under H_0 , and as $n \rightarrow \infty$, the estimators $\widehat{\theta}_j$ in (2.2.3.4) converge in distribution to normally distributed random variable with mean zero and variance $1/n$. It follows that

$$D_m = \sum_{j=1}^m (\sqrt{n}\widehat{\theta}_j)^2 \xrightarrow{d} \chi_m^2.$$

Hence, an asymptotic p-value for the deviance test is

$$\text{p-value} = P(\chi_m^2 > d_m) \tag{2.2.3.10}$$

where d_m is the value of D_m observed on the data.

The asymptotic distribution of the K-statistic under H_0 can be derived by noticing that, asymptotically, K_m is distributed as the maximum of m χ_1^2 distributed random variables. Therefore, letting k_m be the observed value of K_m on the data, its asymptotic p-value is

$$\text{p-value} = 1 - P(\chi_1^2 \leq k_m)^m. \tag{2.2.3.11}$$

The respective proof can be found in [Appendix A.1](#).

The null hypothesis is rejected when the p-value is smaller than a pre-specified significance level α . For example, a discovery claim at 2σ significance corresponds to a level α of approximately 0.05. A rejection of the null hypothesis implies the postulated model g deviates significantly from the true model f .

2.3.2 Model selection

An important step of our analysis is that of selecting the basis functions to be included in our estimators (2.2.3.5) and (2.2.3.6). Here, we rely on a model selection process based on the Bayesian information criterion (BIC) for selection [e.g., Mukhopadhyay, 2017, Algeri, 2021], and can be summarised as follows:

- i. Choose a suitably large value M (usually 10).
- ii. Obtain the estimates $\hat{\theta}_1, \dots, \hat{\theta}_M$ as in (2.2.3.4).
- iii. Rearrange $\hat{\theta}_j^2$ in decreasing order, i.e.,

$$\hat{\theta}_{(1)}^2 \geq \hat{\theta}_{(2)}^2 \geq \dots \geq \hat{\theta}_{(M)}^2,$$

with $\hat{\theta}_{(j)}^2$ denoting the j th largest squared coefficient.

- iv. Choose the largest m that maximises

$$BIC(m) = \sum_{j=1}^m \hat{\theta}_{(j)}^2 - \frac{m \log n}{n}. \quad (2.2.3.1)$$

The procedure outlined here is data-dependent, that is, the value of m selected may vary over different samples. It is therefore necessary to account for the randomness associated with the model selection process. This can be done as described in Section 2.3.3.

2.3.3 Post-selection inference adjustments

Traditional inference is typically constructed under the assumption that the model under study has been selected independently from the data available. In many practical scenarios, however, a data-driven selection procedure is typically implemented, and thus, classical inferential results fail to hold due to the randomness associated with the selection process. While one can easily overcome this problem by relying on data splitting [e.g.

Moran, 1973, Cox, 1975], or bootstrapping [e.g. Algeri and Zhang, 2021], in our setting, it is possible to identify suitable post-selection adjustments for the p-values in (2.2.3.10) and (2.2.3.11).

For what concerns the deviance statistic, a post-selection adjusted p-value can be constructed as described in Algeri [2021]. Specifically, let

$$D_{(m)} = \sum_{j=1}^m \hat{\theta}_{(j)}^2$$

be the deviance statistic obtained as the sum of the squares of the m largest estimated coefficients, with m selected via the BIC criterion in (2.2.3.1), and denote with $d_{(m)}$ be its value observed on the data. An adjusted p-value for the test in (2.2.3.8) is

$$\text{p-value}_{adj,M} = P(\chi_M^2 > d_{(m)}). \quad (2.2.3.1)$$

Notice that, conversely from (2.2.3.10), in (2.2.3.1) the observed value of the deviance statistic is compared to a χ_M^2 rather than χ_m^2 . In virtue of its conservativeness, we refer to the correction in (2.2.3.1) as "naive correction".

An alternative approach consists of applying the usual Bonferroni's correction [e.g. Miller, 1977] and typically used in the context of multiple hypothesis testing. In this case, the adjusted deviance p-value is given by

$$\text{p-value}_{adj,B} = \max\left\{M \cdot P(\chi_m^2 > d_{(m)}), 1\right\} \quad (2.2.3.2)$$

Finally, the post-selection K-statistic is specified as

$$K_{(m)} = \max_{j=1,\dots,m} n\hat{\theta}_{(j)}^2,$$

where m is the value which maximises the BIC in (2.2.3.1), and let $k_{(m)}$ be the value of

$K_{(m)}$ observed on the data. A post-selection adjustment for the respective p-value can be implemented as formalised in Theorem 1. The respective proof is given in the Appendix A.2.

Theorem 1. *As $n \rightarrow \infty$, an upper bound for the limit of $P(K_{(m)} \geq k_{(m)} | H_0)$ is*

$$p\text{-value}_{adj,K} = 1 - P(\chi_1^2 \leq k_{(m)})^M. \quad (2.2.3.3)$$

A comparison in terms of power and type I error of tests based on (2.2.3.1), (2.2.3.2) and (2.2.3.3) is postponed to Section 2.4.3.

2.4 Statistical data analysis of RT Cru

Here we describe how smooth tests can be implemented to thoroughly study the X -ray spectra of RT Cru. As outlined in Section 2.1, our strategy is that of conducting a separate analysis on each of the 9 wavelength bands listed in Table 2.1.

Specifically, in Section 2.4.1, we first demonstrate how the shape of the background can be extracted on the basis of the information contained in the source-free, background-only dataset. Second, in Section 2.4.2.2, we apply the same method to the source spectrum. This step allows us to determine whether a significant difference can be established between the source spectrum and the background spectrum, without imposing any distributional assumption on potential signals. Third, if no difference can be detected, in Section 2.4.2.3 we complement our analysis by means of Likelihood Ratio Tests (LRTs) to assess for the presence of the spectral lines listed in Table 2.1. For both smooth tests and the LRTs, adequate adjustments for multiple hypothesis testing are implemented in order to control for the probability of a false discovery across all the regions of the spectrum being tested. The statistical properties of the testing procedures considered are investigated in Section 2.4.3. Finally, since no spectral line is detected, we set upper limit on the counts necessary for a detection in each of the regions considered.

2.4.1 Data-driven background corrections

When assessing the validity of a postulated background model and implementing adequate data-driven corrections, it is necessary to ensure the size of the source-free sample considered is sufficiently large to reduce the uncertainty associated with the estimation of the background model. Therefore, here we proceed to combine the regions of interest W_r in Table 2.1 into five “calibration” intervals C_w , $w = 1, \dots, 5$, and defined as in Table 2.4.

The choice of considering larger regions ensures that, for each of the newly constructed calibration intervals C_3 - C_5 considered, the corresponding source-free sub-sample includes at least 20,000 events.

The background is assumed to be flat over the entire search area and thus we proceed by testing this assumption on each sub-region by means of the tools described in Section 2.3. The adjusted p-values to test the uniformity of the background, for each calibration region C_w , $w = 1, \dots, 5$ have been computed as in (2.2.3.1), (2.2.3.2) and (2.2.3.3). The results are summarised in Table 2.4.

In three out of the five calibration regions in Table 2.4, the data distribution is consistent with the uniform model (the adjusted p-values are all equal to one). Whereas on C_2 and C_4 , the flat background model is rejected. Therefore, we proceed to implement a data-driven correction for them as described in Section 2.3.2. The newly estimated background densities are

$$\begin{aligned}\hat{b}_{C_2}(x) &= 0.2424 - 0.0038x, & x \in [5.5, 10.2] \text{ \AA}, \\ \hat{b}_{C_4}(x) &= 1.3994 - 0.0250x, & x \in [14.6, 17.4] \text{ \AA}.\end{aligned}\tag{2.2.4.1}$$

Our ultimate goal is to detect signals over the regions of interest, W_r , defined as in Table 2.1, it is, therefore, necessary to convert the re-calibrated background densities obtained for C_2 and C_4 into corrections for the background distribution over W_r , $r = 1, \dots, 9$.

Specifically, let B_{W_r} be the true (unknown) background distribution on region W_r and let b_{W_r} be its density. We denote with $\hat{b}_{W_r}$ our estimate of b_{W_r} . Since no deviation from uniformity was observed over C_1 , C_3 and C_5 , no background update was performed on the

	Wavelength range in Å	m	Bonferroni (Sidak)	K (Sidak)	Naive (Sidak)
C_1	1.65-2.05	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
C_2	5.5-10.2	1	3.7687e-10 (1.8843e-09)	3.7687e-10 (1.8843e-09)	3.6797e-06 (1.8398e-05)
C_3	11.5-13.0	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
C_4	14.6-17.4	1	0.0004 (0.0020)	0.0004 (0.0020)	0.0796 (0.3396)
C_5	18.5-23.0	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)

Table 2.4: Summary of our background correction results for each calibration region. The first two columns are the combined calibration regions and corresponding combined wavelength ranges. The third column reports the number, m , of coefficients selected out of $M = 10$, for each of the combined regions considered. The p-values adjusted for post-selection using the Bonferroni, K, and naive corrections and corrected for multiple hypothesis testing via Sidak (see 2.2.4.1) are given in the fourth, fifth, and sixth columns, respectively.

regions W_1, W_3, W_6, W_7, W_8 and W_9 . Hence we set,

$$\hat{b}_{W_r}(x) = \frac{1}{u_r - l_r}, \text{ for } r = 1, 3, 6, 7, 8, 9,$$

with l_r and u_r be, respectively, the lower and upper bounds of the wavelength range of region W_r . Whereas, we exploit (2.2.4.1) to derive the newly calibrated background models for regions W_2, W_4 and W_5 . Since W_2 and C_2 coincide, we set $\hat{b}_{W_2}(x) = \hat{b}_{C_2}(x)$ and defined in (2.2.4.1). Whereas, the background models for regions W_4 and W_5 are

$$\begin{aligned} \hat{b}_{W_4}(x) &= 2.4749 - 0.0441x, \quad x \in [14.6, 15.15] \text{ Å}, \\ \hat{b}_{W_5}(x) &= 1.1900 - 0.0212x, \quad x \in [16.2, 17.4] \text{ Å}, \end{aligned} \tag{2.2.4.2}$$

and they have been derived as described in Appendix A.3.

2.4.2 Signal detection via smooth tests and LRTs

2.4.2.1 Testing multiple regions and adequate corrections

Let F_r be the true distribution from which the observations in the source sample were generated over region W_r , $r = 1, \dots, 9$, and denote with f_r its density. We assume that, if no

Regions of interest (W_r)	m	Bonferroni (Sidak)	K (Sidak)	Naive (Sidak)
W_1	3	0.0001 (0.0011)	0.0071 (0.0397)	0.0045 (0.0621)
W_2	3	1.0816e-18 (1.0817e-17)	2.7907e-15 (2.9976e-14)	3.3306e-15 (2.4980e-14)
W_3	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_4	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_5	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_6	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_7	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_8	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)
W_9	0	1.0000 (1.0000)	1.0000 (1.0000)	1.0000 (1.0000)

Table 2.5: Summary of our signal detection results for each region of interest using smooth tests. The second column shows the number, m , of coefficients selected out of $M = 10$, for each of the nine regions considered. The p-values adjusted for post-selection using the Bonferroni, K, and naive corrections and corrected for multiple hypothesis testing via Sidak (see 2.2.4.1) are given in the third, fourth and fifth columns, respectively.

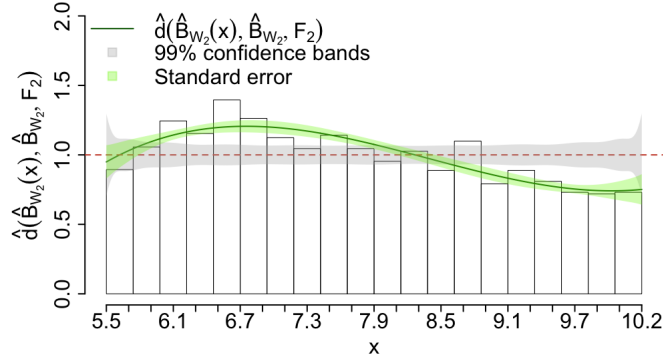


Figure 2.3: Comparison density plot and histogram of the transformed physics data, $u_i = \hat{B}_{W_2}(x_i)$, $i = 1, \dots, 3311$, for region W_2 . The hypothesised model is the corrected background density $\hat{b}_{W_2}(x)$ in (2.2.4.2). The estimated comparison density in (2.2.4.2) is displayed using a green solid line. The green bands are the estimated standard errors of $\hat{d}(\hat{B}_{W_2}(x); \hat{B}_{W_2}, F_2)$ obtained by simulating samples from the estimator in (2.2.4.2). The red dashed line corresponds to the null hypothesis that $d(\hat{B}_{W_2}(x); \hat{B}_{W_2}, F_2) = 1$, i.e., the data distribution on W_2 is not significantly different from the estimated background model $\hat{b}_{W_2} = \hat{b}_{C_2}$ in (2.2.4.1). The grey bands correspond to the 99% confidence bands under the null. The plot suggests that there exists a significant departure of the true data distribution from $\hat{B}_{W_2}(x)$, within the range of $x \in [6.1, 7.9]\text{\AA}$.

Regions of interest (W_r)	Local p-values	Sidak's correction
W_3	0.4810	0.9899
W_4	0.1143	0.5724
W_5	0.3247	0.9359
W_6	0.0385	0.2402
W_7	0.2612	0.8799
W_8	0.5000	0.9922
W_9	0.5000	0.9922

Table 2.6: Local P-values and adequate multiple hypothesis testing adjustments when testing for spectral lines via LRTs.

signal is present, the true density f_r is well approximated by our background estimate $\hat{b}_{W_r}$ obtained as in Section 2.4.1. Hence, in Section 2.4.2.2, we employ smooth tests to assess if significant deviations from $\hat{b}_{W_r}$ occur on any of the nine regions considered. For regions where no significant departure from the background model is detected, we will proceed with a thorough search for spectral lines via LRTs, as described in Section 2.4.2.3.

Notice that, in both our smooth tests and LRTs analyses, multiple tests are conducted simultaneously over different regions. It is therefore necessary to correct the resulting p-values for multiple hypotheses testing in order to ensure that the probability of a false discovery across the entire spectrum is no larger than the predetermined α . Hereinafter, we will refer to the latter as the *global significance* level.

Under the hypothesis of independence of the tests being conducted, α relates to the *local significance* level, α_r , of incorrectly rejecting the background-only hypothesis on region W_r in that

$$\alpha = 1 - \prod_{r=1}^R (1 - \alpha_r).$$

The reader is directed to [Algeri \[2016\]](#) for a self-contained review on the problem of multiple hypothesis testing in searches for new physics also referred to in physics literature as *look-elsewhere effect*.

In our setup, since the regions in which our spectrum is divided are non-overlapping, we

can assume independence among the tests performed separately on each of them. Hence, we can adequately adjust the corresponding p-values by relying on Sidak's correction [e.g., [Kuehl, 2000](#), Sec 3.4]. Specifically, letting p_r be the (smooth test or LRT) p-value obtained for region r , the corresponding Sidak's correction is

$$p_r^{Sidak} = 1 - (1 - p_r)^R \quad (2.2.4.1)$$

with R being the total number of regions being tested. Notice that the same corrections have also been implemented in [Table 2.4](#) when assessing the validity of flat backgrounds on the source-free sample.

2.4.2.2 Signal detection via smooth tests

Similarly to [Section 2.4.1](#), we rely on the deviance and the K-statistics (suitably adjusted for post-selection, as described in [Section 2.3.3](#)) to perform our tests.

The discovery results on each of the regions of interest are summarised in [Table 2.5](#). The second column of the [Table 2.5](#) corresponds to truncation point, m , selected via the *BIC* criteria in [\(2.2.3.1\)](#), when considering a maximum of $M = 10$ coefficients. The Sidak adjusted p-values calculated as in [\(2.2.4.1\)](#) with $R = 9$ are reported in parenthesis. Recall that our estimator and test statistics are constructed by sorting the estimated coefficients $\hat{\theta}_{(j)}$ in [\(2.2.3.4\)](#). Hence, choosing m to be the point of truncation implies that inference and estimation are performed considering only the m largest estimated coefficients.

For regions W_1 and W_2 , the Bonferroni and K adjusted p-values are smaller than the global significance level $\alpha = 0.05$ even after implementing Sidak's correction. Whereas, the p-values adjusted for post-selection by means of the naive method, only detect significant deviations over region W_2 . As discussed in detail in [Section 2.4.3](#), this result is not surprising since the naive approach is the most conservative among the three methods considered. The adjusted p-values for the remaining regions are all equal to one. This implies that our smooth tests analysis allow us to claim that deviations from the background occur only in

regions W_1 and W_2 .

The results obtained on the region W_1 are consistent with those of [Luna and Sokoloski \[2007\]](#). These are known features, arising in the inner accretion disk perhaps, and their detection here is a confirmation that the method is working.

To gain a better understanding of the nature of the deviation from the background model detected on region W_2 , we rely on the so-called *Comparison Density plot* or *CD-plot* [e.g., [Algeri and Zhang, 2021](#)] shown in Figure 2.3. The CD-plot allows us to visualise where the data distribution deviates significantly from the hypothesised distribution (in our case, the recalibrated background density). It displays the estimated comparison density (dark green solid line) and which, for region W_2 , specifies as

$$\hat{d}(u; \hat{B}_{W_2}, F_2) = 0.9500 + 2.1577u - 5.2749u^2 + 2.9179u^3 \quad (2.2.4.2)$$

where $u = \hat{B}_{W_2}(x)$. Whereas, the green bands are the standard errors of $\hat{d}(\hat{B}_{W_2}(x); \hat{B}_{W_2}, F_2)$ obtained by simulating from the estimator in (2.2.4.2) as described in [Algeri and Zhang \[2021\]](#). The grey bands correspond to the 99% confidence bands under the null hypothesis of background only. If the estimated comparison density is within the confidence bands, over the entire range considered, we conclude that there is no significant departure from the background model. Conversely, we expect significant deviations to occur in regions where the estimate lies outside the confidence bands. It is worth emphasising that the CD-plot provides us a representation of the data in the quantile domain; that is, it displays the transformed data $u_i = \hat{B}_{W_2}(x_i)$, $i = 1, \dots, 3311$, and their estimated density. Such representation ensures that the most substantial departures of the data distribution from the expected model are magnified and those due to random fluctuations are smoothed out. More details on the construction and discussion of the CD-plot can be found in [Algeri and Zhang \[2021, Algorithm 1\]](#) and [Algeri \[2020a, Section V A\]](#).

For the specific case of Region W_2 , the CD-plot in Figure 2.3 suggests that significant departures from $\hat{b}_{W_2}$ occur within the range of $x \in [6.1, 7.9]\text{\AA}$. This detection, however,

cannot be attributed to any known spectral features, and it corresponds to the signature of the Chandra optics Iridium absorption edge and is detectable because we are not assuming a particular spectral model here. It is worth pointing out that the departure below one at $x \gtrsim 9.1 \text{ \AA}$ is due to the fact that, since the $\hat{d}(\hat{B}_{W_2}(x); \hat{B}_{W_2}, F_2)$ is the estimate of a density function, its integral is ≈ 1 (adequate corrections to ensure that the integral of the resulting estimate is exactly 1 can be implemented as described, for instance, in [Algeri and Zhang \[2021\]](#)). It follows that any peak or departure above 1 is compensated by a departure below one over the remaining portion of the search region (in our case, for $x \gtrsim 9.1 \text{ \AA}$).

2.4.2.3 Signal detection via LRTs

To further investigate the possibility of undetected spectral lines in regions W_3 - W_9 we proceed by implementing an inferential analysis based on LRTs.

As noted in the introduction, the main advantage of smooth tests is that they allow us to detect deviations from the background occurring at the m “most significant” directions identified by the BIC criterion in (2.2.3.1). Therefore, they are completely model-independent, that is, they do not require us to specify the position of the spectral lines or a model for their shape. Nonetheless, when such information is available, one can assess if deviations from the background occur in the direction of the specified signal model by means of the LRT, which is known to be the most powerful test when a model for the signal is provided.

We begin by assuming that, if a spectral line is present on region W_r , its density is¹

$$s_r(x, \tilde{\mu}_r) = k(\tilde{\mu}_r) \left\{ 1 + \left[\frac{(x - \tilde{\mu}_r)}{0.05} \right]^2 \right\}^{-2.5} \quad (2.2.4.3)$$

¹This is the so-called line response function (LRF) that describes the response of a grating to an infinitesimally narrow line. It is empirically modeled as a modified Lorentzian, also called a Beta-profile; see Equation 9.1 of the *Chandra* Observatory Proposers’ Guide, https://cxc.harvard.edu/proposer/POG/html/chap9.html#tth_sEc9.3.3

with

$$\tilde{\mu}_r \sim N(\mu_r, 0.005^2)$$

where $\tilde{\mu}_r$, $r = 3, \dots, 9$, is the expected signal location (see column 4 of Table 2.1) and $k(\tilde{\mu}_r)$ is the normalising terms depends on the random variable $\tilde{\mu}_r$.

We can then specify a suitable model for regions W_3 - W_9 as

$$q_r(x, \eta_r, \tilde{\mu}_r) = (1 - \eta_r) \hat{b}_{W_r}(x) + \eta_r s_r(x, \tilde{\mu}_r) \quad (2.2.4.4)$$

where $\hat{b}_{W_r}$ is the background model adequately corrected, as needed, following the method described in Section 2.4.2, whereas η_r is the relative intensity of the expected signal. To test if a spectral line with density s_r is present on region W_r we test the hypotheses

$$H_0 : \eta_r = 0 \quad \text{versus} \quad H_1 : \eta_r \in (0, 1], \quad r = 3, \dots, 9 \quad (2.2.4.5)$$

by means of the test statistic

$$\lambda_r = -2 \sum_{i=1}^n \left\{ \log \hat{b}_{W_r}(x_i) - \max_{\eta_r \in (0, 1]} \log q_r(x_i, \eta_r, \tilde{\mu}_r) \right\}. \quad (2.2.4.6)$$

Notice that the parameterisation in (2.2.4.4) implicitly assumes that, under the alternative hypothesis, the true density, f_r , is equal to q_r . Therefore, tests based on such model are unable to capture any deviation from $\hat{b}_{W_r}(x)$ other than the one in the direction of $s_r(x, \tilde{\mu}_r)$ (hence the analysis in Section 2.4.2.2).

Under suitable regularity conditions [e.g., Algeri, 2020b], the null distribution of λ_r in (2.2.4.6) can be approximated by a χ^2 [Wilks, 1938]. However, since η_r in (2.2.4.4) lies on the boundary of its parameter space, under H_0 , the χ^2 approximation is no longer valid. Nonetheless, in this setting, the null asymptotic distribution of the LRT is known to be a zero 50% of the times and it is χ_1^2 the remaining 50% [Chernoff, 1954]. Therefore, it is sufficient to divide by a factor of two the “usual” p-value obtained using the χ_1^2

approximation.

The results of the LRT for each of the seven regions considered are summarised in Table 2.6. In the second column, we report the local p-values, that is, the p-values obtained without adjusting for multiple hypothesis testing whereas the global p-values adjusted by means of Sidak's correction in (2.2.4.1) with $R = 7$ are reported in the third column. All the regions apart from W_6 have local p-values larger than 0.05. For region W_6 , the local p-value 0.0385. However, once adequately adjusted via Sidak's correction, the p-value for W_6 is no longer significant. This result emphasises the importance of implementing adequate corrections when conducting multiple test simultaneously to avoid false discovery claims.

2.4.3 Statistical properties and construction of upper limits

In order to confirm the validity of the analyses performed in Sections 2.4.2.2-2.4.2.3, it is important to investigate the statistical properties (power and probability of type I error) for each of the inferential procedure considered. We will then exploit the power curves obtained to construct suitable upper limits as prescribed in Kashyap [2010].

We investigate the power of the proposed statistical tests by simulating Monte Carlo samples from q_r in (2.2.4.4) for different intensity levels η_r . The size of each simulated dataset is the same as that of the original sample observed on the seven regions considered (i.e., n_r with $r = 3, \dots, 9$ in Table 2.1). Using the simulated data, we implement the methods described in Section 2.3 as well as the LRT procedure discussed in Section 2.4.2.3. When the true value of η_r is non-zero, the null hypothesis in (2.2.4.5) is correctly rejected if the p-value is smaller than the pre-specified significance level. We repeat this process 5000 times, and at each replicate, we record the number of rejections of the null. The proportion of rejections corresponds to the simulated power for the (non-zero) values of η_r fixed when conducting the simulation. We repeat the simulation for different values of η_r in order to generate a series of power curves, one for each region considered. The latter is displayed in Figure 2.4. The same analysis is also repeated by adjusting the respective p-values

for multiple hypothesis testing via Sidak’s correction in (2.2.4.1) at each replicated of the simulation. As expected, adjusting for multiple hypothesis testing leads to a reduction of the power. Not surprisingly, the power of the LRT is higher than any other procedure considered. That is because, smooth tests do not require the specification of the signal model and simply assess for the presence of deviations from the background.

The probabilities of type I error for each of the procedures considered (obtained by setting $\eta_r = 0$) are reported in Tables A.1-A.2. The simulated type I error using Bonferroni adjustment for the deviance statistic often exceeds the significance level of 0.05 in the context of local analysis. It also exceeds the significance level $\alpha_r = 0.0073$ in the global analysis; that is, when setting the probability of false discovery across all seven regions to be $\alpha = 0.05$. Conversely, the K-statistic and the naive post-selection adjustment for the deviance perform well in controlling the specified significance level, even though the naive adjustment appears to be excessively conservative (the respective probability of type I error is always zero in the global analysis). This is also reflected in the power curves reported in Figure 2.4. The naive approach is the most conservative among the three, whereas Bonferroni exhibits the highest power.

Finally, upper limits are constructed by inverting the power curves of the LRT at 50% and 90% and multiplying the resulting η_r value for the sample size, n_r (see Table 2.1) of the discovery region W_r , for $r = 3, \dots, 9$. The results are summarised in Table 2.7. We can interpret the Sidak adjusted upper limits as the number of samples from the expected signal needed to achieve the specified power when testing simultaneously regions W_3 - W_9 . For example, for region W_3 , our 50% upper limit computed using the LRT after Sidak correction is 40 (39.42). This tells us that if a spectral line at position 12.131 Å was present, we would need 40 events in this location (out of the 730 observed in the entire W_3 region) to be able to detect such spectral line with power 50%, while simultaneously looking for spectral lines in the regions $W_4, \dots, W_9$. Whereas, if we were interested in designing a future observation targeting solely region W_3 , our 50% upper limit computed

Regions (W_r)	50% upper limits via LRT		90% upper limits via LRT	
	Local	Sidak adjusted	Local	Sidak adjusted
W_3	29.93	39.42	48.91	53.29
W_4	20.00	26.43	32.36	39.52
W_5	24.02	30.14	35.32	43.80
W_6	22.62	28.08	34.71	39.39
W_7	17.90	24.17	29.71	35.98
W_8	17.84	24.80	30.30	36.25
W_9	37.83	21.87	63.57	76.83

Table 2.7: 50% and 90% upper limits on different regions using the LRT, with and without Sidak’s correction. The 50% upper limits are calculated by the proportion of the expected lines η to achieve the power 0.5 (solid line in Figure 2.4) times the sample sizes n_r . Similarly, the 90% upper limits are calculated by the proportion of the expected lines η to achieve the power 0.9 times the sample sizes n_r .

	Shifted lines [Å]	Local p-values	Sidak’s correction	LRT 50% Upper limits	LRT 90% Upper limits	Sidak’s LRT 50% Upper limits	Sidak’s LRT 90% Upper limits
W_4	14.908	0.5000	0.9921	20.00	32.60	26.92	39.77
W_5	16.930	0.0017	0.0118	24.54	34.85	30.14	43.80

Table 2.8: Local p -values for the LRT and adequate multiple hypothesis testing adjustments when investigating the presence of the 16.93 Å emission line in W_5 .

using the (local) LRT is 30 (29.93). This tells us that if a spectral line at position 12.131 Å was present, we would need 30 events at such a location to detect it with power 50%, and assuming that no other test on other regions is conducted at the same time. Similar interpretations can be given to the 90% upper limits and for other regions.

For the sake of comparison, the upper limits obtained by means of smooth tests are reported in Tables A.3-A.4. Not surprisingly, since smooth tests do not rely on the specification of a model for the signal, they are more conservative than the LRT. For example, for region W_3 , the 50% upper limits computed using the Bonferroni, K-statistic and the naive methods, and adjusted via Sidak for multiple hypothesis testing lead to 53, 64, and 68 events, respectively.

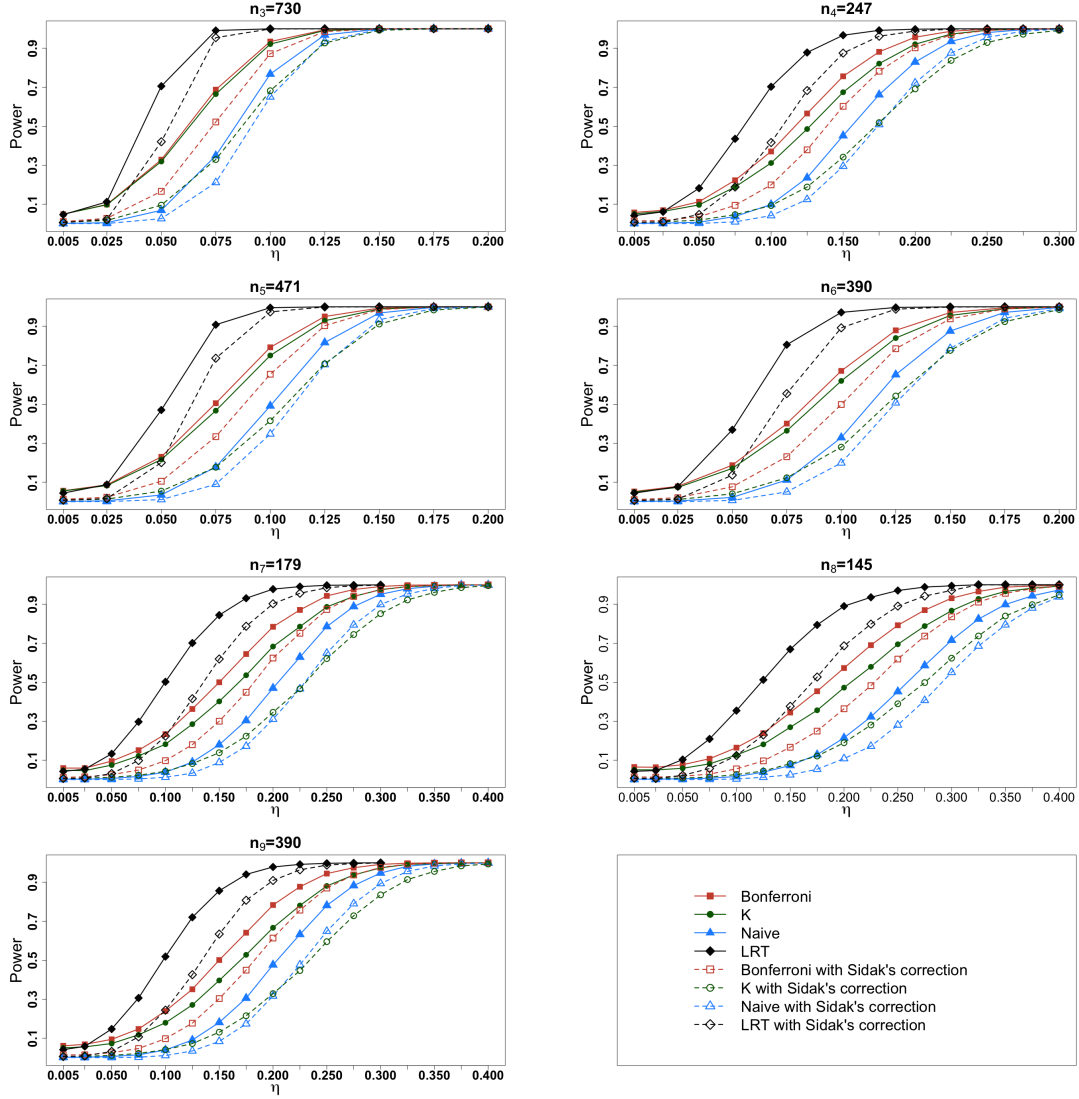


Figure 2.4: Simulated power curves for regions W_3 - W_9 . Red-square and blue-triangle lines correspond to the power curves for the deviance adjusted for post-selection via Bonferroni and the naive approach, respectively. The green-circle lines are the power curves for the K-statistic suitably adjusted for post-selection. The black-diamond lines are the power curves for the LRT. The solid lines correspond to the local power curves, with $\alpha_r = 0.05$, for each $r = 3, \dots, 9$. The dashed lines are the power curves adjusted for multiple testing using Sidak's correction to ensure a global significance of $\alpha = 0.05$ (local significance $\alpha_r = 0.0073$, for each $r = 3, \dots, 9$).

2.5 Summary and discussion

We have developed a novel method to detect weak signals distinct from a smooth background in high-resolution photon-counting spectra. This approach anticipates difficulties likely to be encountered in the coming era of calorimeter spectra. The method is implemented to work with unbinned photon lists that allow the full available spectral resolution to be used, though a modification to use binned spectra is viable from an algorithmic perspective, and it is the subject of future work.

The statistical methodology presented here is particularly advantageous at high resolution because a precise specification of the source model spectrum is often not possible, as the information available in the data usually exceeds that in the models proposed. Here we show that one can indeed exploit this phenomenon by modeling and estimating the “gap” between the (potentially misspecified) model available and the true spectrum using smooth functions like shifted Legendre polynomials. On this note, it is worth emphasizing that, as proven in [Algeri \[2020a\]](#), the closer the postulated model is to the truth, the more accurate (less biased) is the estimate of the latter. It follows that, in principle, one could avoid specifying a model for the spectrum and estimate it by means of smooth functions. Nonetheless, if a model is available (even if misspecified), it should be used in order to reduce the gap between the proposed model and the truth.

The implementation currently ignores spectral calibration products like the effective area and the redistribution matrices, and therefore cannot be applied to CCD resolution spectra. Furthermore, the method relies on a comparison between the smooth model description of the source-free background and the source+background datasets, so it cannot be applied to cases where the background is contaminated by the source or where the background is not smoothly varying.

We first note that our method easily detects the presence of significant source emission in passband W_1 . This is not surprising, as these lines have been identified and analyzed by several studies (e.g., [Luna and Sokoloski \[2007\]](#) resolved it clearly in HETGS+ACIS-S

spectra; and [Danezhkar \[2021a\]](#) successfully modeled the triplet in the same dataset that we use). The chance that a random fluctuation can produce a detectable departure from the background is assessed as $p \lesssim 10^{-2}$ after accounting for multiple hypothesis tests (see [Table 2.5](#)). This serves as a validation of the method, in that a line complex known to exist is correctly found.

An important characteristic of our method is that it is not limited to narrow lines. If the source spectrum has a different shape compared to the background spectrum, we expect that to be detected, i.e., the method allows for a differentiation between the continuum and the background. This was achieved serendipitously in passband W_2 , where the existence of a feature is flagged with high significance ($p \ll 10^{-10}$; see [Table 2.5](#)). The detected feature (see [Figure 2.2](#)) is a characteristic of the response of the *Chandra* mirror coating, exhibiting an edge due to the absorption of incident photons by Iridium. Such features are usually not visible in residuals in standard spectral analyses because the modeling directly incorporates the sensitivity of the telescope system and suitably weights the incident photon spectrum. Because our method does not include such calibration products, it can be used to locate and study such features, incorporating known calibration (as is done for line emission in [Equations 2.2.4.3, 2.2.4.4](#)), thus potentially providing independent measures of uncertainty on the calibration.

We have explored the possibility of detecting emission from some prominent soft X-ray emission lines ([Table 2.1](#); see also [Section 2.2.1](#)) and find that in no case are any of the lines we searched for detectable. Here we compare the model predicted counts obtained using the spectral model of [Danezhkar \[2021a\]](#) with a nominal estimate of background corrected counts estimate obtained via a Bayesian estimate (as in [Freeman 2002](#)), with the 50% upper limits obtained using smooth tests and post-selection adjustments ([Section 2.4](#); [Table 2.7](#)):

W_3 : This passband is centered on the Ne X resonance line at 12.14 Å. The model predicts ≈ 16 counts in this region, and a measurement of the counts over the width of the LRF yields an estimate of ≈ 20 counts, with a 68% uncertainty interval of $\approx \pm 10$.

This would normally be considered a good match between model and data, but the upper limit at this location is ~ 30 counts, increasing to ~ 40 counts when corrections due to multiple hypotheses testing are included, and we conclude that the line is not detectable. This assessment is supported by a visual inspection of the observed counts spectrum, where an enhancement in counts at the location of the line is not apparent.

W_4 and W_5 : The dominant features in these bands are expected to be from the Fe XVII lines at 15.014 Å, 17.051 Å and 17.096 Å. Note that these lines have peak emissivity at ≈ 5 MK, whereas the best-fit spectral model suggests a plasma temperature of ≈ 14 MK. The emissivities at such high temperatures are reduced by a factor of 30, but they remain the strongest lines from a thermal spectrum and have approximately similar intensities. The predicted counts from the best-fit model is ≈ 10 counts in each, and the measured values are ≈ 20 and ≈ 6 counts in W_4 and W_5 respectively. These estimates are *prima facie* inconsistent with theoretical expectations, though the posterior distributions of the intensities do overlap. The local upper limits are 20 and 24 counts, increasing to 26 and 30 counts at the 50% level (see Table 2.7) after accounting for multiple hypotheses, for W_4 and W_5 respectively. This suggests that the 15 Å line is a borderline detectable feature, but the 17 Å lines are not detected.

However, notice that in the counts spectrum in Figure 2.2, there is a relatively strong feature at ≈ 16.93 Å, about 2100 km s^{-1} blueward of 17.051 Å (marked with a brown cross-hatch). We find that the feature is detected with a significance of $p \lesssim 0.002$ locally and at $p \lesssim 0.012$ after multiple hypothesis correction (see Table 2.8), with a 50% upper limit < 25 counts. The estimated signal strength for this feature is $16.9^{+26.9}_{-8.5}$ counts (the bounds represent 68% HPD uncertainty interval as in Table 2.2), consistent with the putative upper limit. Inner accretion disk velocities of $\sim 10^3 \text{ km s}^{-1}$ are plausible for symbiotic variables [see, e.g., Reimers and Cassatella, 1985], though the required shift here is twice as much. A hot spot that can achieve

such a large blue shift would be located at a height of $\approx 0.06 R_{\odot}$ above a $1.4 M_{\odot}$ white dwarf, which is not infeasible. But if this were a blue-shifted Fe XVII $\lambda 17.051$ line, we should expect its $15 \text{ \AA}$ counterpart to also be similarly blue shifted. The location of this putative blue-shifted $15 \text{ \AA}$ line is also marked with a brown cross-hatch in the W_4 panel in Figure 2.2. We find no detectable feature at this location, with a multiple-hypothesis corrected 50% upper limit at < 27 counts (Table 2.8; the estimated counts are $0 \leq 8.2$). We thus conclude that the feature at $16.93 \text{ \AA}$ is *not* a blue-shifted thermal line. We discuss its possible origin further below.

W_6 : The strongest feature in this passband is the O VIII resonance line at $18.96 \text{ \AA}$. As can be seen in Figure 2.2, there is an apparent enhancement at where the line is expected to be, though the predicted model counts are more than $2\times$ smaller compared to the estimated counts, and indeed the predicted counts value falls outside the 68% uncertainty bounds (Table 2.2). Interestingly, the local upper limit shows that the deviation is significant with $p < 0.05$, but when corrections due to multiple hypothesis tests are included, there is no significant evidence for a detection. The 50% upper limit is < 28 , and we conclude that the data are not of suitable quality to constrain the Oxygen emission.

W_7 - W_9 : These three bands include the He-like density sensitive triplet of O VII at 21.6 , 21.8 , and $22.1 \text{ \AA}$. We find no evidence for a detection of any of the lines, with the corrected $p \approx 1$ in all three cases, and with 50% upper limits in the $21 - 24$ counts range. However, this case provides a cautionary illustration of the pitfalls of ignoring detectability. The nominal estimates of the counts in each of the lines shows that the intercombination line at $21.8 \text{ \AA}$ has a brightness of ≈ 10 counts, higher than both the resonance and forbidden line brightnesses. Without the upper limits analysis demonstrating that none of the lines are detectable, it is easy to over-interpret the high brightness estimate of the intercombination line as indicative of a very high density ($\gg 10^{13} \text{ cm}^{-3}$; see, e.g., Smith 2001) in the emitting plasma. While the presence of

such high densities cannot be formally excluded, the current dataset is of insufficient quality to place a definitive constraint. Observations with new observatories like XRISM, Athena, or LEM are necessary to resolve this question.

The hard emission in the 6-7 keV range likely originates close to the accretor. The 6.4 keV Fe K α line could originate from a region of the boundary layer of the accretion disk or in a bright spot on the disk; The 6.7 keV and the 7 keV, can be attributed to He-like and H-like iron emission lines, and could be produced via photoionization and collisional ionization/excitation mechanisms in the hot plasma, either in the vicinity of the WD or the inner-jet region, or both (see, e.g., [Kennea 2009](#), [Karovska 2010](#), [Eze 2014](#)). We speculate that the same photoionization or reflection spectrum photon reprocessing could be the source of the line feature detected at 16.93 Å. Several plausible candidate ionic species ranging from Fe XIII to Fe XIX exist in the XSTAR [[Kallman, 1999](#)] line list within $\approx 500 \text{ km s}^{-1}$ of the detected line. The reflection spectrum model `xillver` [[García, 2013](#)] shows the existence of emission localized at 16.93 Å for flat incident spectra ($\Gamma \approx 1-1.2$), moderate Fe abundances ($A_{\text{Fe}} \approx 0.5-1$), ionization ($\log \xi \approx 2-3$). Intriguingly, a feature at the same location was found by [Hemphill \[2021\]](#) in the *Chandra* LETG spectrum of the ultracompact X-ray binary 4U 1626–67; they are also unable to identify the line, and suggest a photoionized emission as the origin. Unlike the double-peaked emission that [Hemphill](#) find that is reminiscent of an accretion-disk origin, the profile of the line in RT Cru is sharply single-peaked, suggesting an origin in a hot spot or an inner jet. A detailed study of the origin of this line, and a search for other such lines is beyond the scope of both this article and the *Chandra* dataset; we expect that observations obtained with calorimeter resolutions will confirm the presence of such lines and lead to a better understanding of the emission from RT Cru and other symbiotics.

In conclusion, we present a method designed to correctly analyze high-resolution spectra affected by high background. The method is currently implemented for photon lists, but extensions to binned spectra such as those obtained by XMM/RGS are possible in principle

and are the subject of future work. We first characterize the background and determine whether the background-contaminated source spectrum differs significantly in its shape over any passband. This allows us to detect both line emission as well as continuum variations. We properly correct for multiple hypothesis tests, and where a spectral line is not detected, we place upper limits on the source intensity. From a statistical perspective, in addition to the novel analytical framework presented here, we introduce a new test statistic to perform smooth tests and adequate post-selection inferential adjustments. The latter are obtained by simply inverting the power function of our post-selection adjusted smooth tests and obtained via Monte Carlo simulations.

We apply this method to the *Chandra* LETGS+HRC-S spectrum of the symbiotic star RT Cru. Because of the high background, traditional techniques designed to detect and identify lines and compute fluxes are unfeasible for these data. Consequently, a global fit was carried out by [Danežkar \[2021a\]](#) using a power-law and a thermal component, supplemented by Gaussian lines to model the Fe K α region. We first validate our method by confirming the detection of the Fe-line triplet in the 6-7 keV region, and further demonstrate the ability of the method to find variations in the continuum spectrum by detecting the Iridium M edge present due to the mirror coating in all *Chandra* observations.

Since the clear detection of low energy spectral lines would be an important diagnostic for the emission mechanisms that operate in symbiotic systems, we have employed smooth models and smooth tests to correct, when needed, the uncertain background model and to determine whether source spectra show distinct differences in shape compared to the background in several wavelength bands where spectral lines are expected to be present. None of the expected thermal lines are detected, and the upper limits are all larger than the predicted model counts. With the analysis of the O VII He-like triplet, we illustrate how checking for detectability allows us to avoid mistakenly claiming that the emitting plasma is at high density. We thus conclude that the fitted model considered by [Danežkar \[2021a\]](#) is adequate in its overall characteristics to explain the emission from RT Cru.

We serendipitously and unambiguously detect emission in a line-like feature located at $16.93 \text{ \AA}$ (0.732 keV), and conclude that it cannot be attributed to thermal plasma emission. We speculate that it is derived from a hitherto unmodeled photoionization of reflection spectrum component. Future observations with missions like XRISM [[Ishisaki, 2022](#)], Athena [[Barret, 2022](#)], etc., using calorimeter-level spectral resolutions and high effective areas are necessary to detect and model such processes.

Chapter 3

A new class of asymptotically distribution-free smooth tests

3.1 Overview

Let X be a continuous random variable with cumulative distribution function (CDF) Q and probability density function (PDF) q with support $\mathcal{X} \subseteq \mathbb{R}$. We are interested in assessing whether the unknown distribution Q belongs to a family of distribution functions $G_{\boldsymbol{\beta}}$, with PDF $g_{\boldsymbol{\beta}}$, differentiable with respect to $\boldsymbol{\beta} \in \mathcal{B} \subseteq \mathbb{R}^p$. To tackle this problem, we may consider an alternative model of the form

$$g_{\boldsymbol{\beta}}(x) \left\{ 1 + \sum_{j=1}^m \theta_{j\boldsymbol{\beta}} h_{j\boldsymbol{\beta}}(x) \right\} \tag{3.3.1.1}$$

which incorporates the null density, $g_{\boldsymbol{\beta}}$, as a special case. The functions $\{h_{j\boldsymbol{\beta}}\}_{j=1}^{\infty}$ form an orthonormal basis in $L^2(G_{\boldsymbol{\beta}})$; hence

$$\int_{\mathcal{X}} h_{i\boldsymbol{\beta}}(x) h_{j\boldsymbol{\beta}}(x) dG_{\boldsymbol{\beta}}(x) = \mathbb{1}_{\{i=j\}},$$

where $\mathbb{1}_{\{\cdot\}}$ denotes the indicator function. Assuming that the density ratio $\frac{q}{g_\beta} \in L^2(G_\beta)$, as $m \rightarrow \infty$, the term in the curly brackets in (3.3.1.1) provides an orthonormal expansion for $\frac{q}{g_\beta}$. Hence, the coefficients of such an expansion correspond to

$$\theta_{j\beta} = \int_{\mathcal{X}} h_{j\beta}(x) \frac{q(x)}{g_\beta(x)} dG_\beta(x) = \int_{\mathcal{X}} h_{j\beta}(x) dQ(x), \quad \text{for all } j = 1, \dots, m.$$

A smooth test aims to assess the validity of g_β by testing the null hypothesis:

$$H_0 : \theta_{1\beta} = \dots = \theta_{m\beta} = 0, \text{ for some } \beta \in \mathcal{B} \quad (3.3.1.2)$$

against smooth alternatives of the form (3.3.1.1), where at least one of the coefficients $\theta_{1\beta}, \dots, \theta_{m\beta}$ is nonzero. One could also rely on an exponential tilt to express the deviations of the alternative from g_β through the orthonormal set $\{h_{j\beta}\}_{j=1}^m$, as in the original formulation of smooth tests proposed by Neyman [1937]. Other formulations may involve normalizing the expansion in (3.3.1.1) to ensure that the resulting density is non-negative [e.g., Gajek, 1986]. Here, we rely on the alternative in (3.3.1.1), proposed by Barton [1953], by virtue of its simplicity, and we implicitly assume that the $\theta_{j\beta}$ coefficients are suitably constrained to ensure (3.3.1.1) is non-negative¹. Nevertheless, the methods described in what follows can be adapted to other specifications of the alternative model for which the testing problem can be formulated as in (3.3.1.2).

Several test statistics have been proposed in the literature for testing (3.3.1.2). Prominent examples include the score test statistic [Neyman, 1937, Barton, 1953], which consists of the sum of squares of the estimated coefficients in the expansion in (3.3.1.1), and typically employed for testing simple hypotheses; the generalized score test statistic² [Javitz,

¹For example, as discussed in Barton [1953], when $h_{j\beta}$ are the normalized shifted Legendre polynomials evaluated at the probability integral transform $G_\beta(x)$ and $m = 1$, non-negativity of (3.3.1.1) is achieved whenever $|\theta_{1\beta}| < \frac{1}{3}$; whereas, if $m = 2$, we must have $|\sqrt{5}\theta_{2\beta}| \geq |\sqrt{3}\theta_{1\beta}|$.

²Also known as the generalized smooth test statistic or efficient score test statistic.

1975, Thomas and Pierce, 1979] used for testing composite hypotheses; and the order selection test statistic [Aerts et al., 1999] which naturally accounts for the selection of m in (3.3.1.1). Their asymptotic null distributions are known, but may require a very large sample size to reach such limits [Klar, 2000]. Data-driven smooth tests [Ledwina, 1994] focus on the situation in which the basis functions to be included in the expansion in (3.3.1.1) are determined by data-driven selection criteria – such as BIC [e.g., Ledwina, 1994, Kallenberg and Ledwina, 1995, 1997b, Inglot et al., 1997, Inglot and Ledwina, 2006a]. Under the null model G_{β} , classical smooth test statistics are asymptotically distribution-free.

For all the above-mentioned tests, asymptotic distribution-freeness arises as a consequence of the inherent structure of the test statistics, while the choice of basis $\{h_{j\beta}\}_{j=1}^{\infty}$ remains somewhat arbitrary. We propose a shift of perspective in which a careful construction of the basis ensures that any test statistic based on its elements is asymptotically distribution-free by design. Such a construction relies on the *Khmaladze-2 (K2) transform*³ introduced by Khmaladze [2016], which enables both accurate and computationally efficient inference.

The K2 transform uses unitary operators to map model-dependent projections into a “standard projection” that is arbitrarily chosen by the user. This standard projection does not vary with the model being tested and is asymptotically distribution-free under the null hypothesis. In just over a decade, the K2 transformation has enabled several advances in the goodness-of-fit literature across various contexts—from testing discrete and multivariate parametric distributions [Khmaladze, 2013, 2016] to regression [Khmaladze, 2021, 2017] and grouped data analysis [Algeri and Khmaladze, 2024a]. By enabling its use in the context of smooth tests, this work extends its usefulness beyond goodness-of-fit. Specifically, it provides applied scientists with a new suite of distribution-free tests that

³The name “Khmaladze-2 transform” is used to distinguish the transformation employed in this manuscript and proposed in Khmaladze [2016] from an earlier “Khmaladze transform” introduced by the same author in Khmaladze [1982].

offer a middle ground between goodness-of-fit tests and directional tests such as Neyman-Pearson.

While asymptotically distribution-freeness is especially desirable when testing complex distributions and/or when computationally intensive parameter estimation is needed — as is often the case in physics and astronomy [e.g., [Balázs, 2017](#), [Cusin, 2018a, 2019](#), [Zhang, 2026](#), [Algeri, 2026](#)] — in other settings, applied scientists may choose to derive the null distributions of the test statistics by means of resampling methods. In [Section 3.3](#), we discuss a computationally efficient alternative to the classical parametric bootstrap, called the *projected bootstrap*. Specifically, one can avoid re-estimating the unknown parameter β on each bootstrap replicate by exploiting a projection of the underlying empirical process. The efficacy of this approach when deriving the distribution of classical goodness-of-fit statistics estimated via maximum likelihood has been investigated numerically by [Algeri \[2022\]](#). Here, a formal proof is provided to confirm the validity of this approach in the context of smooth tests and for general Z -estimators.

3.2 Smooth tests and the function-parametric empirical process

Let $X_1, \dots, X_n$ be independent and identically distributed (IID) random variables with CDF Q . Consider the Hilbert space $L^2(G_\beta) = \{h : \langle h, h \rangle_{G_\beta} < \infty\}$, with inner product:

$$\langle h, h' \rangle_{G_\beta} = \int_{\mathcal{X}} h(x)h'(x)dG_\beta(x).$$

The *function-parametric* empirical process $v_{\beta,n}(h)$ indexed by function h in $L^2(G_\beta)$ is defined as:

$$v_{\beta,n}(h) = \int_{\mathcal{X}} h(x)dv_{\beta,n}(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [h(X_i) - \mathbb{E}_{G_\beta}[h(X_i)]]$$

and

$$v_{\beta,n}(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [\mathbb{1}_{\{X_i \leq x\}} - G_{\beta}(x)]$$

is the classical empirical process. Hence, $v_{\beta,n}(h) = v_{\beta,n}(x)$ when $h(z) = \mathbb{1}_{\{z \leq x\}}$.

Most statistics proposed in the literature to test (3.3.1.2) can be specified as functionals of the process $v_{\beta,n}(h)$ and its projections. In particular, let

$$\mathbf{u}_{\beta} = \nabla_{\beta} \ln g_{\beta} \quad \text{and} \quad \Gamma_{\beta} = \langle \mathbf{u}_{\beta}, \mathbf{u}_{\beta}^T \rangle_{G_{\beta}}$$

be, respectively, the score function and the Fisher information matrix of G_{β} . Define the orthonormalized score function, with coordinates in $L^2(G_{\beta})$, as

$$\mathbf{b}_{\beta} = [b_{\beta_1}, \dots, b_{\beta_p}]^T = \Gamma_{\beta}^{-1/2} \mathbf{u}_{\beta},$$

where $\Gamma_{\beta}^{-1/2}$ denotes the principal square root matrix of Γ_{β}^{-1} . Denote with $\{\tilde{h}_{j\beta}\}_{j=1}^m$ the “residuals” of $\{h_{j\beta}\}_{j=1}^m$ in (3.3.1.1) after an orthogonal projection onto $\mathbf{b}_{\beta}$, i.e.,

$$\tilde{h}_{j\beta} = h_{j\beta} - \mathbf{b}_{\beta}^T \langle \mathbf{b}_{\beta}, h_{j\beta} \rangle_{G_{\beta}} = h_{j\beta} - \sum_{k=1}^p b_{\beta_k} \langle h_{j\beta}, b_{\beta_k} \rangle_{G_{\beta}}, \quad \text{for } j = 1, \dots, m. \quad (3.3.2.1)$$

Deviations within the parametric family $\{G_{\beta}, \beta \in \mathcal{B}\}$ occur in the direction given by the the score function $\mathbf{u}_{\beta}$. Therefore, since the functions $\tilde{h}_{j\beta}$ are obtained by projecting $h_{j\beta}$ onto the orthogonal complement of the tangent space given by the span of $\mathbf{u}_{\beta}$, they capture deviations outside the hypothesized parametric family.

The most widely used statistics for testing (3.3.1.2) is the generalized score test statistic, [cfr. Javitz, 1975, Thomas and Pierce, 1979], defined as

$$\hat{S}_{m,n} = \sum_{i=1}^m \sum_{j=1}^m v_{\hat{\beta}_n,n}(\tilde{h}_{i\hat{\beta}_n})(\Sigma_{m,\hat{\beta}_n}^{-1})_{ij} v_{\hat{\beta}_n,n}(\tilde{h}_{j\hat{\beta}_n}) \quad (3.3.2.2)$$

where $\Sigma_{m,\hat{\beta}_n}^{-1}$ is the inverse of estimated variance-covariance matrix, $\Sigma_{m,\hat{\beta}_n}$, of elements

$$(\Sigma_{m,\hat{\beta}_n})_{ij} = \mathbb{E}_{G_{\beta}} \left[v_{\beta,n}(\tilde{h}_{i\beta}) v_{\beta,n}(\tilde{h}_{j\beta}) \right] \Big|_{\beta=\hat{\beta}_n},$$

which we assume to be a continuous function in β , and $\hat{\beta}_n$ denotes a consistent estimator of β (see condition (A1) in Section 3.3).

When m , the number of basis functions to be used in (3.3.1.1) and (3.3.2.2), is determined on the basis of the data observed, the so-called order selection test statistic [cfr. Aerts et al., 1999] can be used to incorporate the choice of the order m directly into its formulation. Specifically, let m be the maximizer of the selection criterion

$$\hat{S}_{m,n} - C_{\alpha,n}m, \quad \text{with } m = 0, \dots, M_n, \quad (3.3.2.3)$$

in which $\hat{S}_{m,n}$ is the generalized score test statistic in (3.3.2.2) or its non-normalized counterpart:

$$\hat{S}_{m,n} = \sum_{j=1}^m v_{\hat{\beta}_n,n}^2(\tilde{h}_{j\hat{\beta}_n}). \quad (3.3.2.4)$$

Let M_n denote the maximum number of basis functions to be considered, which could either be fixed or grow to infinity as $n \rightarrow \infty$. Selecting m using the criterion in (3.3.2.3) corresponds to choosing the first m basis functions among $\{h_{j\beta}\}_{j=1}^{M_n}$ to be included in the expansion in (3.3.1.1).

The underlying idea of a test based on (3.3.2.3) is to reject the null hypothesis in (3.3.1.2) if the criterion in (3.3.2.3) is larger than zero for some m in $\mathcal{M}_n = \{1, \dots, M_n\}$ – that is, when an alternative model of the form in (3.3.1.1) is favored over G_{β} . Equivalently, G_{β} is rejected when the *order selection test statistic* defined as

$$\hat{T}_n = \max_{m \in \mathcal{M}_n} \left\{ \frac{\hat{S}_{m,n}}{m} \right\}, \quad (3.3.2.5)$$

is such that $\widehat{T}_n \geq C_{\alpha,n}$, for some constant $C_{\alpha,n}$ that controls the significance level α of the test.

The criterion in (3.3.2.3) can be generalized so that basis functions with indexes j in any subset B of $\{1, \dots, M_n\}$, not necessarily ordered from $j = 1, \dots, m$, are selected [Thas, 2010, pp.103-107]. In this case, the set of indices B is chosen to maximize

$$\widehat{S}_{B,n} - C_n|B|, \quad \text{with } B \subseteq \mathcal{M}_n, \quad (3.3.2.6)$$

where $|B|$ denotes the cardinality of B and $\widehat{S}_{B,n}$ is the counterpart of $\widehat{S}_{m,n}$ with the summations in (3.3.2.2) and (3.3.2.4) taken over the indexes in B . The corresponding *subset selection test statistic* specifies as

$$\widetilde{T}_n = \max_{B \subseteq \mathcal{M}_n: B \neq \emptyset} \left\{ \frac{\widehat{S}_{B,n}}{|B|} \right\}. \quad (3.3.2.7)$$

Compared to $\widehat{T}_n$ in (3.3.2.5), $\widetilde{T}_n$ offers greater flexibility in the selection of the basis functions and thus accommodates a wider range of possible alternative models to be employed in the expansion in (3.3.1.1).

The limiting distribution of the subset selection statistic in (3.3.2.7) cannot be easily derived. Therefore, a computationally efficient algorithm is required to simulate its null distribution, especially when M_n is large. In the next section, we introduce an alternative to the classical parametric bootstrap to perform this task.

3.3 Smooth tests via projected bootstrap

The classical parametric bootstrap requires re-estimating the unknown parameter β on each bootstrap sample to account for the variability introduced by estimation. In some instances, however, a repeated estimation can make the procedure computationally intensive. The so-called “projected bootstrap” [Algeri, 2022, Algeri and Khmaladze, 2024a]

overcomes this limitation by exploiting the projection structure induced by parameter estimation to avoid repeating the estimation of the parameter on each bootstrap replicate. As shown in what follows, such a projection arises in the context of smooth tests rather organically.

Let the true (but unknown) parameter vector under the null be β_0 , that is, if H_0 in (3.3.1.2) holds, then $Q \equiv G_{\beta_0}$. Let $\mathcal{N} \subset \mathcal{B}$ be the closure of a neighborhood of the true parameter value β_0 . Denote with $\mathcal{L}(G_\beta)$ the subspace of $L^2(G_\beta)$ given by

$$\mathcal{L}(G_\beta) = \left\{ h \in L^2(G_\beta) : \langle h, \mathbf{1} \rangle_{G_\beta} = 0 \right\}.$$

In what follows, we focus our attention on the subset of functions in $\mathcal{L}(G_\beta)$ that depend on β , with $\beta \in \mathcal{N}$. Such functions primarily intervene in our study by either characterizing the alternative in (3.3.1.1) or defining the estimating equations for β . We assume such functions to be continuously differentiable in β and to satisfy the usual regularity conditions used in standard asymptotic arguments [cf. Van der Vaart, 2000, Ch.5] and which imply the following:

(A1) Let $\hat{\beta}_n$ be a consistent root of the system of estimating equations

$$v_{\hat{\beta}_n, n}(\psi_{\hat{\beta}_n}) = 0, \tag{3.3.3.1}$$

where ψ_β is a p -dimensional vector function with linearly independent components in $\mathcal{L}(G_\beta)$. Denote with $\nabla_\beta h_\beta$ the gradient of h_β taken with respect to β . The Taylor expansion

$$v_{\hat{\beta}_n, n}(h_{\hat{\beta}_n}) = v_{\beta_0, n}(h_{\beta_0}) + (\hat{\beta}_n - \beta_0)^T \nabla_\beta v_{\beta, n}(h_\beta) \Big|_{\beta=\beta_0} + o_P(1) \tag{3.3.3.2}$$

is valid.

(A2) For any $h_\beta \in \mathcal{L}(G_\beta)$ and $\beta \in \mathcal{N}$,

$$\nabla_\beta \int_{\mathcal{X}} h_\beta(x) dG_\beta(x) = \int_{\mathcal{X}} \nabla_\beta [h_\beta(x) dG_\beta(x)].$$

To ease the intuition, in what follows, we assume (A1)-(A2) directly.

As first proven by Khmaladze [1980], when replacing β with $\hat{\beta}_n$ in $v_{\beta,n}(h_\beta)$, the resulting process is asymptotically equal to a projection of the latter evaluated at the true parameter value β_0 :

Proposition 1 (Khmaladze projection, Khmaladze [1980]). *If (A1)-(A2) hold, then, under H_0 ,*

$$v_{\hat{\beta}_n,n}(h_{\hat{\beta}_n}) = v_{\beta_0,n}(\Pi h_{\beta_0}) + o_P(1), \quad (3.3.3.3)$$

where

$$\Pi h_\beta = h_\beta - \psi_\beta^T \langle \mathbf{u}_\beta, \psi_\beta^T \rangle_{G_\beta}^{-1} \langle \mathbf{u}_\beta, h_\beta \rangle_{G_\beta} \quad (3.3.3.4)$$

is a projection of h_β .

Proof. We give a proof adapted to the notation and setting of the present manuscript. Under Assumption (A2), we have,

$$\begin{aligned} \frac{1}{\sqrt{n}} \nabla_\beta v_{\beta,n}(h_\beta) \Big|_{\beta=\beta_0} &= \frac{1}{\sqrt{n}} \int_{\mathcal{X}} \nabla_\beta [h_\beta(x) dv_{\beta,n}(x)] \Big|_{\beta=\beta_0} \\ &= \frac{1}{\sqrt{n}} \int_{\mathcal{X}} [\nabla_\beta h_\beta(x)] \Big|_{\beta=\beta_0} dv_{\beta_0,n}(x) - \int_{\mathcal{X}} h_{\beta_0}(x) [\nabla_\beta g_\beta(x)] \Big|_{\beta=\beta_0} dx \\ &= - \langle \mathbf{u}_{\beta_0}, h_{\beta_0} \rangle_{G_{\beta_0}} + o_P(1), \end{aligned} \quad (3.3.3.5)$$

in which the last equality holds by the law of large numbers. Substituting the right-hand side of such an equality into (3.3.3.2) gives:

$$v_{\hat{\beta}_n,n}(h_{\hat{\beta}_n}) = v_{\beta_0,n}(h_{\beta_0}) - \sqrt{n}(\hat{\beta}_n - \beta_0)^T \langle \mathbf{u}_{\beta_0}, h_{\beta_0} \rangle_{G_{\beta_0}} + o_P(1).$$

Since the components of ψ_β belong to $\mathcal{L}(G_\beta)$, the asymptotic equality above also holds for $v_{\hat{\beta}_{n,n}}(\psi_{\hat{\beta}_n})$ and equating it to zero leads to:

$$\sqrt{n}(\hat{\beta}_n - \beta_0) = \langle \mathbf{u}_{\beta_0}, \psi_{\beta_0}^T \rangle_{G_{\beta_0}}^{-1} v_{\beta_0,n}(\psi_{\beta_0}) + o_P(1).$$

Combining the last two expressions yields:

$$v_{\hat{\beta}_{n,n}}(h_{\hat{\beta}_n}) = v_{\beta_0,n}(h_{\beta_0}) - v_{\beta_0,n}(\psi_{\beta_0}^T) \langle \mathbf{u}_{\beta_0}, \psi_{\beta_0}^T \rangle_{G_{\beta_0}}^{-1} \langle \mathbf{u}_{\beta_0}, h_{\beta_0} \rangle_{G_{\beta_0}} + o_P(1),$$

which can be equivalently expressed as in (3.3.3.3)-(3.3.3.4) by linearity of Π . Additionally, it is easy to verify that $\Pi\Pi h_\beta = \Pi h_\beta$, thus, Π is a projection operator. ■

In general, the projection given by Π is not orthogonal; that is because

$$\Pi\psi_\beta = 0 \quad \text{and} \quad \langle \mathbf{u}_\beta, \Pi h_\beta \rangle_{G_\beta} = 0$$

while neither $\Pi\mathbf{u}_\beta$ nor $\langle \psi_\beta, \Pi h_\beta \rangle_{G_\beta}$ are zero. From the above expressions, however, it is clear that Π defines an orthogonal projection when $\psi_\beta \equiv \mathbf{u}_\beta$, that is, when the estimation is conducted via maximum likelihood.

In the context of smooth tests, Proposition 1 implies that the processes $v_{\hat{\beta}_{n,n}}$ and $v_{\beta_0,n}$, indexed respectively by $\tilde{h}_{j\hat{\beta}_n}$ and $\Pi\tilde{h}_{j\beta_0}$, are asymptotically equivalent. Furthermore, the functions $\tilde{h}_{j\beta}$ are orthogonal to $\mathbf{u}_\beta$ by construction (see Equation (3.3.2.1)). It follows that

$$\Pi\tilde{h}_{j\beta_0} = \tilde{h}_{j\beta_0} - \psi_{\beta_0}^T \langle \mathbf{u}_{\beta_0}, \psi_{\beta_0}^T \rangle_{G_{\beta_0}}^{-1} \langle \mathbf{u}_{\beta_0}, \tilde{h}_{j\beta_0} \rangle_{G_{\beta_0}} = \tilde{h}_{j\beta_0}.$$

This fact, combined with the asymptotic equality in (3.3.3.3), brings us to the following proposition.

Proposition 2. Assume that (A1) and (A2) hold for each $\tilde{h}_{j\beta}$, $j \in \mathcal{M}_n$, then

$$v_{\beta_0,n}(\tilde{h}_{j\beta_0}) = v_{\beta_0,n}(\Pi\tilde{h}_{j\beta_0}) = v_{\hat{\beta}_{n,n}}(\tilde{h}_{j\hat{\beta}_n}) + o_P(1). \quad (3.3.3.6)$$

To understand the computational advantages entailed by Proposition 2, denote with $\hat{\beta}_{\text{obs}}$ the estimate of β obtained on the observed data by solving (3.3.3.1). In the parametric bootstrap, such an estimate plays the same role as β_0 in (3.3.3.6). Let $\hat{\beta}_n^{(b)}$ be the parameter estimate obtained on the b -th bootstrap sample generated from $G_{\hat{\beta}_{\text{obs}}}$. Equation (3.3.3.6) implies that the empirical process indexed by functions $\tilde{h}_{j\hat{\beta}_n^{(b)}}$ is asymptotically equal to the process indexed by $\tilde{h}_{j\hat{\beta}_{\text{obs}}}$. In other words, the projection structure characterizing the functions $\tilde{h}_{j\hat{\beta}_{\text{obs}}}$ makes the effect of re-estimating the parameter asymptotically negligible. Thus, for sufficiently large n , the user must evaluate the process $v_{\hat{\beta}_{\text{obs}},n}(\tilde{h}_{j\hat{\beta}_{\text{obs}}})$ over different bootstrap samples but does not need to repeat the estimation of β at each replicate. It follows that this *projected bootstrap* procedure can always make the simulation more efficient. As demonstrated in the next section with an example, the projected bootstrap is particularly advantageous when parameter estimation is time-consuming.

Finally, the consistency of the projected bootstrap in recovering the true distribution of the function-parametric empirical process $v_{\beta_0,n}(\tilde{h}_{j\beta_0})$ is guaranteed under the same conditions needed for the classical parametric bootstrap [cfr. Babu and Rao, 2004].

3.3.1 Empirical illustrations of smooth tests via the projected bootstrap

Let G_β be the asymmetric Laplace distribution with unknown asymmetry parameter β . The corresponding PDF, g_β , can be expressed as:

$$g_\beta(x) = \begin{cases} \frac{\sqrt{2}}{\sigma} \frac{\beta}{1+\beta^2} \exp\left(-\frac{\sqrt{2}\beta}{\sigma}(x-\theta)\right), & \text{for } x \geq \theta \\ \frac{\sqrt{2}}{\sigma} \frac{\beta}{1+\beta^2} \exp\left(\frac{\sqrt{2}}{\sigma\beta}(x-\theta)\right), & \text{for } x < \theta \end{cases}. \quad (3.3.3.1)$$

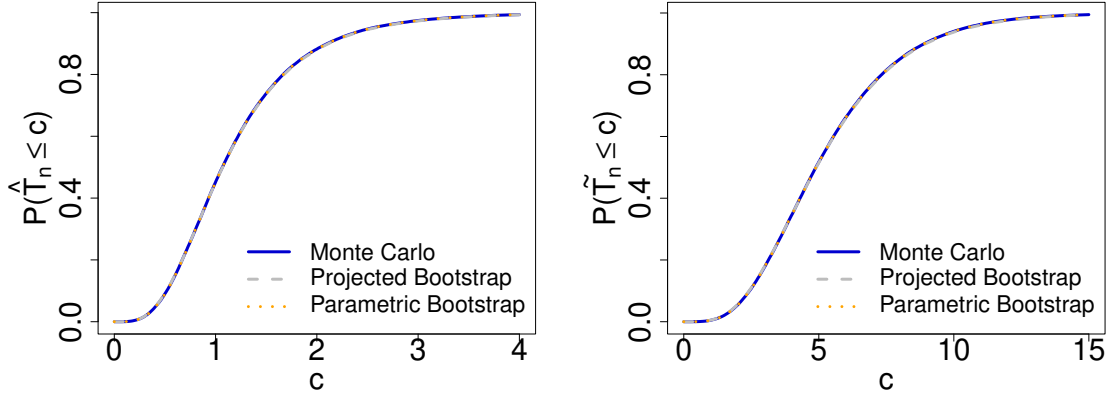


Figure 3.1: Comparing the simulated null distributions of the order selection test statistic in (3.3.2.5) (Left), and subset selection test statistic in (3.3.2.7) (Right) via the parametric bootstrap (orange dotted lines), the projected bootstrap (grey dashed lines), and Monte Carlo (blue solid lines).

In what follows, the parameters θ and σ are set equal to 0 and 2, respectively, and are assumed to be known.

A dataset $\mathbf{x}_n = (x_1, \dots, x_n)$ of $n = 100$ observations is generated from the density specified in (3.3.3.1), with the true value of the asymmetry parameter set to be $\beta_0 = 0.1$. In the present simulation study, such a dataset plays the same role as the observed data sample in a real-data analysis. The maximum likelihood estimator $\hat{\beta}_n$ of β based on $\mathbf{x}_n$ is obtained by solving the following equation [Kotz et al., 2002]:

$$1 - \frac{2\beta^2}{(1 + \beta^2)} + \frac{\sqrt{2}}{\sigma} \left[\frac{1}{n\beta} \sum_{j=1}^n (x_j - \theta)^- - \frac{\beta}{n} \sum_{j=1}^n (x_j - \theta)^+ \right] = 0, \quad (3.3.3.2)$$

where

$$(x_j - \theta)^- = -\min(0, x_j - \theta) \quad \text{and} \quad (x_j - \theta)^+ = \max(0, x_j - \theta).$$

Given the complexity of finding the solution of (3.3.3.2), it is anticipated that using the projected bootstrap – in which the estimator is obtained only once rather than recalculated

on each bootstrap sample – will expedite the simulation procedure.

To test the hypothesis in (3.3.1.2), we consider the order selection statistic in (3.3.2.5) and the subset selection statistic in (3.3.2.7). Their null distributions are simulated via three methods: the classical parametric bootstrap, the projected bootstrap, and Monte Carlo with data generated from the true distribution. By incorporating a simulation of Monte Carlo samples, we aim to determine whether the two bootstrap procedures considered can accurately recover the true null distribution of the test statistics for a sample size of $n = 100$. All simulations are conducted using 100,000 replicates. The orthonormal basis functions $\{h_{j\beta}\}_{j=1}^{M_n}$ are chosen as compositions of the normalized shifted Legendre polynomials h_j on $[0, 1]$ with the CDF G_β , i.e., $h_{j\beta} = h_j \circ G_\beta$. The maximum number of basis functions M_n considered is 10.

The results of the simulation, presented in Figure 3.1, demonstrate that the simulated null distributions of the order selection and subset selection test statistics are consistent across all three methods considered. This indicates that both bootstrap procedures successfully recover the true null distribution of the test statistics, even with a sample size as small as 100 observations. In terms of computational efficiency, however, the projected bootstrap method is significantly faster than the classical parametric bootstrap. Specifically, simulating the null distributions of the order selection and subset selection statistics using 100,000 replicates required approximately 9 minutes of CPU time using the projected bootstrap, less than a fourth of the time needed by the classical parametric bootstrap, which required approximately 37 minutes.

The p-values for testing (3.3.1.2) are also computed by comparing the value of the order selection and subset selection statistics calculated on $\mathbf{x}_n$ with their null distribution simulated by means of the projected bootstrap. For the order selection statistic, the p-value obtained is 0.843, and for the subset selection statistic, it amounts to 0.963.

3.4 Distribution-free smooth tests via K2 transform

In general, statistics given by functionals of the process $v_{\beta,n}(\tilde{h}_{j\beta})$ are not distribution-free. Thus, while the projected bootstrap introduced in Section 3.3 can reduce the computational burden of simulating their null distribution, a different simulation must be implemented for each model being tested.

This section demonstrates that another route to distribution-freeness is available to practitioners. Instead of combining the values of the process $v_{\beta,n}(\tilde{h}_{j\beta})$ in a rather specific – and possibly forceful – manner to construct, at most, a few distribution-free statistics, the distribution-free property can be retrieved by ensuring that the empirical process $v_{\beta,n}(\tilde{h}_{j\beta})$ is itself asymptotically distribution-free under the null. Such an approach guarantees that the limiting null distribution of all its functionals is also distribution-free, thereby providing the user with an entire family of asymptotically distribution-free statistics. The tool that enables such a construction is the so-called Khmaladze-2 (K2) transform introduced by Khmaladze [2016].

In classical nonparametric goodness-of-fit testing, the probability integral transform is the transformation commonly employed to map the empirical process into the uniform empirical process, which is known to be asymptotically distributed as a standard Brownian bridge. Distribution-freeness is therefore achieved by means of a simple change of variable. One can think of the K2 transform also as a change of variable, but in functional space. In the parametric setting, such a change of variable allows us to retrieve asymptotic distribution-freeness by mapping a wide range of different projected Brownian motions, arising when estimating different distributions, into the same “standard” projected Brownian motion of choice. Moreover, when applied in the context of smooth tests, the K2 transform implicitly guides the construction of an orthonormal basis for $L^2(G_\beta)$ that guarantees the distribution-freeness of the empirical process indexed by functions in such a basis, as well as that of all its functionals.

Let F_γ be a “reference distribution” that has the same support $\mathcal{X}$ as G_β . For the

moment, assume that γ and β have the same dimension; an extension to the case where γ has fewer parameters than β will be discussed in Section 3.4.1. The distribution F_γ constitutes the starting point of our construction and should be chosen to be simple – that is, easy to simulate from, differentiate, and evaluate. Consider the Hilbert space $L^2(F_\gamma) = \{\phi : \langle \phi, \phi \rangle_{F_\gamma} < \infty\}$, with inner product

$$\langle \phi, \phi' \rangle_{F_\gamma} = \int_{\mathcal{X}} \phi(x) \phi'(x) dF_\gamma(x),$$

and the empirical process

$$v_{\gamma,n}(\phi) = \int_{\mathcal{X}} \phi(x) dv_{\gamma,n}(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [\phi(X_i) - \mathbb{E}_{F_\gamma}[\phi(X_i)]]$$

indexed by such functions. Let $\mathbf{a}_\gamma = [a_{\gamma_1}, \dots, a_{\gamma_p}]^T$ be the orthonormalized score function of F_γ . Denote with $\mathcal{L}(F_\gamma)$ and $\mathcal{L}_\perp(F_\gamma)$ the subspaces of $L^2(F_\gamma)$ given by

$$\mathcal{L}(F_\gamma) = \left\{ \phi \in L^2(F_\gamma) : \langle \phi, \mathbf{1} \rangle_{F_\gamma} = 0 \right\}, \quad \mathcal{L}_\perp(F_\gamma) = \left\{ \tilde{\phi} \in \mathcal{L}(F_\gamma) : \langle \tilde{\phi}, \mathbf{a}_\gamma \rangle_{F_\gamma} = 0 \right\}.$$

For any given orthonormal basis $\{\phi_{j\gamma}\}_{j=1}^\infty$ of $L^2(F_\gamma)$, with elements in $\mathcal{L}(F_\gamma)$, the empirical process $v_{\gamma,n}$ indexed by the residuals

$$\tilde{\phi}_{j\gamma} = \phi_{j\gamma} - \mathbf{a}_\gamma^T \langle \mathbf{a}_\gamma, \phi_{j\gamma} \rangle_{F_\gamma},$$

is asymptotically distributed, under F_γ , as a projected Brownian motion with mean and covariance given by

$$\begin{aligned} \langle \tilde{\phi}_{j,\gamma}, \mathbf{1} \rangle_{F_\gamma} &= 0 \quad \text{and} \\ \langle \tilde{\phi}_{i\gamma}, \tilde{\phi}_{j\gamma} \rangle_{F_\gamma} &= \mathbf{1}_{\{i=j\}} - \sum_{k=1}^p \langle a_{\gamma_k}, \phi_{i\gamma} \rangle_{F_\gamma} \langle a_{\gamma_k}, \phi_{j\gamma} \rangle_{F_\gamma}, \end{aligned}$$

respectively. Such a limiting distribution is the “standard” distribution that will be recovered through the K2 transform. In particular, since a Gaussian process is fully characterized by its mean and covariance functions, the K2 transform enables the construction of an orthonormal basis $\{h_{j\beta}\}_{j=1}^\infty$ for $L^2(G_\beta)$, hereinafter referred to as *K2 orthonormal basis*, whose residuals, $\{\tilde{h}_{j\beta}\}_{j=1}^\infty$ satisfy

$$\langle \tilde{h}_{j\beta}, \mathbf{1} \rangle_{G_\beta} = \langle \tilde{\phi}_{j\gamma}, \mathbf{1} \rangle_{F_\gamma} = 0 \quad \text{and} \quad \langle \tilde{h}_{j\beta}, \tilde{h}_{s\beta} \rangle_{G_\beta} = \langle \tilde{\phi}_{j\gamma}, \tilde{\phi}_{s\gamma} \rangle_{F_\gamma}.$$

It follows that the processes $v_{\beta,n}(\tilde{h}_{j\beta})$ and $v_{\gamma,n}(\tilde{\phi}_{j\gamma})$, as well as all their functionals, have the same limiting distribution under G_β and F_γ , respectively, thereby enabling the construction of a large class of asymptotically distribution-free statistics for testing (3.3.1.2).

For example, let $\hat{\beta}_n$ and $\hat{\gamma}_n$ denote the estimators of β and γ , respectively. Consider the order-selection and subset-selection statistics defined in (3.3.2.5) and (3.3.2.7), in which $\hat{S}_{m,n}$ is chosen to be the (unnormalized) generalized score statistic in (3.3.2.4), with $\tilde{h}_{j\hat{\beta}_n}$ corresponding to the residuals of the K2 orthonormal basis. Under G_β , such statistics have the same limiting distribution as the statistics

$$\max_{m \in \mathcal{M}_n} \sum_{j=1}^m \frac{v_{\gamma,n}^2(\tilde{\phi}_{j\hat{\gamma}_n})}{m} \quad \text{and} \quad \max_{B \subseteq \mathcal{M}_n: B \neq \emptyset} \sum_{j \in B} \frac{v_{\gamma,n}^2(\tilde{\phi}_{j\hat{\gamma}_n})}{|B|}, \quad (3.3.4.1)$$

under F_γ . Since the latter can be chosen arbitrarily, the limiting null distribution of the statistics in (3.3.4.1) can be easily simulated by means of the projected bootstrap described in Section 3.3. This also implies that, when testing different models, the standard limiting null distribution of the corresponding processes can be obtained using a single simulation conducted under F_γ .

The construction of the K2 orthonormal basis and validation of its properties are described in detail in Section 3.4.1; whereas, its effectiveness in retrieving distribution-freeness in finite samples is investigated in Section 3.4.2 through a suite of simulation studies.

3.4.1 On the construction of the K2 orthonormal basis

Let $l_{\gamma,\beta}$ be the function defined by

$$l_{\gamma,\beta}(x) = \sqrt{\frac{f_{\gamma}(x)}{g_{\beta}(x)}}, \quad x \in \mathcal{X}.$$

The multiplication operator $\phi \mapsto l_{\gamma,\beta}\phi$ defines an isometry from $L^2(F_{\gamma})$ to $L^2(G_{\beta})$.

It can be easily verified that for all j 's,

$$\langle l_{\gamma,\beta}\phi_{j\gamma}, \mathbf{1} \rangle_{G_{\beta}} \neq 0, \quad \text{and} \quad \langle l_{\gamma,\beta}\phi_{j\gamma}, l_{\gamma,\beta}\phi_{s\gamma} \rangle_{G_{\beta}} = \langle \phi_{j\gamma}, \phi_{s\gamma} \rangle_{F_{\gamma}}.$$

It follows that the functions $l_{\gamma,\beta}\phi_{j\gamma}$ have the same covariance as $\phi_{j\gamma}$ under F_{γ} , but do not have the same mean. Thus, they belong to $L^2(G_{\beta})$ but not to its subspace $\mathcal{L}(G_{\beta})$.

To rectify this, consider a linear operator, K , such that, when applied to $l_{\gamma,\beta}\phi_{j\gamma}$, the resulting functions have mean zero under G_{β} but, at the same time, their covariance is preserved, that is,

$$\langle Kl_{\gamma,\beta}\phi_{j\gamma}, \mathbf{1} \rangle_{G_{\beta}} = 0, \quad \langle Kl_{\gamma,\beta}\phi_{j\gamma}, Kl_{\gamma,\beta}\phi_{s\gamma} \rangle_{G_{\beta}} = \langle \phi_{j\gamma}, \phi_{s\gamma} \rangle_{F_{\gamma}}. \quad (3.3.4.1)$$

The second condition in (3.3.4.1) is satisfied for any K which is unitary. For what concerns the first condition, note that

$$\langle l_{\gamma,\beta}\phi_{j\gamma}, l_{\gamma,\beta} \rangle_{G_{\beta}} = \langle \phi_{j\gamma}, \mathbf{1} \rangle_{F_{\gamma}} = 0;$$

hence, if $Kl_{\gamma,\beta} = \mathbf{1}$, then

$$\langle Kl_{\gamma,\beta}\phi_{j\gamma}, \mathbf{1} \rangle_{G_{\beta}} = \langle Kl_{\gamma,\beta}\phi_{j\gamma}, Kl_{\gamma,\beta} \rangle_{G_{\beta}} = \langle l_{\gamma,\beta}\phi_{j\gamma}, l_{\gamma,\beta} \rangle_{G_{\beta}} = 0.$$

A unitary operator K satisfying the above requirements is the reflection

$$K = I - 2 \frac{l_{\gamma, \beta} - \mathbf{1}}{\|l_{\gamma, \beta} - \mathbf{1}\|^2} \langle l_{\beta, \gamma} - \mathbf{1}, \cdot \rangle_{G_\beta} = I - \frac{l_{\gamma, \beta} - \mathbf{1}}{1 - \langle l_{\gamma, \beta}, \mathbf{1} \rangle_{G_\beta}} \langle l_{\beta, \gamma} - \mathbf{1}, \cdot \rangle_{G_\beta}, \quad (3.3.4.2)$$

where I is an identity operator. The properties of K are summarized in Proposition 3 and verified in Appendix B.1.

Proposition 3 (Khmaladze 2016). *The operator K is unitary, and satisfies*

$$Kl_{\gamma, \beta} = \mathbf{1} \quad \text{and} \quad K\mathbf{1} = l_{\gamma, \beta}.$$

Moreover, for any $\zeta \in L^2(G_\beta)$ such that $\langle \zeta, l_{\gamma, \beta} \rangle_{G_\beta} = \langle \zeta, \mathbf{1} \rangle_{G_\beta} = 0$, $K\zeta = \zeta$.

Thus far, we have demonstrated that the functions $Kl_{\gamma, \beta}\phi_{j\gamma} \in \mathcal{L}(G_\beta)$ have the same mean and covariance as the functions $\phi_{j\gamma} \in \mathcal{L}(F_\gamma)$. Therefore, when β is known, test statistics constructed as functionals of the empirical process $v_{\beta, n}(h_{j\beta})$ are asymptotically distribution-free if $h_{j\beta} = Kl_{\gamma, \beta}\phi_{j\gamma}$.

When γ and β are unknown, an additional step is necessary to map functions $\tilde{\phi}_{j\gamma} \in \mathcal{L}_\perp(F_\gamma)$ into functions $\tilde{h}_{j\beta} \in \mathcal{L}_\perp(G_\beta)$ with the same mean and covariance. For instance, if we were to naively choose $\tilde{h}_{j\beta}$ to be

$$\tilde{h}_{j\beta} = Kl_{\gamma, \beta}\phi_{j\gamma} - \mathbf{b}_\beta^T \langle \mathbf{b}_\beta, Kl_{\gamma, \beta}\phi_{j\gamma} \rangle_{G_\beta},$$

then, in general,

$$\langle \tilde{h}_{j\beta}, \tilde{h}_{s\beta} \rangle_{G_\beta} = \langle \phi_{j\gamma}, \phi_{s\gamma} \rangle_{F_\gamma} - \langle Kl_{\gamma, \beta}\phi_{j\gamma}, \mathbf{b}_\beta^T \rangle_{G_\beta} \langle \mathbf{b}_\beta, Kl_{\gamma, \beta}\phi_{s\gamma} \rangle_{G_\beta} \neq \langle \tilde{\phi}_{j\gamma}, \tilde{\phi}_{s\gamma} \rangle_{F_\gamma}.$$

In the above expression, equality holds, however, if

$$\langle \mathbf{b}_\beta, Kl_{\gamma, \beta}\phi_{j\gamma} \rangle_{G_\beta} = \langle \mathbf{a}_\gamma, \phi_{j\gamma} \rangle_{F_\gamma}$$

for all j 's. This motivates the construction of an operator, $\mathbf{U}_p$, such that, when applied to the functions $Kl_{\gamma,\beta}\phi_{j\gamma}$, guarantees

$$\langle \mathbf{b}_\beta, \mathbf{U}_p Kl_{\gamma,\beta}\phi_{j\gamma} \rangle_{G_\beta} = \langle \mathbf{a}_\gamma, \phi_{j\gamma} \rangle_{F_\gamma}, \quad (3.3.4.3)$$

and at the same time preserves the mean and covariance structure, i.e.,

$$\langle \mathbf{U}_p Kl_{\gamma,\beta}\phi_{j\gamma}, \mathbf{1} \rangle_{G_\beta} = 0, \quad \langle \mathbf{U}_p Kl_{\gamma,\beta}\phi_{j\gamma}, \mathbf{U}_p Kl_{\gamma,\beta}\phi_{s\gamma} \rangle_{G_\beta} = \langle \phi_{j\gamma}, \phi_{s\gamma} \rangle_{F_\gamma}. \quad (3.3.4.4)$$

Let c_{λ_k} be the k th component of the vector function $Kl_{\gamma,\beta}\mathbf{a}_\gamma$, i.e.,

$$c_{\lambda_k} = Kl_{\gamma,\beta}a_{\gamma_k}, \quad k = 1, \dots, p.$$

While any unitary operator $\mathbf{U}_p$ such that $\mathbf{U}_p \mathbf{1} = \mathbf{1}$ fulfills the requirements in (3.3.4.4), to ensure (3.3.4.3) holds, $\mathbf{U}_p$ must be chosen so that $\mathbf{U}_p c_{\lambda_k} = b_{\beta_k}$, for each $k = 1, \dots, p$.

One could naively attempt to construct $\mathbf{U}_p$ as a composition of p unitary operators, each mapping c_{λ_k} to b_{β_k} ; such an approach, however, would not lead to the desired result. To see that, consider the reflection operator on $L^2(G_\beta)$ defined as

$$\begin{aligned} U_{b_{\beta_k} c_{\lambda_k}} &= I - 2 \frac{b_{\beta_k} - c_{\lambda_k}}{\|b_{\beta_k} - c_{\lambda_k}\|^2} \langle b_{\beta_k} - c_{\lambda_k}, \cdot \rangle_{G_\beta} \\ &= I - \frac{b_{\beta_k} - c_{\lambda_k}}{\mathbf{1} - \langle b_{\beta_k}, c_{\lambda_k} \rangle_{G_\beta}} \langle b_{\beta_k} - c_{\lambda_k}, \cdot \rangle_{G_\beta}, \quad k = 1, \dots, p. \end{aligned}$$

Such an operator is self-adjoint and unitary on $L^2(G_\beta)$. It maps c_{λ_k} to b_{β_k} and b_{β_k} to c_{λ_k} , while leaving all functions orthogonal to both b_{β_k} and c_{λ_k} unchanged. Let $p = 2$, then

$$U_{b_{\beta_2} c_{\lambda_2}} \circ U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_1} = U_{b_{\beta_2} c_{\lambda_2}} b_{\beta_1},$$

and in general

$$U_{b_{\beta_2} c_{\lambda_2}} b_{\beta_1} \neq b_{\beta_1} \quad \text{unless} \quad \langle c_{\lambda_2}, b_{\beta_1} \rangle_{G_\beta} = 0.$$

To address this issue, set $\tilde{c}_{\lambda_1} = c_{\lambda_1}$ and a set of functions $\{\tilde{c}_{\lambda_k}\}_{k=2}^p$, where each $\tilde{c}_{\lambda_k}$ is constructed to be orthogonal to $\mathbf{1}$ and to every b_{β_j} for which $j \leq k-1$. Specifically, we set

$$\tilde{c}_{\lambda_k} = U_{b_{\beta_{k-1}} \tilde{c}_{\lambda_{k-1}}} \circ \cdots \circ U_{b_{\beta_1} \tilde{c}_{\lambda_1}} c_{\lambda_k}. \quad (3.3.4.5)$$

A proof that each $\tilde{c}_{\lambda_k}$ satisfies the required orthogonality conditions is provided in Appendix B.2.

Now, define the operator U_p as

$$U_p = U_{b_{\beta_p} \tilde{c}_{\lambda_p}} \circ \cdots \circ U_{b_{\beta_1} \tilde{c}_{\lambda_1}}.$$

Since U_p is a composition of unitary operators on $L^2(G_\beta)$, it is itself unitary on $L^2(G_\beta)$. Moreover,

$$U_p c_{\lambda_k} = U_{b_{\beta_p} \tilde{c}_{\lambda_p}} \circ \cdots \circ U_{b_{\beta_k} \tilde{c}_{\lambda_k}} \tilde{c}_{\lambda_k} = U_{b_{\beta_p} \tilde{c}_{\lambda_p}} \circ \cdots \circ U_{b_{\beta_{k+1}} \tilde{c}_{\lambda_{k+1}}} b_{\beta_k} = b_{\beta_k},$$

thus, U_p maps c_{λ_k} into b_{β_k} , for each $k = 1, \dots, p$ and, since all the functions b_{β_k} and $\tilde{c}_{\lambda_k}$ are orthogonal to $\mathbf{1}$, $U_p \mathbf{1} = \mathbf{1}$. These properties are formalized in Proposition 4.

Proposition 4. *The operator U_p is unitary on $L^2(G_\beta)$ and satisfies*

$$U_p \mathbf{1} = \mathbf{1}, \quad \text{and} \quad U_p c_{\lambda_k} = b_{\beta_k}, \quad k = 1, \dots, p.$$

From Proposition 4, it follows that each function $U_p K l_{\gamma, \beta} \phi_{j\gamma}$ fulfills (3.3.4.3), thus:

$$\begin{aligned} U_p K l_{\gamma, \beta} \tilde{\phi}_{j\gamma} &= U_p K l_{\gamma, \beta} \left[\phi_{j\gamma} - \mathbf{a}_\gamma^T \langle \mathbf{a}_\gamma, \phi_{j\gamma} \rangle_{F_\gamma} \right] \\ &= U_p K l_{\gamma, \beta} \phi_{j\gamma} - \mathbf{b}_\beta^T \langle \mathbf{b}_\beta, U_p K l_{\gamma, \beta} \phi_{j\gamma} \rangle_{G_\beta}. \end{aligned}$$

We now have access to all the elements needed to define the K2 orthonormal basis at the core of the proposed distribution-freeness construction:

Proposition 5. *The set of functions $\{U_p K l_{\gamma, \beta} \phi_{j\gamma}\}_{j=1}^{\infty}$ forms an orthonormal basis for $L^2(G_\beta)$. Define, for each $j \geq 1$,*

$$h_{j\beta} = U_p K l_{\gamma, \beta} \phi_{j\gamma}, \quad \text{and} \quad \tilde{h}_{j\beta} = U_p K l_{\gamma, \beta} \tilde{\phi}_{j\gamma}.$$

Then, the empirical processes $v_{\beta, n}(\tilde{h}_{j\beta})$ and $v_{\gamma, n}(\tilde{\phi}_{j\gamma})$ have the same limiting distribution under G_β and F_γ , respectively.

Proof. The first claim follows from the unitary nature of the operators U_p , K , and $l_{\gamma, \beta}$. The second claim is a direct consequence of the asymptotic Gaussianity of $v_{\beta, n}(\tilde{h}_{j\beta})$ and $v_{\gamma, n}(\tilde{\phi}_{j\gamma})$ and the construction of the $\{\tilde{h}_{j\beta}\}_{j=1}^{\infty}$ basis, which ensures that the empirical process indexed by such functions has, under G_β , the same mean and covariance as the empirical process indexed by functions $\{\tilde{\phi}_{j\gamma}\}_{j=1}^{\infty}$ under F_γ . ■

The approach described above can be generalized to cases where the dimension of γ is smaller than p , say s . In this setting, one can simply expand the orthonormal set of score functions $\{a_{\gamma_k}\}_{k=1}^s$ to a larger orthonormal set $\{a_{\gamma_k}\}_{k=1}^p$ in $L^2(F_\gamma)$, ensuring that all elements remain orthogonal to each other and the constant function $\mathbf{1}$. This extension can be accomplished, for instance, by selecting $p - s$ additional functions from another orthonormal basis in $L^2(F_\gamma)$ outside the span of $\{\mathbf{1}, a_{\gamma_1}, \dots, a_{\gamma_s}\}$, and applying the Gram-Schmidt orthogonalization procedure.

3.4.2 Simulation Studies

Consider a dataset of $n = 100$ observations generated from a distribution Q with density

$$q(x) = 0.3d_1(x; \mu_1, \sigma_1) + 0.5d_2(x; \mu_2, \sigma_2) + 0.2d_3(x), \quad x \in \mathcal{X} = [-10, 10],$$

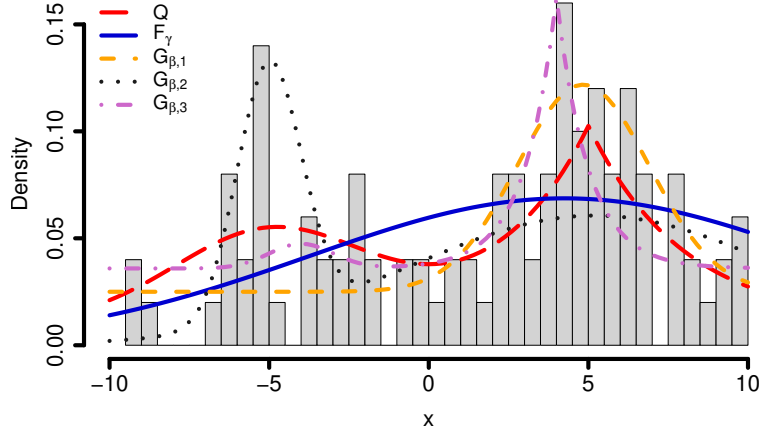


Figure 3.2: The histogram of the simulated dataset is shown together with the densities of the true data-generating model Q , the reference distribution F_γ , and the hypothesized distributions $G_{\beta,1}, G_{\beta,2}, G_{\beta,3}$. The unknown parameters β and γ are estimated via maximum likelihood.

where d_1 and d_2 denote, respectively, the densities of truncated normal and truncated Laplace random variables; d_3 is the uniform density. The values of the location parameters are $\mu_1 = -5$, $\mu_2 = 5$ and scale parameters are $\sigma_1 = \sigma_2 = 3$. We aim to test the hypothesis in (3.3.1.2) for three different specifications of the null density, i.e.,

$$g_{\beta,1}(x) = 0.5d_1(x; \beta_1, \beta_2) + 0.5d_3(x);$$

$$g_{\beta,2}(x) = 0.3d_1(x; -5, 1) + 0.7d_1(x; \beta_1, \beta_2);$$

$$g_{\beta,3}(x) = \beta_1d_1(x; -4, 1) + \beta_2d_2(x; 4, 1) + (1 - \beta_1 - \beta_2)d_3(x),$$

where $\beta = (\beta_1, \beta_2)$ is the unknown parameter vector to be estimated. Their corresponding CDFs are denoted by $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$, respectively. The reference distribution, F_γ , is chosen to be a truncated normal distribution over $\mathcal{X}$ with unknown parameter γ corresponding to its mean and variance. Figure 3.2 shows the histogram of the dataset considered, along with the densities of $q, g_\gamma, g_{\beta,1}, g_{\beta,2}$, and $g_{\beta,3}$ estimated via maximum

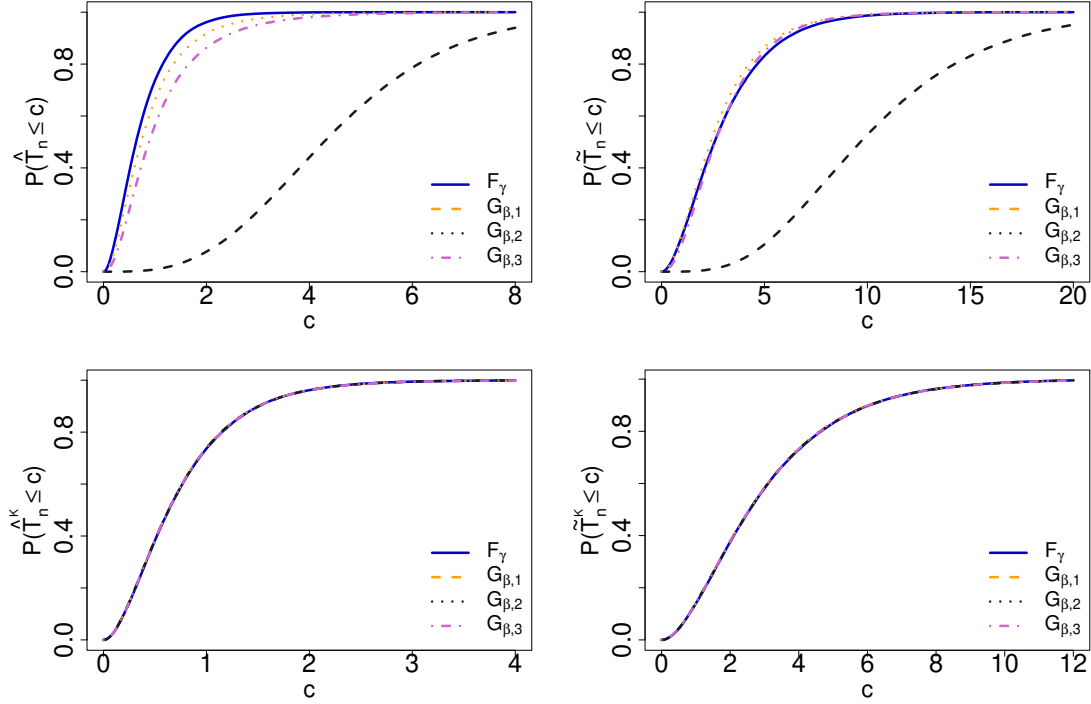


Figure 3.3: The simulated null distributions of the order selection statistics (left panels) and the subset selection test statistics (right panels), using basis functions obtained by composing the normalized shifted Legendre polynomials with the null CDFs (upper panels) and the K2 transform (lower panels), under F_γ , $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$.

likelihood.

Consider the case in which the basis functions used in (3.3.1.1) are constructed as compositions of the normalized shifted Legendre polynomials on $[0,1]$ and the CDF under the null, i.e.,

$$\begin{aligned}
 h_{1,\beta}(x) &= \sqrt{12}[G_{\beta,s}(x) - 0.5], \\
 h_{2,\beta}(x) &= \sqrt{5}[6G_{\beta,s}^2(x) - 6G_{\beta,s}(x) + 1], \\
 h_{3,\beta}(x) &= \sqrt{7}[20G_{\beta,s}^3(x) - 30G_{\beta,s}^2(x) + 12G_{\beta,s}(x) - 1], \\
 h_{4,\beta}(x) &= 3[70G_{\beta,s}^4(x) - 140G_{\beta,s}^3(x) + 90G_{\beta,s}^2(x) - 20G_{\beta,s}(x) + 1], \quad \text{etc.}
 \end{aligned} \tag{3.3.4.1}$$

with $s = 1, 2, 3$ when testing $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$, respectively. For each of the three null models considered, we simulated the null distributions of the statistics $\hat{T}_n$ in (3.3.2.5) and $\tilde{T}_n$ in (3.3.2.7) constructed using the above-mentioned sets of basis functions, up to $M_n = 6$, and with $\hat{S}_{m,n}$ chosen to be the unnormalized generalized score statistic in (3.3.2.4). The simulation was performed using the projected bootstrap, described in Section 3.3.1, with 100,000 replicates. For comparison, the distribution of the statistics in (3.3.4.1) under F_γ has also been simulated following the same sampling scheme and choosing the basis functions ϕ_j to be a composition of normalized shifted Legendre polynomials on $[0, 1]$ and F_γ ; hence, for $j = 1, \dots, 4$ they correspond to the right-hand sides of the equations in (3.3.4.1) with F_γ in place of $G_{\beta,s}$. The resulting simulated null distributions are shown in the upper panels of Figure 3.3, where the blue solid curve corresponds to the statistics under F_γ , and the yellow dotted, black dashed, and purple dash-dotted curves correspond to those constructed under $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$, respectively. As shown in the upper-left panel of Figure 3.3, the null distributions of the order selection statistic $\hat{T}_n$ under $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$ differs substantially from that constructed under F_γ , and given by the first expression in (3.3.4.1). In the case of the subset selection test statistic $\tilde{T}_n$, the null distributions under $G_{\beta,1}$, and $G_{\beta,3}$ are rather similar. They also match the null distribution of the same statistic, constructed under F_γ , and given by the right expression in (3.3.4.1). However, all these distributions differ substantially from that of $\tilde{T}_n$ under $G_{\beta,2}$. These discrepancies are expected given that, in general, the unnormalized generalized score statistic in (3.3.2.4) is not distribution-free.

To assess the impact of the K2 transform, we repeated the same experiment but for the case in which the basis functions used in (3.3.1.1) when testing $G_{\beta,1}$, $G_{\beta,2}$, $G_{\beta,3}$ are the K2-transformed basis functions, obtained as described in Section 3.4.1, and corresponding to the images of the K2 transform applied to the basis functions ϕ_j given by compositions of Legendre polynomials and F_γ . In what follows, we denote the order selection and subset selection statistics constructed using the K2-transformed basis functions by $\hat{T}_n^K$ and $\tilde{T}_n^K$,

	$\alpha = 0.001$				$\alpha = 0.05$				$\alpha = 0.1$			
	$\hat{T}_n$	$\tilde{T}_n$	$\hat{T}_n^K$	$\tilde{T}_n^K$	$\hat{T}_n$	$\tilde{T}_n$	$\hat{T}_n^K$	$\tilde{T}_n^K$	$\hat{T}_n$	$\tilde{T}_n$	$\hat{T}_n^K$	$\tilde{T}_n^K$
F_γ	0.859	0.851	–	–	1.000	1.000	–	–	1.000	1.000	–	–
$G_{\beta,1}$	0.008	0.091	0.053	0.025	0.351	0.551	0.519	0.333	0.533	0.688	0.662	0.473
$G_{\beta,2}$	0.016	0.038	0.643	0.695	0.238	0.424	0.967	0.969	0.357	0.566	0.985	0.986
$G_{\beta,3}$	0.007	0.028	0.037	0.035	0.164	0.285	0.371	0.351	0.286	0.402	0.498	0.482

Table 3.1: Power comparison between classical order selection and subset selection test statistics with their K2-transformed counterparts. The null distributions are $G_{\beta,1}, G_{\beta,2}, G_{\beta,3}$; the reference distribution is F_γ ; the true data-generating distribution is Q ; and significance levels are set at 0.001, 0.05, and 0.1.

respectively. As shown in the lower panels of Figure 3.3, for both such statistics, the simulated null distributions under $G_{\beta,1}$, $G_{\beta,2}$, and $G_{\beta,3}$ are indistinguishable from the distribution of the corresponding statistics in (3.3.4.1) simulated under F_γ . It follows that, for all models and statistics considered in this example, a sample size of 100 observations is sufficient to retrieve the distribution-free property even when two parameters are estimated.

To evaluate the performance of the testing procedure in (3.3.1.2) under different choices of basis functions, we computed the simulated power using both order selection and subset selection statistics and calculated it as the proportion of bootstrap replications under Q for which the value of these statistics exceeds the critical value obtained from the simulated null distribution at the prescribed significance levels.

Table 3.1 presents the resulting power when testing $G_{\beta,1}, G_{\beta,2}, G_{\beta,3}$ using the statistics $\hat{T}_n, \tilde{T}_n, \hat{T}_n^K$, and $\tilde{T}_n^K$ at significance levels $\alpha = 0.001, 0.05$, and 0.1. The power when testing the reference distribution F_γ using $\hat{T}_n$ and $\tilde{T}_n$ is also reported. When testing $G_{\beta,2}$ and $G_{\beta,3}$, the K2-based test statistics $\hat{T}_n^K$ and $\tilde{T}_n^K$ exhibit substantially higher power than $\hat{T}_n$ and $\tilde{T}_n$. Notably, when testing $G_{\beta,2}$ at a significance level of $\alpha = 0.001$, $\tilde{T}_n^K$ attains power as high as 0.695, whereas $\tilde{T}_n$ achieves only 0.038. When testing $G_{\beta,1}$, $\hat{T}_n^K$ shows higher power than $\hat{T}_n$ while the power of $\tilde{T}_n^K$ is lower than that of $\tilde{T}_n$. This reflects the fact that K2-transformed statistics can exhibit higher power against certain alternatives, though such improvements in power should not be assumed to be true in general.

3.5 Case study: analyzing an X-ray spectrum from RT Cru

In X-ray astronomy, spectral analysis is essential for understanding the fundamental properties of stars, galaxies, and other celestial objects. In particular, the presence of spectral lines in X-ray spectra provides valuable insights into an object's chemical composition, distance from Earth, temperature, motion, surrounding environments, and other important attributes.

Here, we focus on the study of a high-resolution spectrum from the star RT Cru and obtained in November 2015 by the *Chandra* X-ray Observatory [Swartz et al., 2010]. RT Cru is of particular astronomical significance because it belongs to the rare class of X-ray-emitting symbiotic systems – crucial for studying Type Ia supernovae⁴ and, more broadly, investigating the expansion of the Universe. In Zhang [2023], smooth tests were primarily employed to assess the departure from uniformity in a background-only spectrum, that is, a spectrum in which no spectral lines are present. Likelihood ratio tests were then used to test for the presence of spectral lines in a spectrum that was known to contain at least three spectral lines in the wavelength region between 1.65 Å and 2.05 Å (where 1Å = 10⁻¹⁰m). Focusing on the latter set of data, here we use the methodology described in Section 3.4 to assess the validity of the parametric model:

$$g_{\beta}(x) = (1 - \beta_1 - \beta_2 - \beta_3)b(x) + \sum_{r=1}^3 \beta_r s_r(x), \quad x \in \mathcal{X}, \quad (3.3.5.1)$$

where $\mathcal{X} = [1.65, 2.05]$; $b(x)$ is a uniform background density on $\mathcal{X}$; and the functions $s_r(x)$ model each of the the three expected spectral lines. They consist of a convolution of a

⁴A Type Ia supernova is a powerful nuclear explosion that occurs when a small, dense star called a white dwarf gathers too much mass from a nearby companion star. Once it exceeds its stability limit, the white dwarf undergoes an uncontrollable burst of nuclear fusion.

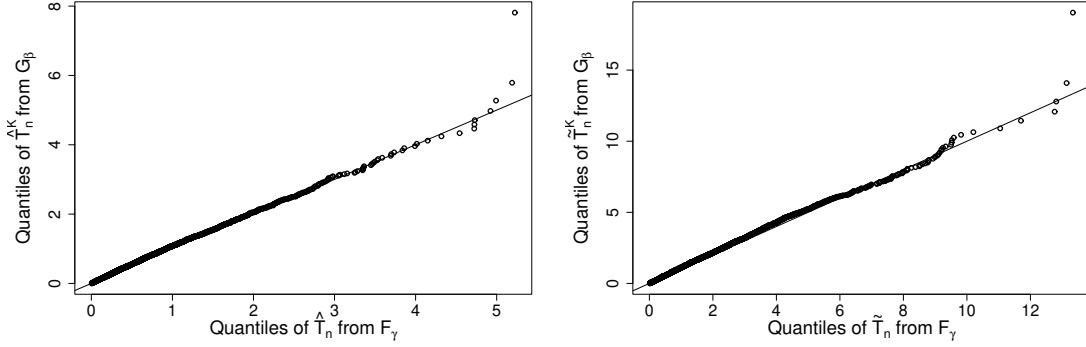


Figure 3.4: Left: QQ plots of the simulated order selection test statistics in (3.3.2.5) under G_β and in (3.3.4.1) under F_γ . Right: QQ plots of the simulated subset selection test statistics in (3.3.2.7) under G_β and in (3.3.4.1) under F_γ . For both statistics computed under G_β , the functions $h_{j\beta}$ are elements of the $K2$ basis constructed as in Proposition 5.

normal density with a Moffat function⁵ [Moffat, 1969] and specify as

$$s_r(x) \propto \int_{-\infty}^{+\infty} \frac{\exp\left\{-\frac{1}{2}\left(\frac{w_0 - \mu_r}{\sigma_r}\right)^2\right\}}{\left[1 + \left(\frac{x - w_0}{0.05}\right)^2\right]^{2.5}} dw_0, \quad x \in \mathcal{X}, \quad (3.3.5.2)$$

where μ_r, σ_r are known parameters, with $\mu_1 = 1.78499, \mu_2 = 1.85247$, and $\mu_3 = 1.94365$, and $\sigma_1 = \sigma_2 = \sigma_3 = 0.0025$. The unknown parameters β_r represent the relative intensities of the spectral lines and are estimated via maximum likelihood.

We use the approach described in Section 3.4 to test the validity of (3.3.5.1). The reference distribution F_γ considered is a simplified version of (3.3.5.1). In particular, it consists of a convex combination of the uniform distribution and truncated normal distributions over $\mathcal{X}$. Its density is:

$$f_\gamma(x) = (1 - \gamma_1 - \gamma_2 - \gamma_3)b(x) + \sum_{r=1}^3 \gamma_r p_r(x, \mu_r, 0.05), \quad x \in \mathcal{X}$$

⁵The Moffat function is equivalent to the density of a location-scale Student's t -distribution with a location of 0, a scale of 0.025, and 4 degrees of freedom.

where p_r is the density of a truncated normal over $\mathcal{X}$, with μ_r being the known positions of the spectral lines and listed after (3.3.5.2). The unknown parameter $\gamma = (\gamma_1, \gamma_2, \gamma_3)$ is estimated via the maximum likelihood. The basis functions for F_γ are chosen to be compositions of normalized shifted Legendre polynomials on $[0, 1]$ with F_γ . These functions are employed as a starting point to construct the K2-transformed basis functions $h_{j\beta}$ for $L^2(G_\beta)$, whose residuals are subsequently used when calculating the order selection and subset selection statistics in (3.3.2.5) and (3.3.2.7), respectively, with $\hat{S}_{m,n}$ as in (3.3.2.4) and $\{h_{j\beta}\}_{j=1}^{M_n}$ as in Proposition 5. Hence, consistently with the notation used in the previous section, we denote them with $\hat{T}_n^K$ and $\tilde{T}_n^K$. Their limiting null distributions are obtained by simulating 100,000 realizations of the statistics (3.3.4.1) under F_γ through the projected bootstrap (cf. Section 3.3). The maximum number of basis functions used is $M_n = 6$. To assess the accuracy of the approximation, we also simulated the distributions of $\hat{T}_n^K$ and $\tilde{T}_n^K$ under G_β . As shown by the QQ-plots in Figure 3.4, the null distributions of such statistics under G_β are indistinguishable from those of the statistics in (3.3.4.1) under F_γ .

Finally, the p-values from the order selection and subset selection test statistics are 0.463 and 0.454, respectively, indicating that the hypothesized model in (3.3.5.1) fits the observed spectrum well.

From an astrophysical standpoint, the Gaussian peaks $s_r(x)$, $r = 1, 2, 3$, in (3.3.5.1) correspond to iron lines⁶ from various ionization states. Specifically, s_1 and s_2 correspond to the Fe XXV and Fe XXVI lines, which occur in extremely hot conditions where the iron has lost most of its electrons. In contrast, s_3 corresponds to the Fe K α line – a fluorescent signal emitted by iron atoms that still hold most of their electrons, indicating that some X-rays are being reflected by nearby, cooler, denser material. Hence, by failing to reject the model in (3.3.5.1), our test is in agreement with the claim of Danehkar [2021b] of a multi-phase environment in RT Cru, where a very hot, highly ionized plasma – responsible

⁶Each chemical element produces a unique set of spectral lines that arise from changes in atomic energy levels. Iron stands out because it has an exceptionally large number of these lines spanning ultraviolet, visible, and infrared light. As a result, iron is extremely valuable for modeling and interpreting observed spectra in many scientific studies.

for the Fe XXV and Fe XXVI lines – coexists with cooler, denser material that produces the Fe $K\alpha$ fluorescence.

3.6 Summary and discussion

This article introduces a new class of smooth test statistics whose asymptotic distribution-freeness is achieved by relying on the K2 transform. The latter consists of a change of variable in functional space, which enables the construction of an empirical process with a standard asymptotic null distribution. In the context of smooth tests, such a transformation is especially valuable in that it yields a new family of orthonormal bases in $L^2(G_\beta)$. Test statistics based on elements of such bases are shown to be asymptotically distribution-free even when the parameters are estimated, and the sample size is only moderately large.

The projected bootstrap is also discussed as a computationally efficient alternative to the classical parametric bootstrap. In particular, the projection structure induced by parameter estimation allows us to simulate the null distribution of the test statistics of interest without re-estimating the model parameters at each bootstrap replicate. Simulation experiments show that the computational gain attained by the projected bootstrap can be substantial compared to the parametric bootstrap, especially when the estimation of the parameters is CPU-intensive.

While the present manuscript focuses on the univariate setting, the proposed framework can be easily adapted to test multivariate parametric models. In particular, when G_β is D -dimensional distribution, the reference distribution, F_γ could be chosen to be a product of D univariate distributions $F_{\gamma,d}$, $d = 1, \dots, D$. The K2 orthonormal basis in $L^2(G_\beta)$ can then be constructed by applying the K2 transform to a tensor product of bases in $L^2(F_{\gamma,d})$ (e.g., [Algeri \[2021\]](#)).

Extensions to smooth tests for regression [[Ingлот and Ledwina, 2006b](#), [Rayner et al., 2022](#)] are also possible. In this case, instead of relying on the classical empirical process for i.i.d. data to express the statistics of interest as functionals from it (see [Section 3.2](#)),

the random measure to be used is the weighted empirical process proposed in [Khmaladze \[2017\]](#).

Chapter 4

On validating angular power spectral models for the stochastic gravitational-wave background without distributional assumptions

4.1 Overview

Estimating and testing models for the angular power spectrum is required in many astrophysical and cosmological settings, including the cosmic microwave background [et al. \[2020\]](#), galaxy sky surveys [Ahumada \[2020\]](#), and weak gravitational lensing surveys [Jeffrey \[2021\]](#). The focus here is on the stochastic gravitational-wave background (SGWB) resulting from the superposition of gravitational-wave (GW) signals that are too weak or too numerous to be resolved individually. While the SGWB and its anisotropies have not yet been detected by second-generation GW detectors such as the Advanced LIGO [Aasi \[2015\]](#) and Advanced Virgo [Acernese \[2014\]](#), upper limits have been placed on the GW energy

density of the background as well as its angular power spectra [Abbott \[2021a,b\]](#).

Anisotropies in the SGWB are expected to arise from a variety of cosmological and astrophysical phenomena [Contaldi \[2017\]](#), [Jenkins \[2019\]](#), [Jenkins and Sakellariadou \[2019\]](#), [Pitrou \[2020\]](#). They can provide information on the astrophysical distribution of GW sources within galaxies, formation of large-scale structure in the universe, and different GW production mechanisms [Cusin \[2017, 2018a,b, 2019\]](#). GWs generated by cosmological phase transitions necessarily have anisotropies analogous to those of the cosmic microwave background and may provide information regarding primordial inhomogeneities during an early inflationary era [Geller \[2018\]](#), [Bertacca \[2020\]](#). GW signals emitted by neutron stars within our galaxy can trace out the galactic plane [Talukder \[2014\]](#), [Mazumder et al. \[2014\]](#), while cosmic strings may cause anisotropies at small angular scales [Jenkins and Sakellariadou \[2018\]](#).

Theoretical models of SGWB anisotropy typically do not predict the realization of the anisotropy on the sky; instead, they predict the distribution of the SGWB power across different angular scales described by the SGWB angular power spectrum. It is, therefore, critical to measure the SGWB angular power spectrum in order to establish a comparison with theoretical models. In practice, one starts by cross-correlating the strain time-series data measured by two GW detectors located at two different locations on Earth. Under the assumption of a stationary SGWB, one can then leverage the rotating antenna pattern of the detector pair (induced by the rotation of the Earth) to estimate the SGWB anisotropy across the sky from the GW data [Allen and Romano \[1999\]](#), [Thrane \[2009\]](#), [Abbott \[2021a\]](#). Expanding the estimated anisotropy in terms of spherical harmonics, and then squaring and summing the expansion coefficients, results in estimators of the SGWB angular power spectrum (see Sec. 4.2). These estimators can then be directly compared with their theoretical counterparts.

Estimators of the SGWB anisotropy, such as coefficients in the spherical harmonic expansion, are obtained by averaging over many time segments of the strain data and are,

therefore, asymptotically multivariate Gaussian distributed. This feature has been used in past parameter estimation analyses whose objective was to estimate individual spherical harmonic modes [Tsukada \[2023\]](#) or to estimate a specific anisotropy realization on the sky, such as the galactic plane [Agarwal \[2022\]](#). Estimators of the angular power spectrum are obtained by squaring the SGWB anisotropy estimators; hence, they are not Gaussian distributed. Nevertheless, some studies have assumed Gaussianity of the angular power spectrum [Agarwal \[2023\]](#), but such an assumption has been shown to lead to biases.

As shown in Appendix [C.1](#), the estimators of the angular power spectrum follow a generalized χ^2 distribution, which cannot be accurately approximated by the Gaussian distribution. Moreover, using the generalized χ^2 likelihood is nontrivial because, in general, the density does not have a closed-form expression. Alternatively, one could devise a scheme that does not require any assumptions regarding the angular power spectrum’s underlying distribution. The latter approach is used here to implement and adapt the statistical procedure outlined in our companion paper [Algeri \[2025\]](#) to the context of the SGWB.

The remainder of the chapter is organized as follows. Sec. [4.2](#) describes the search methods used to estimate the angular power spectra from GW detector data. Sec. [4.3](#) presents a comprehensive explanation of how the distribution-free statistical framework outlined in our companion paper [Algeri \[2025\]](#) can be tailored here to the specifics of an SGWB analysis. Sec. [4.3.3](#) investigates the statistical properties of the proposed procedure via simulation studies. In Sec. [4.4](#), the method is applied to GW data from Advanced LIGO and Advanced Virgo’s third observing run (O3), comparing it with a theoretical model for the angular power spectrum arising from the superposition of compact binary coalescences [Cusin \[2018a, 2019\]](#), [Yang \[2023\]](#). Our results demonstrate that this framework can reliably estimate the unknown parameters of the hypothesized models for the angular power spectrum and provides a powerful and computationally efficient goodness-of-fit test to assess the validity of such models.

4.2 Spherical harmonics search methods for the SGWB

GW data from the Advanced LIGO and Advanced Virgo detectors may be used to generate estimators of the SGWB angular power spectrum. Each GW detector, labeled by $I = 1, 2$, produces a strain time series given by

$$s_I(t) = h_I(t) + n_I(t),$$

with $n_I(t)$ denoting the detector noise and $h_I(t)$ is the detector response to a GW signal. The GW signal can be decomposed in terms of GW modes $h_A(f, \hat{\Omega})$ defined by their polarization A , frequency f , and propagation direction $\hat{\Omega}$:

$$h_I(t) = \int_{-\infty}^{\infty} df \int_{S^2} d\hat{\Omega} h_A(f, \hat{\Omega}) F_I^A(\hat{\Omega}, t) e^{i2\pi f(t - \hat{\Omega} \cdot \vec{x}_I(t)/c)}.$$

Here, $F_I^A(\hat{\Omega}, t)$ is the time-dependent detector response function, $\vec{x}_I(t)$ gives the time-dependent location of interferometer I , and S^2 denotes integration over the 2-sphere. If $\hat{X}_I(t)$ and $\hat{Y}_I(t)$ are unit vectors pointing along the arms of detector I , the detector response function is

$$F_I^A(\hat{\Omega}, t) = \frac{1}{2} \left[\hat{X}_I^a(t) \hat{X}_I^b(t) - \hat{Y}_I^a(t) \hat{Y}_I^b(t) \right] e_{ab}^A(\hat{\Omega}),$$

in which we assume Einstein summation over the spatial indices a, b . The gravitational-wave polarization tensors $e_{ab}^A(\hat{\Omega})$ are defined as in [Thrane \[2009\]](#). We assume the SGWB is unpolarized, Gaussian, and stationary. Additionally, by assuming perturbations to the spacetime metric are real, the Fourier coefficients $h_A(f, \hat{\Omega})$ satisfy $h_A(-f, \hat{\Omega}) = h_A^*(f, \hat{\Omega})$, where $*$ denotes complex conjugation. They also have zero means, $\langle h_A(f, \hat{\Omega}) \rangle = 0$, and the expected value of their two-point correlation is

$$\langle h_A^*(f, \hat{\Omega}) h_{A'}(f', \hat{\Omega}') \rangle = \frac{1}{4} \mathcal{P}(f, \hat{\Omega}) \delta_{AA'} \delta(f - f') \delta^2(\hat{\Omega}, \hat{\Omega}'),$$

where $\mathcal{P}(f, \hat{\Omega})$ gives the spectral and angular distribution of the SGWB [Thrane \[2009\]](#), $\delta_{AA'}$ denotes the Kronecker delta function in polarization, and the remaining δ 's denote Dirac delta functions in frequency and angular position on the sky. We also assume that $\mathcal{P}(f, \hat{\Omega})$ can be factored into its separate spectral and angular components:

$$\mathcal{P}(f, \hat{\Omega}) = H(f) \mathcal{P}(\hat{\Omega}),$$

where $H(f)$ is a dimensionless quantity assumed to be a power law both for its simplicity and its ability to approximate most interesting SGWB models [Abbott \[2017a\]](#), [Allen and Romano \[1999\]](#), [Abbott \[2007\]](#),

$$H(f) = \left(\frac{f}{f_{\text{ref}}} \right)^{\alpha-3}.$$

We set the reference frequency $f_{\text{ref}} = 25$ Hz while the spectral index α may be assigned a value depending on the type of SGWB model being considered. For the astrophysical SGWB resulting from merger events between compact objects such as black holes and neutron stars, $\alpha = 2/3$ [Abbott \[2021a\]](#).

Next, decompose the angular distribution of the background $\mathcal{P}(\hat{\Omega})$ into its spherical harmonic components $Y_{\ell m}$ and coefficients $a_{\ell m}$:

$$\mathcal{P}(\hat{\Omega}) = \sum_{\ell=0}^{\ell_{\text{max}}} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{\Omega}).$$

Note that these definitions assume that each GW frequency bin has the same anisotropy. This assumption can be relaxed, and the analysis can be pursued in separate frequency bins (or bands) f_b . In this case, the spherical harmonic coefficients will become frequency dependent and we denote them as $a_{f_b, \ell m}$.

The value of ℓ_{max} determines the highest-order spherical harmonic modes that are considered in the analysis. This value may be chosen differently for different spectral

models of the SGWB, and while $\ell_{max} = 4$ is typically used when $\alpha = 2/3$ [Abbott \[2021a\]](#), $\ell_{max} = 8$ is used here. This choice of ℓ_{max} is more appropriate for the frequency range over which we will evaluate the angular power spectrum. It has been shown that while $\ell_{max} = 4$ is an optimal choice for making a GW detection of the chosen SGWB model, it corresponds to an angular resolution that is lower than what is resolvable by the Advanced LIGO-Virgo detectors [Floden \[2022\]](#), which justifies considering spherical harmonic modes corresponding to a higher angular resolution.

Next, apply a short-term Fourier transform of the time series output $s_I(t)$ to obtain

$$\tilde{s}_I(f, t) \equiv \int_{t-\tau/2}^{t+\tau/2} dt' e^{-i2\pi f t'} s_I(t').$$

Here, τ is chosen to be 192 seconds, which is long enough to be much greater than the light travel time between detectors but short enough so as to minimize effects due to changes in the detectors' response functions induced by the Earth rotation [Thrane \[2009\]](#), [Abbott \[2021a\]](#). The cross-correlation between detectors I and J is

$$C(t; f) = \frac{2}{\tau} \tilde{s}_I^*(t; f) \tilde{s}_J(t; f),$$

where, to simplify the notation, the I, J indices corresponding to the two detectors from the cross-spectra terms have been dropped. Assuming that the noise of the two detectors is uncorrelated [Thrane \[2009\]](#), the expectation value of the cross-correlation spectrum can be expressed as

$$\langle C(t; f) \rangle = \sum_{\ell=0}^{\ell_{max}} \sum_{m=-\ell}^{\ell} H(f) \gamma_{\ell m}(t; f) a_{f_b, \ell m},$$

where frequency f belongs to the band f_b . The overlap reduction function, $\gamma_{\ell m}(t; f)$, is a geometric factor that accounts for the relative orientation and separation of the two detectors in a given pair [Thrane \[2009\]](#). The covariance matrix of the cross-correlation

spectra is given by

$$\begin{aligned} N_{ft,f't'} &= \langle C(t; f) C^*(t'; f') \rangle - \langle C(t; f) \rangle \langle C^*(t'; f') \rangle \\ &\approx \delta(t - t') \delta(f - f') P_I(t; f) P_J(t; f), \end{aligned} \quad (4.4.2.1)$$

Here, $P_I(t; f)$ and $P_J(t; f)$ correspond to the one-sided power spectral densities of the two GW detectors' outputs. Assuming the present GW signal powers (both cross- and auto-correlated) are much less than the detector noise power, we obtain the approximation in Equation (4.4.2.1). For a given set of $a_{f_b, \ell m}$, the likelihood function for the cross-correlation spectra in the frequency band f_b is proportional to

$$\begin{aligned} &\exp\left(-\sum_{f,t} [C(t; f) - H(f) \sum_{\ell m} \gamma_{\ell m}(t; f) a_{f_b, \ell m}]^*\right. \\ &\quad \left. N_{ft,ft}^{-1} [C(t; f) - H(f) \sum_{\ell m} \gamma_{\ell m}(t; f) a_{f_b, \ell m}]\right) \end{aligned} \quad (4.4.2.2)$$

and the summation is over $f \in f_b$ and $t \in [0, T)$. The maximizers of this likelihood function have the form

$$\hat{a}_{f_b, \ell m} = \sum_{\ell' m'} (\Gamma_{f_b}^{-1})_{\ell m, \ell' m'} \hat{X}_{f_b, \ell' m'} \quad (4.4.2.3)$$

where

$$\hat{X}_{f_b, \ell m} = \sum_{f \in f_b, t \in [0, T)} \gamma_{\ell m}^*(t; f) \frac{H(f)}{P_I(t; f) P_J(t; f)} C(t; f) \quad (4.4.2.4)$$

$$\Gamma_{f_b, \ell m, \ell' m'} = \sum_{f \in f_b, t \in [0, T)} \gamma_{\ell m}^*(t; f) \frac{H^2(f)}{P_I(t; f) P_J(t; f)} \gamma_{\ell' m'}(t; f). \quad (4.4.2.5)$$

Here, Γ is the Fisher information matrix [Thrane \[2009\]](#). Note that if Equation (4.4.2.2)

holds, each $\hat{a}_{f_b, \ell m}$ is a linear combination of complex Gaussian variables; thus, they also follow a complex Gaussian distribution with mean $\langle \hat{a}_{f_b, \ell m} \rangle = a_{f_b, \ell m}$ (discussed further below) and covariance $\text{Cov}(\hat{a}_{f_b, \ell m}, \hat{a}_{f_b, \ell' m'}) = (\Gamma_{f_b}^{-1})_{\ell m, \ell' m'}$. Indeed the same is true, at least approximately, even if Equation (4.4.2.2) does not hold, but the summation over t in Equation (4.4.2.4) is taken over a large number of segments (as is the case with the Advanced LIGO and Advanced Virgo observing runs).

Equation (4.4.2.4) defines a map of the gravitational-wave background convolved with the antenna pattern of the detector pair, the so-called dirty map. The covariance matrix of the dirty map is given in Equation (4.4.2.5). Since Equation (4.4.2.3) represents the deconvolution of the GW signal from the detector response, it is the so-called clean map. The covariance matrix of the clean map is obtained by inverting the covariance matrix of the dirty map.

The estimates $\hat{a}_{f_b, \ell m}$ are evaluated by summing over discrete bins in the frequency band f_b . Broadband analyses perform this sum over the entire available range of frequencies to which the detectors are most sensitive Abbott [2021a]. However, in the analysis presented here, we obtain multiple samples of $\hat{a}_{f_b, \ell m}$ corresponding to sums over different frequency bands. The subscript f_b used in Equation (4.4.2.2) and onwards denotes the frequency range over which each set of $\hat{a}_{f_b, \ell m}$, $\Gamma_{f_b, \ell m, \ell' m'}$, and $\hat{X}_{f_b, \ell m}$ is evaluated. Similar notation is employed throughout the chapter whenever a quantity is evaluated over some discrete range of frequencies rather than considering f_b to be a continuous variable.

From a statistical perspective, maximum likelihood is fundamentally a method for estimating deterministic quantities. This means the $\hat{a}_{f_b, \ell m}$ in Equation (4.4.2.3) serve as maximum likelihood estimators of the spherical harmonic coefficients, $a_{f_b, \ell m}$, that uniquely characterize the SGWB of the only realization of the universe available. Therefore, similarly to what is implicitly done in many statistical analyses, the $a_{f_b, \ell m}$ are not treated as random variables [e.g., Thrane, 2009, Abbott, 2017b, 2019, 2021a]. This is a reasonable

assumption in practical scenarios, and treating the $a_{f_b, \ell m}$ as fixed avoids the risk of incorrectly specifying their underlying distribution, which can substantially bias the resulting inference.

The proposed statistical methodology requires that multiple replicates of $\hat{a}_{f_b, \ell m}$ are available. To ensure that is the case, we partition the total observation time T into S segments, each of duration ΔT , so that $T = \Delta T \cdot S$. The third observing run (O3) includes data from both Advanced LIGO detectors as well as the Advanced Virgo detector, collected from April 1, 2019, to March 27, 2020, with a month-long pause in October 2019. Aside from this pause, the duty cycles of the three detectors were 77%, 75%, and 76% for LIGO Livingston, LIGO Hanford, and Virgo, respectively [Davis \[2021\]](#). For this study, we subdivide the O3 data set into $S = 15$ segments, each corresponding to approximately 10 days of detector live time. We also divide the frequency range 20-180 Hz into $B = 8$ discrete frequency bins, each 20 Hz wide. We assume the same time-domain and frequency-domain data quality cuts that were applied during other O3 anisotropic SGWB search analyses [Abbott \[2021a\]](#).

For each $s = 1, \dots, S$, define $\hat{a}_{f_b, s, \ell m}$, $\hat{X}_{f_b, s, \ell m}$, and $\Gamma_{f_b, s, \ell m, \ell' m'}$ using Equations (4.4.2.3)-(4.4.2.5), but sum over t in the interval $[(s-1)\Delta T, s\Delta T)$ instead of $[0, T)$. It can be easily verified that $\hat{a}_{f_b, s, \ell m}$ are independent (across s) complex Gaussian variables, each with mean $\langle \hat{a}_{f_b, s, \ell m} \rangle = a_{f_b, \ell m}$ and covariance $\text{Cov}(\hat{a}_{f_b, s, \ell m}, \hat{a}_{f_b, s, \ell' m'}) = (\Gamma_{f_b, s}^{-1})_{\ell m, \ell' m'}$. Because the detector sensitivity varies over time – due to factors such as transient environmental noise, detector maintenance, and modifications – we do not assume that $\Gamma_{f_b, s}^{-1}$ is identical across different time segments s .

Since the detector network consists of two Advanced LIGO detectors and one Advanced Virgo detector, there are three pairs of detectors to consider, or three ‘baselines’. Each baseline has its own dirty map and Fisher matrix, and these may be summed together to obtain one overall dirty map and one overall Fisher matrix corresponding to the entire detector network [Thrane \[2009\]](#).

Now, consider the estimator of the squared angular power in each mode ℓ [Thrane \[2009\]](#), [Romano and Cornish \[2017\]](#),

$$\hat{A}_{f_b,s,\ell} = \frac{1}{1+2\ell} \sum_{m=-\ell}^{\ell} \left[|\hat{a}_{f_b,s,\ell m}|^2 - (\Gamma_{f_b,s}^{-1})_{\ell m, \ell m} \right]. \quad (4.4.2.6)$$

Denote with $\hat{\mathbf{A}}_{f_b,s}$ the $\ell_{\max} \times 1$ -dimensional vector with components $\hat{A}_{f_b,s,\ell}$ for all $\ell = 1, \dots, \ell_{\max}$. For each $s = 1, \dots, S$, the estimator $\hat{\mathbf{A}}_{f_b,s}$ has mean $\langle \hat{\mathbf{A}}_{f_b,s} \rangle$ and covariance matrix $\Sigma_{f_b,s}$, whose (ℓ, ℓ') -th element is

$$\begin{aligned} \Sigma_{f_b,s,\ell\ell'} &= \frac{1}{(1+2\ell)(1+2\ell')} \sum_{m,m'} \left(|(\Gamma_{f_b,s}^{-1})_{\ell m, \ell' m'}|^2 \right. \\ &\quad \left. + 2 \operatorname{Re}[a_{f_b,\ell m}^* (\Gamma_{f_b,s}^{-1})_{\ell m, \ell' m'} a_{f_b,\ell' m'}] \right). \end{aligned} \quad (4.4.2.7)$$

The derivation of Equation (4.4.2.7) is given in Appendix C.2. Previous works [e.g., [Agarwal, 2023](#)] have omitted the second term in the numerator of Equation (4.4.2.7), implicitly making the assumption that $a_{f_b,\ell m}$'s are complex Gaussian random variables with zero mean. However, this assumption is not consistent with estimating the coefficients $a_{f_b,\ell m}$ (as realized in our universe) via the maximum likelihood technique – that is, likelihood maximization implicitly assumes the coefficients to be the (deterministic) limit to which the maximum likelihood estimates converge. In this work, this inconsistency is avoided by treating $a_{f_b,\ell m}$ as fixed.

Finally, it is worth mentioning that one could always replace $\Gamma_{f_b,s}^{-1}$ in Equation (4.4.2.7) with its regularized counterpart if a regularization scheme were applied to invert the Fisher matrix [Cf. [Thrane, 2009](#), [Agarwal, 2023](#), [Panda, 2019](#)]. Such regularization is typically implemented via singular value decomposition to mitigate small singular values arising from directions in the sky to which the detector network is insensitive [[Thrane, 2009](#), [Abbott, 2021a](#)]. However, this procedure introduces a bias that propagates into $\Sigma_{f_b,s}$ in nontrivial ways [[Yang, 2023](#)]. To avoid this bias, the Fisher matrix is not regularized here.

4.3 Statistical methods

The main goal is to develop an inferential strategy for estimating angular power spectral models and propose a statistical test to assess whether the proposed model is consistent with the data derived from the estimator in Equation (4.4.2.6).

Let $\boldsymbol{\theta} \in \mathbb{R}^p$ be an unknown parameter vector, and denote with $\mathbf{A}_{f_b}(\boldsymbol{\theta})$ the proposed model for the angular power spectrum. Now, assume that the $\hat{\mathbf{A}}_{f_b,s}$ vectors are independent across all frequency bins and data segments. Hence, in this context, ‘large sample’ asymptotic results refer to the situation in which $N = \ell_{\max} \cdot B \cdot S \rightarrow \infty$, but, since typically the $\ell_{\max}$ elements of the vectors $\hat{\mathbf{A}}_{f_b,s}$ are not independent for different values of $\ell = 1, \dots, \ell_{\max}$, the effective sample size is (at least) $B \cdot S$. Formally, we aim to estimate $\boldsymbol{\theta}$ and test the hypotheses

$$H_0 : \langle \hat{\mathbf{A}}_{f_b,s} \rangle = \mathbf{A}_{f_b}(\boldsymbol{\theta}) \quad \text{vs.} \quad H_a : \langle \hat{\mathbf{A}}_{f_b,s} \rangle \neq \mathbf{A}_{f_b}(\boldsymbol{\theta}) \quad (4.4.3.1)$$

without assuming $\hat{\mathbf{A}}_{f_b,s}$ follows any specific distribution.

4.3.1 Parameter and covariance estimation

Begin by estimating the true value of the parameter vector $\boldsymbol{\theta}$, hereinafter denoted by $\boldsymbol{\theta}_0$, using the vectors $\hat{\mathbf{A}}_{f_b,s}$. For the moment, assume that the covariance matrices $\Sigma_{f_b,s}$ are known, and define the *sphered errors* as

$$\boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}) = \tilde{\mathbf{A}}_{f_b,s} - \tilde{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}), \quad (4.4.3.1)$$

with

$$\tilde{\mathbf{A}}_{f_b,s} = \Sigma_{f_b,s}^{-1/2} \hat{\mathbf{A}}_{f_b,s} \quad \text{and} \quad \tilde{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}) = \Sigma_{f_b,s}^{-1/2} \mathbf{A}_{f_b}(\boldsymbol{\theta}), \quad (4.4.3.2)$$

where $\Sigma_{f_b,s}^{-1/2}$ is the inverse square-root matrix of $\Sigma_{f_b,s}$ and can be obtained via standard procedures [Higham, 2008, Horn and Johnson, 2012]. The transformation in Equation

(4.4.3.2) is known as *whitening* or *sphering* and enables the construction of a statistically orthonormalized version of the angular power spectrum $\hat{\mathbf{A}}_{f_b,s}$. In other words, under H_0 in Equation (4.4.3.1), $\tilde{\mathbf{A}}_{f_b,s}$ has mean given by $\tilde{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta})$, its components are uncorrelated, and each has a unit variance.

The true value of the unknown parameter $\boldsymbol{\theta}_0$ can be estimated by solving the *generalized nonlinear least squares* problem:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \sum_{b,s} \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta})^T \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}). \quad (4.4.3.3)$$

The optimization in Equation (4.4.3.3) is the same as a maximization of the likelihood of $\hat{\mathbf{A}}_{f_b,s}$ under the assumption of Gaussianity. Yet, its underlying rationale differs in that it aims to minimize the squared differences of the sphered errors, regardless of their distribution. In general, such a minimization does not enjoy a closed-form solution and is typically solved using iterative numerical methods [Nocedal and Wright \[1999\]](#).

The solution of Equation (4.4.3.3), can be shown to be consistent under H_0 . That is, if the model $\mathbf{A}_{f_b}(\boldsymbol{\theta})$ is correctly specified, and if it is identifiable – that is, $\mathbf{A}_{f_b}(\boldsymbol{\theta}) = \mathbf{A}_{f_b}(\boldsymbol{\theta}')$ implies $\boldsymbol{\theta} = \boldsymbol{\theta}'$ [Cf. [Davidson and MacKinnon, 2004](#), Ch. 7] – then, $\hat{\boldsymbol{\theta}}$ converges in probability [for a definition see [Van der Vaart, 2000](#), Ch. 2] to $\boldsymbol{\theta}_0$. Furthermore, under additional classical regularity conditions [Cf. [Davidson and MacKinnon, 2004](#), Ch. 7], $\hat{\boldsymbol{\theta}}$ is approximately Gaussian distributed with mean $\boldsymbol{\theta}_0$ and its variance approaches zero as $N \rightarrow \infty$. This can be seen by taking the derivative of the objective function in Equation (4.4.3.3) with respect to $\boldsymbol{\theta}$, which shows that the generalized least squares estimator $\hat{\boldsymbol{\theta}}$ is a solution to the system of p estimating equations:

$$\frac{1}{\sqrt{N}} \sum_{b,s} \dot{\mathbf{A}}_{f_b,s}(\hat{\boldsymbol{\theta}})^T \boldsymbol{\epsilon}_{f_b,s}(\hat{\boldsymbol{\theta}}) = \mathbf{0}, \quad (4.4.3.4)$$

where $\dot{\mathbf{A}}_{f_b,s}(\hat{\boldsymbol{\theta}})$ denotes the $\ell_{\max} \times p$ matrix with columns corresponding to the partial

derivatives of $\tilde{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta})$ evaluated at $\hat{\boldsymbol{\theta}}$, i.e.,

$$\dot{\mathbf{A}}_{f_b,s}(\hat{\boldsymbol{\theta}}) = \Sigma_{f_b,s}^{-1/2} \left[\frac{\partial}{\partial \theta_1} \mathbf{A}_{f_b}(\boldsymbol{\theta}), \dots, \frac{\partial}{\partial \theta_p} \mathbf{A}_{f_b}(\boldsymbol{\theta}) \right] \Big|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}},$$

in which θ_j , $j = 1, \dots, p$ are the elements of $\boldsymbol{\theta}$. A first-order Taylor expansion of the left-hand side of Equation (4.4.3.4) around $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0$ leads to:

$$\frac{1}{\sqrt{N}} \sum_{b,s} \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)^T \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}_0) - \mathbf{R}_p \sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1) = \mathbf{0} \quad (4.4.3.5)$$

where the notation $o_p(1)$ is used to indicate random terms that converge to zero in probability as $N \rightarrow \infty$ and $\mathbf{R}_p$ is a $p \times p$ matrix given by

$$\mathbf{R}_p = \frac{1}{N} \sum_{b,s} \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)^T \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0). \quad (4.4.3.6)$$

By rearranging the terms in Equation (4.4.3.5), we obtain the asymptotic representation:

$$\sqrt{N}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = \frac{1}{\sqrt{N}} \mathbf{R}_p^{-1} \sum_{b,s} \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)^T \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}_0) + o_p(1). \quad (4.4.3.7)$$

By the central limit theorem, it follows that $\hat{\boldsymbol{\theta}}$ is approximately Gaussian distributed with mean $\boldsymbol{\theta}_0$ and variance-covariance matrix $\mathbf{R}_p^{-1}/N$.

When $\Sigma_{f_b,s}^{-1}$ is unknown but a uniformly (for all s) consistent estimator is available, the statistical properties of $\hat{\boldsymbol{\theta}}$ described above remain valid provided that this estimator is substituted for $\Sigma_{f_b,s}^{-1}$ in Equation (4.4.3.2) [Cf. Davidson and MacKinnon, 2004, Ch. 7].

In our case, such an estimator for $\Sigma_{f_b,s}^{-1}$ can be obtained by estimating $\Sigma_{f_b,s}$ from Equation (4.4.2.7), and a corresponding ‘plug-in’ estimator for $\Sigma_{f_b,s}$ is now briefly described. Notice that

$$\hat{a}_{f_b,\ell m} = \sum_{s=1}^S \sum_{\ell', m'} (\Gamma_{f_b}^{-1})_{\ell m, \ell' m'} \hat{X}_{f_b,s, \ell' m'}, \quad (4.4.3.8)$$

and hence $\hat{a}_{f_b, \ell m}$ depends on the number of segments S . In Appendix C.3, it is established that $\hat{a}_{f_b, \ell m}$ converges in probability to $a_{f_b, \ell m}$ as $S \rightarrow \infty$ ¹. The estimator has elements $(\hat{\Sigma}_{f_b, s})_{\ell, \ell'}$ obtained by replacing $a_{f_b, \ell m}^*$ and $a_{f_b, \ell m}$ with $\hat{a}_{f_b, \ell m}^*$ and $\hat{a}_{f_b, \ell m}$ in Equation (4.4.2.7). A theoretical analysis confirming the positive definiteness of these covariance estimators and a detailed proof that the inverse $\hat{\Sigma}_{f_b, s}^{-1}$ is a uniformly consistent estimator of $\Sigma_{f_b, s}^{-1}$ can be found in Appendix C.3.

4.3.2 Distribution-free testing of the angular power spectrum

In its essence, the statistical formulation of the problem provided in Sec. 4.3.1 is that of a regression problem in which the estimated angular power spectrum, $\hat{\mathbf{A}}_{f_b, s}$, plays the role of the outcome whose mean we aim to describe through the model $\mathbf{A}_{f_b, s}(\boldsymbol{\theta})$. We can, therefore, employ constructs from the theory of goodness-of-fit for regression to assess the validity of such a model.

The main building blocks needed to devise a test for the hypotheses in Equation (4.4.3.1) are the *sphered residuals*. They can be obtained by replacing $\boldsymbol{\theta}$ in Equation (4.4.3.1) with the solution of Equation (4.4.3.3), i.e.,

$$\hat{\boldsymbol{\epsilon}}_{f_b, s} = \tilde{\mathbf{A}}_{f_b, s} - \tilde{\mathbf{A}}_{f_b, s}(\hat{\boldsymbol{\theta}}). \quad (4.4.3.1)$$

In principle, one could test the hypotheses in Equation (4.4.3.1) using classical goodness-of-fit statistics constructed as the sum of some function of the sphered residuals – with Pearson’s χ^2 being possibly the most prominent example among those – and derive their distribution under the hypothesis H_0 using the parametric bootstrap [Cf. Freedman, 1981] based on the generalized χ^2 distribution derived in Appendix C.1. Such statistics, however, are known to exhibit no statistical power² toward infinitely many possible ‘weak’ deviations from the postulated model [e.g., Khmaladze, 1984, Algeri and Khmaladze, 2024b, Acharya

¹Note that as S increases, so does the total observation time T .

²In statistics, the concept of ‘power’ refers to the sensitivity of a test in detecting departures from the model hypothesized under the null hypothesis.

and Kashyap, 2024]. That is, there are infinite possible ways in which the hypothesized model could be wrong, but such statistics would not be able to identify them. Test statistics based on the process of partial sums of the residuals [e.g., Loynes, 1980, Stute, 1997], however, can restore statistical power [e.g., Khmaladze, 1984, Algeri and Khmaladze, 2024b, Acharya and Kashyap, 2024] and are thus recommended.

4.3.2.1 Model testing via partial sums of the residuals

Let ϵ and $\hat{\epsilon}$ denote, respectively, the $N \times 1$ stacked vectors of errors $\epsilon_{f_b,s}(\theta_0)$ and residuals $\hat{\epsilon}_{f_b,s}$ for all frequency bins and data segments, that is,

$$\epsilon = [\epsilon_{f_1,1}^T(\theta_0), \dots, \epsilon_{f_B,1}^T(\theta_0), \dots, \epsilon_{f_1,S}^T(\theta_0), \dots, \epsilon_{f_B,S}^T(\theta_0)]^T,$$

and

$$\hat{\epsilon} = [\hat{\epsilon}_{f_1,1}^T, \dots, \hat{\epsilon}_{f_B,1}^T, \dots, \hat{\epsilon}_{f_1,S}^T, \dots, \hat{\epsilon}_{f_B,S}^T]^T.$$

The process of partial sums of the sphered residuals is

$$w_{k,N} = \frac{1}{\sqrt{N}} \hat{\epsilon}^T \mathbb{I}_{k,N},$$

$$\text{with } \mathbb{I}_{k,N} = [\underbrace{1, \dots, 1}_k, \underbrace{0, \dots, 0}_{N-k}]^T. \quad (4.4.3.2)$$

It consists of normalized cumulative sums of the first k elements of $\hat{\epsilon}$ with $k = 1, \dots, N$.

Following the same argument of Khmaladze [2021], it is possible to show that $w_{k,N}$ is asymptotically equal to a projection of the process of partial sums of the sphered errors given by

$$\frac{1}{\sqrt{N}} \epsilon^T \mathbb{I}_{k,N}. \quad (4.4.3.3)$$

As shown below, such a projection plays a fundamental role in deriving the limiting distribution of $w_{k,N}$.

Consider a first-order Taylor expansion of the residuals in Equation (4.4.3.1) evaluated at $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0$, that is,

$$\hat{\boldsymbol{\epsilon}}_{f_b,s} = \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}_0) - \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1). \quad (4.4.3.4)$$

Substituting the right-hand side of Equation (4.4.3.7) for $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0$ in Equation (4.4.3.4) leads to

$$\begin{aligned} \hat{\boldsymbol{\epsilon}}_{f_b,s} = & \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}_0) \\ & - \frac{1}{N} \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0) \left[\mathbf{R}_p^{-1} \sum_{f_b,s} \dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)^T \boldsymbol{\epsilon}_{f_b,s}(\boldsymbol{\theta}_0) \right] + o_p(1). \end{aligned} \quad (4.4.3.5)$$

Let $\dot{\mathbf{A}}(\boldsymbol{\theta}_0)$ be a $N \times p$ matrix denoting the stacked matrix of $\dot{\mathbf{A}}_{f_b,s}(\boldsymbol{\theta}_0)$ across all frequency bins and time segments, i.e.,

$$\dot{\mathbf{A}}(\boldsymbol{\theta}_0) = \left[\dot{\mathbf{A}}_{f_{1,1}}^T(\boldsymbol{\theta}_0), \dots, \dot{\mathbf{A}}_{f_{B,1}}^T(\boldsymbol{\theta}_0), \dots, \dot{\mathbf{A}}_{f_{B,S}}^T(\boldsymbol{\theta}_0) \right]^T,$$

then, the vector $\hat{\boldsymbol{\epsilon}}$, with elements given by Equation (4.4.3.5), can be written as

$$\begin{aligned} \hat{\boldsymbol{\epsilon}} &= \boldsymbol{\epsilon} - \frac{1}{N} \dot{\mathbf{A}}(\boldsymbol{\theta}_0) \mathbf{R}_p^{-1} \dot{\mathbf{A}}(\boldsymbol{\theta}_0)^T \boldsymbol{\epsilon} + o_p(1) \\ &= \boldsymbol{\epsilon} - \sum_{j=1}^p \boldsymbol{\mu}_{\boldsymbol{\theta}_j} \boldsymbol{\mu}_{\boldsymbol{\theta}_j}^T \boldsymbol{\epsilon} + o_p(1), \end{aligned} \quad (4.4.3.6)$$

where $\boldsymbol{\mu}_{\boldsymbol{\theta}_j}$ corresponds to the j th column of the matrix

$$\frac{1}{\sqrt{N}} \dot{\mathbf{A}}(\boldsymbol{\theta}_0) \mathbf{R}_p^{-1/2}.$$

From Equation (4.4.3.6), it follows that the sphered residuals are asymptotically equal to a projection of the errors orthogonal to the vectors $\{\boldsymbol{\mu}_{\boldsymbol{\theta}_j}\}_{j=1}^p$. Similarly, the process

$w_{k,N}$ can be written as:

$$w_{k,N} = \frac{1}{\sqrt{N}} \epsilon^T \left(\mathbf{I}_N - \sum_{j=1}^p \boldsymbol{\mu}_{\theta_j} \boldsymbol{\mu}_{\theta_j}^T \right) \mathbb{I}_{k,N} + o_p(1), \quad (4.4.3.7)$$

where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix. It follows that, in the limit, $w_{k,N}$ is a projection of the process in Equation (4.4.3.3) orthogonal to the vectors $\{\boldsymbol{\mu}_{\theta_j}\}_{j=1}^p$.

The process in Equation (4.4.3.3) can be shown to be asymptotically distributed as a Brownian motion if H_0 in Equation (4.4.3.1) is true. This result follows from the fact that such a process consists of partial sums of weakly dependent³ quantities [Cf. Philipp and Stout, 1975, Bradley, 2005] with mean zero and finite variance. The condition of weak dependence is satisfied since the subvectors $\epsilon_{f_b,s}(\boldsymbol{\theta}_0)$ of ϵ are all independent from one another. It follows that since $w_{k,N}$ is a projection of the process in Equation (4.4.3.3), its limiting distribution under H_0 is that of a projected Brownian motion. Its mean and covariance are

$$\begin{aligned} \langle w_{k,N} \rangle &\simeq 0 \quad \text{and} \\ \langle w_{k,N} w_{k',N} \rangle &\simeq \frac{1}{N} \mathbb{I}_{k,N}^T \left(\mathbf{I}_N - \sum_{j=1}^p \boldsymbol{\mu}_{\theta_j} \boldsymbol{\mu}_{\theta_j}^T \right) \mathbb{I}_{k',N}. \end{aligned} \quad (4.4.3.8)$$

where the notation $\simeq$ indicates that the functions on both sides are asymptotically equal as $N \rightarrow \infty$.

Test statistics to assess the validity of H_0 in Equation (4.4.3.1) can be constructed by considering functionals of the process $w_{k,N}$ in Equation (4.4.3.2). Two examples are

$$K_N = \max_k |w_{k,N}| \quad \text{and} \quad C_N = \frac{1}{N} \sum_{k=1}^N |w_{k,N}|^2, \quad (4.4.3.9)$$

³Intuitively, a sequence of random variables is said to be ‘weakly dependent’ if the dependence among the elements of the sequence reduces and eventually ceases to exist for elements that are sufficiently far away from one another within the sequence.

which can be seen as the counterparts of the Kolmogorov–Smirnov statistic and the Cramér–von Mises statistic in the context of regression.

The second asymptotic equality in Equation (4.4.3.8) implies that the distribution of $w_{k,N}$ under H_0 depends on the model being tested through the vectors $\{\boldsymbol{\mu}_{\theta_j}\}_{j=1}^p$. Hence, the same is true for the distribution of K_N and C_N . Such distributions, however, are unaffected by the distribution of the errors in $\boldsymbol{\epsilon}$ and, therefore, they only depend on the mean and covariance of $\hat{\mathbf{A}}_{f_b,s}$. This implies that the statistics K_N and C_N always have the same distribution under H_0 , regardless of the distribution of $\hat{\mathbf{A}}_{f_b,s}$ being approximately Gaussian in the limit.

To illustrate this aspect with an example, consider testing the models $\mathbf{A}_{f_b}^{M1}(\boldsymbol{\theta})$ and $\mathbf{A}_{f_b}^{M2}(\boldsymbol{\theta})$, with components given by

$$\begin{aligned}\mathbf{A}_{f_b,\ell}^{M1}(\boldsymbol{\theta}) &= (\theta_0 + \theta_1\ell + \theta_2\ell^2)\bar{f}_b^{2/3}, \\ \mathbf{A}_{f_b,\ell}^{M2}(\boldsymbol{\theta}) &= \exp(\theta_0 + \theta_1\ell + \theta_2\ell^2)\bar{f}_b^{2/3}\end{aligned}\tag{4.4.3.10}$$

for $\ell = 1, \dots, 8$, $f_b = [20, 40], [40, 60], \dots, [160, 180]$, $\boldsymbol{\theta} = (\theta_0, \theta_1, \theta_2)$ and $\bar{f}_b$ is the center of the frequency band f_b .

Consider the sphered errors generated either from a standard multivariate Gaussian distribution or independently from Laplace distribution with mean 0 and variance 1. Let the covariance matrices $\boldsymbol{\Sigma}_{f_b,s}$ be generated from a Wishart distribution with 10 degrees of freedom, and these matrices are assumed to be known. Denote the unsphered errors as

$$\mathbf{u}_{f_b,s}^{(1)} \sim N(\mathbf{0}, \boldsymbol{\Sigma}_{f_b,s}), \quad \mathbf{u}_{f_b,s}^{(2)} \sim L(\mathbf{0}, \boldsymbol{\Sigma}_{f_b,s}),\tag{4.4.3.11}$$

for all $s = 1, \dots, S$. We consider four datasets generated from the four possible additive combinations of models and error terms in Equations (4.4.3.10)-(4.4.3.11) with $\boldsymbol{\theta} = (4, -3, 0.4)$. For each of these datasets, we perform $S = 15$ repeated simulations, resulting in an effective sample size of 120, and a total of $N = 960$. The datasets are generated 10^4

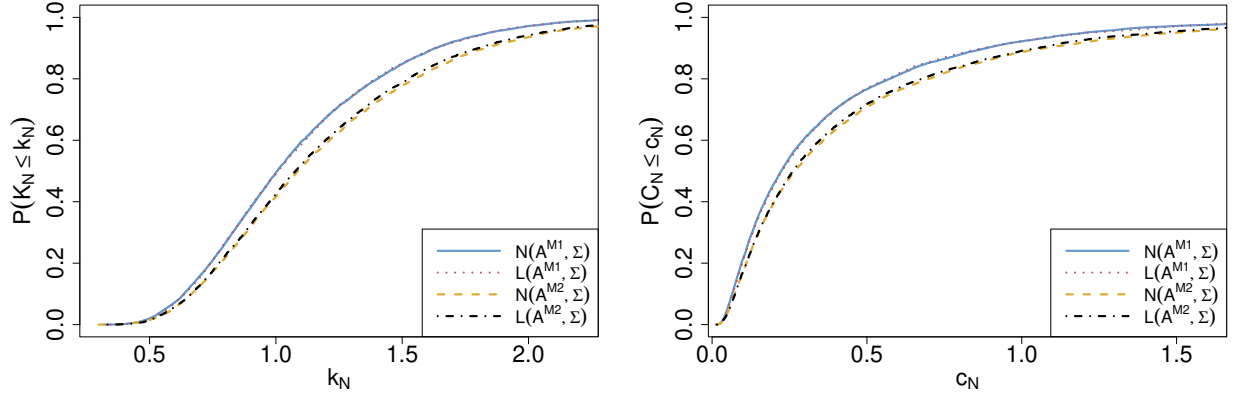


Figure 4.1: Graphs of the simulated null cumulative distribution functions of the Kolmogorov–Smirnov (top) and Cramér–von Mises (bottom) statistics from the four combinations of $\mathbf{A}_{f_b,s}^{M1}$ and $\mathbf{u}_{f_b,s}^{(1)}$ (blue solid); $\mathbf{A}_{f_b,s}^{M1}$ and $\mathbf{u}_{f_b,s}^{(2)}$ (red dotted); $\mathbf{A}_{f_b,s}^{M2}$ and $\mathbf{u}_{f_b,s}^{(1)}$ (yellow dashed); $\mathbf{A}_{f_b,s}^{M2}$ and $\mathbf{u}_{f_b,s}^{(2)}$ (darkgreen dash-dotted).

times; each time, we calculate a single realization of the process $w_{k,N}$ in Equation (4.4.3.7), which yields one observation of the statistics K_N and C_N from Equation (4.4.3.9).

Figure 4.1 shows the simulated distribution of the Kolmogorov–Smirnov and Cramér–von Mises statistics in Equation (4.4.3.9) for each of the four combinations of models and error terms specified in Equations (4.4.3.10)–(4.4.3.11) under H_0 in (4.4.3.1) – that is, the statistics are constructed so that the model being tested and the model used to simulate the data is the same. Observe that the distributions of the test statistics overlap when the underlying models are the same, regardless of the error distributions. This observation aligns with Equations (4.4.3.8)–(4.4.3.9), which demonstrate that the distribution of the statistics depends solely on the model being tested through the vectors $\{\boldsymbol{\mu}_{\theta_j}\}_{j=1}^p$.

4.3.2.2 Construction of distribution-free test statistics

Since the null distribution of the test statistics in Equation (4.4.3.9) depends on the model being tested, it can only be derived on a case-by-case basis for each specific choice of the postulated angular power spectrum model $\mathbf{A}_{f_b}(\boldsymbol{\theta})$. Unfortunately, this process can be

computationally burdensome when testing several different and possibly complicated models. Nevertheless, it is possible to overcome this limitation by relying on the *Khmaladze-2 (K2) transform* [Khmaladze \[2021\]](#).

Given an arbitrary orthonormal set of vectors $\{\mathbf{r}_j\}_{j=1}^p$ in $\mathbb{R}^N$, the K2 transformation allows us to map $w_{N,k}$ into a process of partial sums, hereafter denoted by $v_{k,N}$ with ‘standard’ limiting distribution under the null hypothesis. Specifically, for sufficiently large N , its null distribution is well-approximated by that of a projected Brownian motion with covariance

$$\langle v_{k,N} v_{k',N} \rangle \simeq \frac{1}{N} \mathbb{I}_{k,N}^T \left(\mathbf{I}_N - \sum_{j=1}^p \mathbf{r}_j \mathbf{r}_j^T \right) \mathbb{I}_{k',N}. \quad (4.4.3.12)$$

Thus, for any choice of $\{\mathbf{r}_j\}_{j=1}^p$ independent from $\mathbf{A}_f(\theta)$ and such that $\sum_{j=1}^p \mathbf{r}_j \mathbf{r}_j^T \neq \mathbf{I}_N$, the distribution of $v_{k,N}$ is also independent from such a model. One can then rely on the functionals of the newly constructed process $v_{k,N}$ such as

$$\tilde{K}_N = \max_k |v_{k,N}|, \quad \tilde{C}_N = \frac{1}{N} \sum_{k=1}^N |v_{k,N}|^2, \quad (4.4.3.13)$$

to test the hypotheses in Equation (4.4.3.1) in a distribution-free manner. As demonstrated in Section 4.3.3, this implies one can test different postulated models for the angular power spectrum by relying on a single simulation of the distribution of the statistics in Equation (4.4.3.13).

There are many possible ways to choose the orthonormal set $\{\mathbf{r}_j\}_{j=1}^p$. For example, as proposed by [Khmaladze \[2021\]](#), one could consider

$$\mathbf{r}_1 = (1/\sqrt{N}, \dots, 1/\sqrt{N})^T,$$

letting $\mathbf{r}_2$ be the vector of elements

$$\mathbf{r}_{2,n} = \sqrt{\frac{12N}{N^2-1}} \left(\frac{n}{N} - \frac{N+1}{2N} \right), \quad n = 1, \dots, N; \quad (4.4.3.14)$$

the remaining vectors $\mathbf{r}_3, \dots, \mathbf{r}_p$ can be obtained by the Gram–Schmidt process applied to the powers of $\mathbf{r}_2$. Alternatively, one could construct the orthonormal set $\{\mathbf{r}_j\}_{j=1}^p$ by considering a set of p orthonormal functions with a given inner product. For example, the Hermite polynomials $\{H_j(x)\}_{j=1}^p$ satisfy the condition:

$$\int_{-\infty}^{+\infty} H_i(x) H_j(x) e^{-x^2} dx = \delta_{ij}.$$

We can then specify the elements of the vectors $\mathbf{r}_j$ as

$$r_{j,n} = \frac{1}{\sqrt{c_j}} H_j(x_n) \sqrt{e^{-x_n^2}},$$

where $c_j = \sum_n H_j^2(x_n) e^{-x_n^2}$ are normalizing constants and $x_n = -\zeta + \frac{2\zeta(n-1)}{N-1}$, $n = 1, \dots, N$, represent N equally spaced points over the interval $[-\zeta, \zeta]$ for some appropriately chosen constant $\zeta > 0$.

The main steps involved in the construction of the process $v_{k,N}$, for any given orthonormal set $\{\mathbf{r}_j\}_{j=1}^p$, are outlined in [Algeri \[2025\]](#). Below, we provide a thorough exposition of the technical aspects of such a transformation.

To ease the intuition, let us first consider the simple scenario where the unknown parameter $\boldsymbol{\theta}$ is one-dimensional. This situation is encountered, for example, when the interest is solely in estimating the amplitude of the astrophysical kernel used to generate theoretical values of the angular power spectrum (see Sec. 4.4). Define the operator $U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1}$ on $\mathbb{R}^N$ as

$$U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} = \mathbf{I} - \frac{\langle \boldsymbol{\mu}_{\boldsymbol{\theta}_1} - \mathbf{r}_1, \cdot \rangle}{1 - \langle \boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1 \rangle} (\boldsymbol{\mu}_{\boldsymbol{\theta}_1} - \mathbf{r}_1), \quad (4.4.3.15)$$

where $\mathbf{I}$ is the identity operator. One can easily verify that $U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1}$ maps $\boldsymbol{\mu}_{\boldsymbol{\theta}_1}$ to $\mathbf{r}_1$, $\mathbf{r}_1$ to

$\boldsymbol{\mu}_{\boldsymbol{\theta}_1}$, and leaves vectors orthogonal to both $\boldsymbol{\mu}_{\boldsymbol{\theta}_1}$ and $\mathbf{r}_1$ unchanged. As shown in Appendix C.4, such an operator is unitary and self-adjoint. The *K2-transformed errors* are given by

$$\mathbf{e} = U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\epsilon}$$

and we define the *K2-transformed residuals* as

$$\hat{\mathbf{e}} = U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \hat{\boldsymbol{\epsilon}}.$$

Similarly to Equation (4.4.3.6), the K2-transformed residuals are asymptotically equal to a projection of the K2-transformed errors. Specifically,

$$\begin{aligned} \hat{\mathbf{e}} &= U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\epsilon} - U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\mu}_{\boldsymbol{\theta}_1} \boldsymbol{\mu}_{\boldsymbol{\theta}_1}^T \boldsymbol{\epsilon} + o_p(1) \\ &= \mathbf{e} - \mathbf{r}_1 \mathbf{r}_1^T \mathbf{e} + o_p(1), \end{aligned} \tag{4.4.3.16}$$

where the second equality arises from the fact that

$$\begin{aligned} U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\mu}_{\boldsymbol{\theta}_1} &= \mathbf{r}_1 \\ \text{and } \boldsymbol{\mu}_{\boldsymbol{\theta}_1}^T \boldsymbol{\epsilon} &= \langle U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\mu}_{\boldsymbol{\theta}_1}, U_{\boldsymbol{\mu}_{\boldsymbol{\theta}_1}, \mathbf{r}_1} \boldsymbol{\epsilon} \rangle = \mathbf{r}_1^T \mathbf{e}. \end{aligned}$$

The process of partial sums of the K2-transformed residuals is

$$\begin{aligned} v_{k,N} &= \frac{1}{\sqrt{N}} \hat{\mathbf{e}}^T \mathbb{I}_{k,N} \\ &= \frac{1}{\sqrt{N}} \mathbf{e}^T \left(I_N - \mathbf{r}_1 \mathbf{r}_1^T \right) \mathbb{I}_{k,N} + o_p(1); \end{aligned} \tag{4.4.3.17}$$

its limiting distribution is that of a projected Brownian motion with zero mean and covariance approximately equal to the right-hand side of Equation (4.4.3.12) for $p = 1$.

Let us now consider the situation in which $\boldsymbol{\theta}$ is multidimensional – as would be the case when estimating the amplitude, mean, and standard deviation of a Gaussian astrophysical

kernel (see Sec. 4.4, Equation (4.4.4.5)). Similarly to the one-dimensional case, the goal is to identify an operator, hereafter denoted by $\mathbf{U}_p^T$, such the resulting K2-transformed vector of residuals $\hat{\mathbf{e}} = \mathbf{U}_p^T \hat{\mathbf{e}}$ is asymptotically equal to a projection of the vector of K2-transformed errors $\mathbf{e} = \mathbf{U}_p^T \boldsymbol{\epsilon}$, i.e.,

$$\begin{aligned}\hat{\mathbf{e}} &= \mathbf{U}_p^T \boldsymbol{\epsilon} - \sum_{j=1}^p \mathbf{U}_p^T \boldsymbol{\mu}_{\theta_j} \boldsymbol{\mu}_{\theta_j}^T \boldsymbol{\epsilon} + o_p(1) \\ &= \mathbf{e} - \sum_{j=1}^p \mathbf{r}_j \mathbf{r}_j^T \mathbf{e} + o_p(1).\end{aligned}\tag{4.4.3.18}$$

To serve this purpose, we aim to construct $\mathbf{U}_p^T$ so that it enables mapping the vectors $\{\boldsymbol{\mu}_{\theta_j}\}_{j=1}^p$ to the arbitrarily chosen orthonormal set $\{\mathbf{r}_j\}_{j=1}^p$. A – somewhat naive – first attempt, could be that of combining operators of the form $\mathbf{U}_{\boldsymbol{\mu}_{\theta_j}, \mathbf{r}_j}$, constructed as in Equation (4.4.3.15). Such an approach, however, does not yield the desired result. To see this, consider

$$\mathbf{U}_{\boldsymbol{\mu}_{\theta_1}, \mathbf{r}_1} \mathbf{U}_{\boldsymbol{\mu}_{\theta_2}, \mathbf{r}_2} \boldsymbol{\mu}_{\theta_2} = \mathbf{U}_{\boldsymbol{\mu}_{\theta_1}, \mathbf{r}_1} \mathbf{r}_2 \neq \mathbf{r}_2.$$

To overcome this difficulty, consider a set of vectors $\{\tilde{\mathbf{r}}_j\}_{j=1}^p$ constructed so that each $\tilde{\mathbf{r}}_j$ is orthogonal to $\boldsymbol{\mu}_{\theta_k}$, for all $k < j$. Specifically, we choose $\tilde{\mathbf{r}}_1 = \mathbf{r}_1$ and

$$\tilde{\mathbf{r}}_j = \mathbf{U}_{j-1} \mathbf{r}_j\tag{4.4.3.19}$$

for $j = 2, \dots, p$, with $\mathbf{U}_j$ denoting the product operator:

$$\mathbf{U}_j = \mathbf{U}_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j} \dots \mathbf{U}_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1}\tag{4.4.3.20}$$

with each operator $\mathbf{U}_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j}$ acting on everything on its right. Since the operator $\mathbf{U}_j$ involves a product of unitary operators, it is also unitary. Its adjoint operator is given by its transpose:

$$\mathbf{U}_j^T = \mathbf{U}_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \dots \mathbf{U}_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j}.\tag{4.4.3.21}$$

One can verify that the vectors $(\tilde{\mathbf{r}}_j)_{j=1}^p$ are orthogonal to $\boldsymbol{\mu}_{\theta_k}$, for all $k < j$ by noticing that

$$\begin{aligned}\langle \tilde{\mathbf{r}}_2, \boldsymbol{\mu}_{\theta_1} \rangle &= \langle \mathbf{r}_2, U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \boldsymbol{\mu}_{\theta_1} \rangle = \langle \mathbf{r}_2, \mathbf{r}_1 \rangle = 0, \\ \langle \tilde{\mathbf{r}}_3, \boldsymbol{\mu}_{\theta_1} \rangle &= \langle U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \mathbf{r}_3, U_{\boldsymbol{\mu}_{\theta_2}, \tilde{\mathbf{r}}_2} \boldsymbol{\mu}_{\theta_1} \rangle = \langle U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \mathbf{r}_3, \boldsymbol{\mu}_{\theta_1} \rangle \\ &= \langle \mathbf{r}_3, U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \boldsymbol{\mu}_{\theta_1} \rangle = \langle \mathbf{r}_3, \mathbf{r}_1 \rangle = 0, \\ \langle \tilde{\mathbf{r}}_3, \boldsymbol{\mu}_{\theta_2} \rangle &= \langle U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \mathbf{r}_3, U_{\boldsymbol{\mu}_{\theta_2}, \tilde{\mathbf{r}}_2} \boldsymbol{\mu}_{\theta_2} \rangle = \langle U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \mathbf{r}_3, \tilde{\mathbf{r}}_2 \rangle \\ &= \langle \mathbf{r}_3, \mathbf{r}_2 \rangle = 0,\end{aligned}$$

and proceeding by induction.

The operator $\mathbf{U}_p$ defined in Equation (4.4.3.20), for $j = p$, maps vectors $\mathbf{r}_j$ to $\boldsymbol{\mu}_{\theta_j}$ for all $j = 1, \dots, p$. Specifically,

$$\begin{aligned}\mathbf{U}_p \mathbf{r}_j &= U_{\boldsymbol{\mu}_{\theta_p}, \tilde{\mathbf{r}}_p} \cdots U_{\boldsymbol{\mu}_{\theta_1}, \tilde{\mathbf{r}}_1} \mathbf{r}_j \\ &= U_{\boldsymbol{\mu}_{\theta_p}, \tilde{\mathbf{r}}_p} \cdots U_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j} \tilde{\mathbf{r}}_j \\ &= U_{\boldsymbol{\mu}_{\theta_p}, \tilde{\mathbf{r}}_p} \cdots U_{\boldsymbol{\mu}_{\theta_{j+1}}, \tilde{\mathbf{r}}_{j+1}} \boldsymbol{\mu}_{\theta_j} = \boldsymbol{\mu}_{\theta_j},\end{aligned}\tag{4.4.3.22}$$

where the second equality follows from the definition of $\tilde{\mathbf{r}}_j$ in Equation (4.4.3.19) and the third equality follows from the fact that each operator $U_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j}$ maps $\tilde{\mathbf{r}}_j$ to $\boldsymbol{\mu}_{\theta_j}$. The fourth equality holds since $\boldsymbol{\mu}_{\theta_j}$ is orthogonal to $\boldsymbol{\mu}_{\theta_{j'}}$, with $j' \neq j$ and to all $\tilde{\mathbf{r}}_{j'}$ with $j' > j$, and the operators $U_{\boldsymbol{\mu}_{\theta_j}, \tilde{\mathbf{r}}_j}$ keep the vectors that are orthogonal to both $\boldsymbol{\mu}_{\theta_j}$ and $\tilde{\mathbf{r}}_j$ unchanged. Moreover, since $\boldsymbol{\mu}_{\theta_j} = \mathbf{U}_p \mathbf{r}_j$ then

$$\mathbf{U}_p^T \boldsymbol{\mu}_{\theta_j} = \mathbf{U}_p^T \mathbf{U}_p \mathbf{r}_j = \mathbf{r}_j;$$

hence, $\mathbf{U}_p^T$ maps the vectors $\boldsymbol{\mu}_{\theta_j}$ to $\mathbf{r}_j$ for all $j = 1, \dots, p$, thereby providing the desired

transformation of the residuals in Equation (4.4.3.18) leading to the process

$$\begin{aligned} v_{k,N} &= \frac{1}{\sqrt{N}} \hat{\mathbf{e}}^T \mathbb{I}_{k,N} \\ &= \frac{1}{\sqrt{N}} \mathbf{e}^T \left(\mathbf{I}_N - \sum_{j=1}^p \mathbf{r}_j \mathbf{r}_j^T \right) \mathbb{I}_{k,N} + o_p(1). \end{aligned} \quad (4.4.3.23)$$

Such a process generalizes Equation (4.4.3.17) to the multidimensional setting. Under H_0 , the process $v_{k,N}$ in Equation (4.4.3.23) converges to a projected Brownian motion with covariance approximately equal to the right-hand side of Equation (4.4.3.12).

The limiting null distributions of the test statistics $\tilde{K}_N$ and $\tilde{C}_N$ in Equation (4.4.3.13) can be easily derived numerically. Specifically, let $\{\mathbf{e}^{(b)}\}_{b=1}^B$ be a set of B vectors of errors simulated from a standard multivariate normal. Realizations of the limiting process of $v_{k,N}$ can be obtained by calculating

$$v_{k,N}^{(b)} = \frac{1}{\sqrt{N}} \left[\mathbf{e}^{(b)} - \sum_{j=1}^p \mathbf{r}_j \mathbf{r}_j^T \mathbf{e}^{(b)} \right]^T \mathbb{I}_{k,N}, \quad (4.4.3.24)$$

for $b = 1, \dots, B$. Each $v_{k,N}^{(b)}$ is a realization of a mean-zero Gaussian process with covariance as in Equation (4.4.3.12). Thus, for sufficiently large N , it converges to the same projected Brownian motion as $v_{k,N}$. By replacing $v_{k,N}$ with $v_{k,N}^{(b)}$ in Equation (4.4.3.13), we can obtain B replicates of the test statistics, denoted as $\tilde{K}_N^{(b)}$, respectively. Therefore, letting $\tilde{K}_N^{\text{obs}}$ and $\tilde{C}_N^{\text{obs}}$ be their values obtained on the observed data, the corresponding p-values are:

$$\frac{1 + \sum_{b=1}^B \mathbb{1}\{\tilde{K}_N^{(b)} \geq \tilde{K}_N^{\text{obs}}\}}{1 + B} \quad \text{and} \quad \frac{1 + \sum_{b=1}^B \mathbb{1}\{\tilde{C}_N^{(b)} \geq \tilde{C}_N^{\text{obs}}\}}{1 + B} \quad (4.4.3.25)$$

with $\mathbb{1}$ denoting the indicator function.

4.3.3 Simulation studies

4.3.3.1 Distribution-free property

To demonstrate the validity of the distribution-free property of the method outlined in Section 4.3.2.2, let us revisit the example introduced in Section 4.3.2. Under the same setup, we simulate the null distributions of the statistics $\tilde{K}_N$ and $\tilde{C}_N$ in Equation (4.4.3.13) with $v_{k,N}$ constructed as in Equation (4.4.3.23) for each of the four combinations of models and error structures given in Equations (4.4.3.10)-(4.4.3.11). For the sake of comparison, we also simulate the null distribution of $\tilde{K}_N$ and $\tilde{C}_N$ using the Monte Carlo procedure described at the end of Section 4.3.2.2 and for which the errors are generated from a standard Gaussian. Figure 4.2 shows that regardless of the error structure or model considered, the graphs of the simulated null distributions overlap for both the Kolmogorov-Smirnov and the Cramér-Von Mises statistics. This confirms that the proposed tests are, for sufficiently large N , distribution-free, and the proposed Monte Carlo scheme accurately approximates the limiting null distributions of the test statistics based on $v_{k,N}$. The CPU time required by the latter amounts to 3.74 seconds.

4.3.3.2 Statistical properties

The simulation studies that follow aim to assess the statistical properties of tests based on the statistics $\tilde{K}_N$ and $\tilde{C}_N$ in Equation (4.4.3.13).

To ensure that the probability of a false rejection – known in statistics as the probability of Type I error – does not exceed the pre-determined significance level, we generate 10^4 datasets from the additive combination of the mean function $\mathbf{A}_{f_b,\ell}^{M1}(\boldsymbol{\theta})$ and the Gaussian error $\mathbf{u}_{f_b,s}^{(1)}$ in Equations (4.4.3.10)-(4.4.3.11). We test the validity of $\mathbf{A}_{f_b,\ell}^{M1}(\boldsymbol{\theta})$ by means of the statistics $\tilde{K}_N$ and $\tilde{C}_N$. Table 4.1 shows the simulated probabilities of rejecting $\mathbf{A}_{f_b,\ell}^{M1}(\boldsymbol{\theta})$ when choosing different significance levels. The results indicate that the simulated Type I error probabilities closely match the nominal levels.

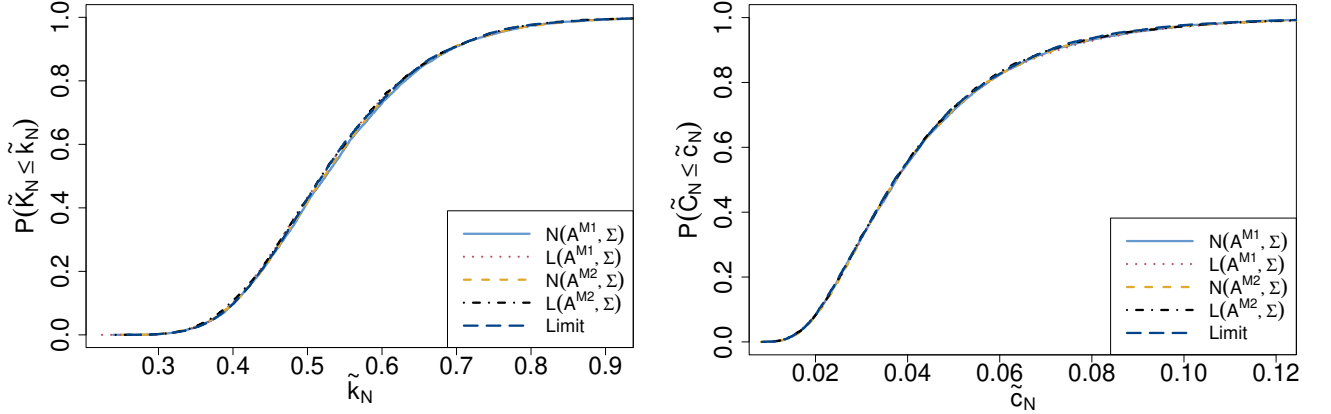


Figure 4.2: Graphs of the simulated null cumulative distribution functions of the Kolmogorov-Smirnov (top) and the Cramér-Von Mises (bottom) statistics in Equation (4.4.3.13) for the four combinations of models and error structures in Equations (4.4.3.10)-(4.4.3.11), alongside their limiting null distribution.

α	0.001	0.01	0.05	0.1
$\tilde{K}_N$	0.0009	0.008	0.046	0.093
$\tilde{C}_N$	0.0007	0.009	0.047	0.092

Table 4.1: Probability of Type I error for $\tilde{K}_N$ and $\tilde{C}_N$ at various significant levels

Next, we investigate the statistical power of the proposed statistics – that is, the probability of correctly rejecting the proposed model when misspecified. In our simulation, we test $\mathbf{A}_{f_b, \ell}^{M1}(\boldsymbol{\theta})$ while generating data using three different models for the mean. Specifically, we consider

$$\begin{aligned}
 G_1 : \langle \hat{\mathbf{A}}_f \rangle_\ell &= (4 - 3\ell + 0.4\ell^2 + 0.02\ell^3) \bar{f}_b^{(2/3)}; \\
 G_2 : \langle \hat{\mathbf{A}}_f \rangle_\ell &= (4 - 3\exp(0.23\ell)) \bar{f}_b^{(2/3)}; \\
 G_3 : \langle \hat{\mathbf{A}}_f \rangle_\ell &= (4 - 3\ell^{2.07}) \bar{f}_b^{(2/3)}.
 \end{aligned} \tag{4.4.3.1}$$

A Monte Carlo simulation involving 10^4 replicates is performed for each of the models in Equation (4.4.3.1) and combined with the Gaussian error structure in Equation (4.4.3.11). The simulated power of the Kolmogorov–Smirnov and Cramér–von Mises test statistics are reported in Tables 4.2 and 4.3, respectively, for different significance levels α . For both

α	0.001	0.01	0.05	0.1
G_1	0.639	0.957	1.000	1.000
G_2	0.407	0.817	0.981	0.998
G_3	0.776	0.989	1.000	1.000

Table 4.2: Statistical power of $\tilde{K}_N$ when testing $\mathbf{A}_{f_b, \ell}^{M1}(\boldsymbol{\theta})$ and generating data from G_1 , G_2 , G_3 at various significant levels

α	0.001	0.01	0.05	0.1
G_1	0.232	0.588	0.895	0.971
G_2	0.105	0.364	0.730	0.881
G_3	0.397	0.753	0.957	0.991

Table 4.3: Statistical power of $\tilde{C}_N$ when testing $\mathbf{A}_{f_b, \ell}^{M1}(\boldsymbol{\theta})$ and generating data from G_1 , G_2 , G_3 at various significant levels

test statistics, the simulated statistical power is close to one when the significance level is 0.05 and is reasonably high even at smaller significance levels.

4.4 Testing spectral models for the SGWB

We now apply the methodology described in Sec. 4.3 to the case of a realistic astrophysical SGWB model and the data obtained by the Advanced LIGO and Advanced Virgo GW detectors in their third observing run (O3), as described in Sec. 4.2.

Let us turn our attention to the theoretical model we wish to test. The energy density of the stochastic gravitational-wave background is defined as

$$\Omega_{\text{GW}}(\hat{\Omega}, f) = \frac{f}{\rho_c} \frac{d^3 \rho_{\text{GW}}}{d^2 \hat{\Omega} df}(\hat{\Omega}, f) = \frac{\bar{\Omega}_{\text{GW}}(f)}{4\pi} + \delta\Omega_{\text{GW}}(\hat{\Omega}, f), \quad (4.4.4.1)$$

where ρ_{GW} is the GW energy density, f is a continuous variable for GW frequency, and ρ_c is the critical density of the universe today. We use f to denote frequency as a continuous variable in contrast to f_b , which represents a discrete frequency band with width 20Hz and

midpoint $\bar{f}_b$. The energy density is split into an isotropic component $\bar{\Omega}_{\text{GW}}$ and anisotropic perturbations $\delta\Omega_{\text{GW}}$ Cusin [2018a, 2019]:

$$\begin{aligned}\bar{\Omega}_{\text{GW}}(f) &= \int dr \partial_r \bar{\Omega}_{\text{GW}}(f, r), \\ \partial_r \bar{\Omega}_{\text{GW}}(f, r) &= \frac{f}{\rho_c} \mathcal{B}(f, r; \boldsymbol{\theta}),\end{aligned}\tag{4.4.4.2}$$

where r denotes the conformal distance and the function $\mathcal{B}(f, r; \boldsymbol{\theta})$ is an astrophysical kernel with free parameters $\boldsymbol{\theta}$ that contains information on the local production of GWs at galaxy scales (we set the speed of light $c = 1$). Following Cusin [2017], the angular power spectrum is then given by

$$A_\ell(f; \boldsymbol{\theta}) = \frac{2}{\pi} \int dk k^2 |\delta\Omega_{\text{GW}, \ell}(k, f; \boldsymbol{\theta})|^2, \text{ and} \tag{4.4.4.3}$$

$$\delta\Omega_{\text{GW}, \ell}(k, f; \boldsymbol{\theta}) = \frac{f}{4\pi\rho_c} \int dr \mathcal{B}(f, r; \boldsymbol{\theta}) [b(r) \delta_{m,k}(r) j_\ell(kr)], \tag{4.4.4.4}$$

where k is the wavenumber, j_ℓ are spherical Bessel functions, and $\delta_{m,k}$ is the dark-matter over-density, related to galaxy overdensity via the bias factor that we assume to be scale-independent and with redshift evolution given by $b(z) = b_0 \sqrt{1+z}$, where $b_0 = 1.5$ if we replace the conformal distance r with redshift z Yang [2023]. Note that Equation (4.4.4.3) is the model to be tested. The astrophysical kernel can be computed from the galaxy distribution and galaxy GW luminosity; for our purposes, we also replace the conformal distance r with redshift z and adopt the empirical parameterization Cusin [2018a, 2019], Yang [2023]

$$\mathcal{B}(f, z; \boldsymbol{\theta}) = A_{\text{max}} f^{-1/3} e^{-(z-z_c)^2/2\sigma_z^2}, \tag{4.4.4.5}$$

with the free parameters $\boldsymbol{\theta} = (A_{\text{max}}, z_c, \sigma_z)$.

A direct detection of the SGWB has not yet been achieved, so the data from O3 are

consistent with noise [Abbott \[2021a\]](#). Therefore, we are unable to use O3 data with the method developed in [Sec. 4.3](#) to validate the theoretical model for the angular power spectrum. Instead, we add a simulated signal (generated from the theoretical model itself) to the noise-only O3 data, and then apply our distribution-free methodology to these simulated data. The addition of signal to noise must be done in terms of maps of GW power rather than the angular power spectrum, since the latter physically represents *squared* power rather than simply power. We may then represent the sum of noise and the injected signal as

$$\hat{a}_{f_b, s, \ell m}^{inj} = \hat{a}_{f_b, s, \ell m} + a_{f_b, \ell m}^M \quad (4.4.4.6)$$

where the first term on the right-hand side of Equation (4.4.4.6) is given by Equation (4.4.2.3) while the index f_b and s correspond to the discrete frequency bands and the segments used to divide the Advanced LIGO-Virgo data, both of which are as described in [Sec. 4.2](#). The number of data segments $S = 15$ was chosen to maximize the number of samples to provide our statistical framework while also ensuring that each segment includes data from each part of the sidereal day. The second term on the right-hand side of Equation (4.4.4.6) is derived from the theoretical model and denoted with the superscript M . Note that the theoretical model does not directly provide values of $a_{f_b, \ell m}^M$; nevertheless, we can draw a set of $a_{f_b, \ell m}^M$ using the given model $A_\ell(f; \boldsymbol{\theta})$. More specifically, a given map element $a_{f_b, \ell m}^M$ is complex-valued. We assume its real and imaginary parts are independent random variables, each drawn from the multivariate normal distribution $\mathcal{N}(0, \frac{1}{2}A_\ell(f; \boldsymbol{\theta}))$. In the case when $m = 0$, each $a_{f_b, \ell 0}^M$ is purely real and drawn instead from $\mathcal{N}(0, A_\ell(f; \boldsymbol{\theta}))$. The frequency values at which we evaluate the theoretical model $A_\ell(f; \boldsymbol{\theta})$ are taken to be the midpoints, $\bar{f}_b$, of the frequency bands considered.

We note that we sample the map only once and must treat that sample as though it is the only one provided by our realization of the universe. To make this sample map consistent with the angular power spectrum from which it was drawn, we introduce a

scaling factor $\tau_{f_b, \ell}$:

$$\tau_{f_b, \ell} = \sqrt{\frac{A_\ell(\bar{f}_b; \boldsymbol{\theta})}{\frac{1}{2\ell+1} \sum_{m=-\ell}^{\ell} |a_{f_b, \ell m}^M|^2}}. \quad (4.4.4.7)$$

We apply the scaling factor to the drawn map elements such that they satisfy the condition below:

$$A_\ell(\bar{f}_b; \boldsymbol{\theta}) = \frac{1}{1+2\ell} \sum_{m=-\ell}^{\ell} |\tau_{f_b, \ell} a_{f_b, \ell m}^M|^2. \quad (4.4.4.8)$$

This coincides with the hypothesis testing problem (4.4.3.1) in Sec. 4.3.

We follow Yang [2023] and apply the scaling factor $\mathcal{K}(f)$ to properly scale the LIGO results $\hat{a}_{f_b, \ell m}$ relative to the values provided by the theoretical model $a_{f_b, \ell m}^M$:

$$\mathcal{K}(f) = \frac{2\pi^2}{3H_0^2} f_{ref}^3 \left(\frac{f}{f_{ref}} \right)^\alpha \quad (4.4.4.9)$$

Here, f_{ref} is the reference frequency of 25Hz used in O3 Abbott [2021a], and f again takes on values of the midpoints of each frequency band $\bar{f}_b$. We also apply the square of this scaling factor to the inverted Fisher matrix in order to obtain a properly scaled covariance matrix for the A_ℓ (Equation (4.4.2.7)). For notational simplicity, for the remainder of this section, when we write $a_{f_b, \ell m}^M$, we assume it to be the version scaled by $\tau_{f_b, \ell}$, and when we write $\hat{a}_{f_b, s, \ell m}$ or $(\Gamma_{f_b, s}^{-1})_{\ell m, \ell' m'}$, we assume them to be the versions scaled by $\mathcal{K}(f)$ and $\mathcal{K}(f)^2$ respectively.

With the model map elements $a_{f_b, \ell m}^M$ and data map elements $\hat{a}_{f_b, s, \ell m}$ properly scaled, we may find their sum $\hat{a}_{f_b, s, \ell m}^{inj}$ to complete the signal injection. The corresponding angular power spectrum $\hat{A}_{f_b, s, \ell}^{inj}$ is calculated according to Equation (4.4.2.6),

$$\hat{A}_{f_b, s, \ell}^{inj} = \frac{1}{1+2\ell} \sum_{m=-\ell}^{\ell} \left[|\hat{a}_{f_b, s, \ell m}^{inj}|^2 - (\Gamma_{f_b, s}^{-1})_{\ell m, \ell m} \right], \quad (4.4.4.10)$$

and the covariance matrix is computed according to Equation (4.4.2.7), with $a_{f_b, \ell m}$ replaced by $\hat{a}_{f_b, \ell m}^{inj}$, which is defined as the average of $\hat{a}_{f_b, s, \ell m}^{inj}$ over the data segments s .

The necessary inputs for the procedure described in Sec. 4.3 are the data $\hat{A}_{f_b, s, \ell}^{inj}$, their corresponding covariance matrices, and model values $A_\ell(\bar{f}_b; \boldsymbol{\theta})$. In this study, we test our procedure on two variations of these quantities.

We compare the model described by Equations (4.4.4.3)-(4.4.4.5) to data injected using the same model, applying the covariance matrix as described above. When generating the injected signal, the parameter values are 1.2×10^{-28} , 0.6, and 0.7 for A_{max} , z_c , and σ_z respectively. The parameter estimates obtained via Eq (4.4.3.3) are 1.19×10^{-28} , 0.590, and 0.694. Next, we use the same covariance matrix and again use a postulated model as given in Equation (4.4.4.5), but in this case we use data which has been injected with a different variation of the model. In particular, we replace the astrophysical kernel with

$$\mathcal{B}(f, z; \boldsymbol{\theta}) = f^{-1/3}(\beta_0 + \beta_1 z), \quad (4.4.4.11)$$

where $\boldsymbol{\theta} = (\beta_0, \beta_1)$. For this alternative model, we set $\beta_0 = 1.2 \times 10^{-28}$ and $\beta_1 = 6.0 \times 10^{-29}$ when generating the injected data. The parameter estimates for the postulated model are then 2.84×10^{-28} , 3.58, and 2.72 for A_{max} , z_c , and σ_z , respectively.

To perform the test in Equation (4.4.3.1), we employ the test statistics $\tilde{K}_N$ and $\tilde{C}_N$ in Equation (4.4.3.13). We simulate their limiting distributions by running 10^5 Monte Carlo simulations of the limiting process in Equation (4.4.3.24). The final results align with our expectations. Specifically, under the first scenario, $\tilde{K}_N$ and $\tilde{C}_N$ lead to p-values of 0.111 and 0.264, respectively, implying that the model is correctly identified as consistent with the data. Under the second scenario, however, the test statistics yield p-values 0.002 and 0.008, respectively, and correctly identify the model as not being consistent with the data.

4.5 Summary and discussion

This paper extends the distribution-free statistical framework presented in a companion paper [Algeri \[2025\]](#) to account for the unique challenges arising when estimating and testing models for the angular power spectrum of the SGWB.

Framing the problem in the context of regression avoids the need to specify a likelihood function for the angular power spectrum. The procedure only requires a model for its mean and covariance matrix. To ensure consistency in our analysis, a new closed-form expression for the latter is derived to account for the variability associated with the maximum likelihood estimation of the spherical harmonic coefficients of the SGWB sky map. Simulation studies and an application to data from the third observing run of Advanced LIGO and Advanced Virgo demonstrated that the proposed testing procedure can detect departures from the hypothesized angular power spectral models when misspecified while adequately controlling for the probability of false discoveries. We note that the current sensitivity of the Advanced LIGO and Advanced Virgo detectors has not allowed for direct detection of the SGWB. To circumvent this limitation, we have used the O3 dataset as a noise realization, and a simulated signal has been injected into it. Given the good performance in such a setting, we expect the proposed framework to be valuable when applied to data collected by more sensitive detectors.

Finally, we note that inconsistencies in the statistical formalism can be found in the literature concerning the statistical analysis of the SGWB. In particular, when the spherical harmonic coefficients of the SGWB sky map are estimated via maximum likelihood [cf. [Thrane, 2009](#)] the user implicitly assumes such quantities – as well as the corresponding sky maps – are deterministic. Therefore, they should not be simultaneously treated as zero-mean complex Gaussian random variables. A statistical analysis that enables the estimation and testing of angular power spectral models while also allowing the spherical harmonic coefficients to be random can be established. It requires, however, a revision of the modeling framework at a fundamental level.

Chapter 5

Conclusion and future work

This dissertation develops a unified inferential framework for astrophysical data analysis that integrates background calibration, model validation, and signal discovery. While each chapter addresses distinct astrophysical problems—from X-ray spectroscopy to gravitational-wave backgrounds—they share a common methodological foundation: the construction of statistically rigorous, computationally efficient, and practically implementable tools that accommodate the complexities inherent in modern astrophysical research.

The proposed framework offers exceptional flexibility and broad applicability across diverse astrophysical problems. Chapter 2 introduces smooth tests that can be employed for both discrete and continuous i.i.d. data, with a direct application to detecting spectral lines in the presence of high background. Chapter 3 then extends smooth tests via the K2 transform, enabling computationally efficient and statistically accurate inference for multiple complex astrophysical models. Building upon the K2 transform’s broad applicability to goodness-of-fit problems, Chapter 4 further develops robust and efficient tests for SGWB angular power spectrum models without relying on specific distributional assumptions.

The effectiveness of the framework is validated comprehensively through both simulation studies and real-data applications. Simulation results in Chapters 3 and 4 confirm that the proposed tests based on the K2 transform can achieve high statistical power while

substantially reducing computational cost and maintaining nominal Type I error rates, even with moderate sample sizes and complex models. When applied to real data, the RT Cru analysis demonstrates how data-driven background corrections combined with appropriate inferential adjustments enable meaningful scientific conclusions even when faced with high backgrounds and limited signals—detecting known features while properly constraining upper limits on undetected lines and serendipitously identifying new phenomena. The application to Advanced LIGO-Virgo O3 data further demonstrates the framework’s readiness for next-generation observations, where increased sensitivity will demand more sophisticated statistical tools.

Several natural extensions for future research emerge from this work. The framework proposed in Chapter 2 can be readily applied to both discrete and continuous data, but many practical applications involve binned data obtained from instruments such as cameras, telescopes, and particle detectors. A natural extension would be to adapt the framework to facilitate signal searches under high background with binned data. Chapter 3 focuses on smooth tests in the classical univariate setting but can be naturally extended to multivariate parametric models. In particular, when the hypothesized distribution is D -dimensional, the reference distribution could be chosen as a product of D univariate distributions. The K2 orthonormal basis can then be constructed by applying the K2 transform to a tensor product of bases (e.g., [Algeri \[2021\]](#)). Extensions to smooth tests for regression [[Inglot and Ledwina, 2006b](#), [Rayner et al., 2022](#)] are also possible using the weighted empirical process proposed in [Khmaladze \[2017\]](#) instead of the classical empirical process for IID data. Chapter 4 identifies a fundamental inconsistency in SGWB statistical analyses: spherical harmonic coefficients estimated via maximum likelihood [[Thrane, 2009](#)] are treated as deterministic, yet they are simultaneously modeled as zero-mean complex Gaussian random variables. While a statistical analysis that enables both estimation and testing while treating coefficients as random can be established, this requires a fundamental revision of the existing framework. Developing such a coherent approach is a natural

direction for future work.

Appendix A

Supplementary Material for Chapter 2

A.1 Proof of the convergence of K-statistic

Let k_m be the observed value of the K-statistic K_m on the data and $W_j \stackrel{i.i.d}{\sim} \chi_1^2$ for $j = 1, \dots, m$, we have

$$\begin{aligned} P(K_m > k_m \mid H_0) &= 1 - P(K_m \leq k_m \mid H_0) \\ &= 1 - P\left(\max_{j=1,\dots,m} n\hat{\theta}_j^2 \leq k_m \mid H_0\right) \\ &\stackrel{d}{\rightarrow} 1 - P\left(\max_{j=1,\dots,m} W_j \leq k_m\right) \\ &= 1 - P\left(\chi_1^2 \leq k_m\right)^m, \end{aligned}$$

corresponding to Equation (2.2.3.11), where the notation $\stackrel{d}{\rightarrow}$ denotes convergence in distribution, and the convergence is for $n \rightarrow \infty$.

A.2 Proof of theorem 1

Consider the event E_m defined as

$$E_m = \left\{ m \text{ maximises the BIC criterion in (2.2.3.1)} \right\}$$

Then, an upper bound for the limit of $P(K_{(m)} \geq k_{(m)} \mid H_0)$ is

$$\begin{aligned} P(K_{(m)} \geq k_{(m)} \mid H_0) &= P(K_m > k_m \cap E_m \mid H_0) \\ &= P\left(\max_{j=1,\dots,m} n\hat{\theta}_{(j)}^2 > k_m \cap E_m \mid H_0\right) \\ &= P\left(\max_{j=1,\dots,m} n\hat{\theta}_{(j)}^2 > k_m \mid E_m \cap H_0\right)P(E_m \mid H_0) \\ &= P\left(\max_{j=1,\dots,M} n\hat{\theta}_{(j)}^2 > k_m \mid E_m \cap H_0\right)P(E_m \mid H_0) \\ &\leq \sum_{m=1}^M P\left(\max_{j=1,\dots,M} n\hat{\theta}_{(j)}^2 > k_m \mid E_m \cap H_0\right)P(E_m \mid H_0) \\ &= P\left(\max_{j=1,\dots,M} n\hat{\theta}_{(j)}^2 > k_m \mid H_0\right) \xrightarrow{d} 1 - P\left(\chi_1^2 \leq k_m\right)^M \end{aligned}$$

where the convergence is for $n \rightarrow \infty$, and follows from (2.2.3.11). This proves Theorem 1 (Equation (2.2.3.3)).

A.3 Corrected background density

The corrected background density for region W_4 (Equation (2.2.4.2)) is calculated as:

$$\begin{aligned} \hat{b}_{W_4}(x) &= \frac{1}{\int_{l_4}^{u_4} \hat{b}_{C_4}(x)} \hat{b}_{C_4}(x) \\ &= \frac{1}{\int_{14.6}^{15.15} (1.3994 - 0.0250x)} (1.3994 - 0.0250x) \\ &= 2.4749 - 0.0441x \end{aligned}$$

Regions of interest (W_r)	Bonferroni	K	Naive	LRT
W_3	0.0511 (± 0.0022)	0.0509 (± 0.0022)	2e-4 ($\pm 1e-4$)	0.0434 (± 0.0020)
W_4	0.0559 (± 0.0023)	0.0500 (± 0.0022)	3e-4 ($\pm 2e-4$)	0.0445 (± 0.0021)
W_5	0.0472 (± 0.0021)	0.0455 (± 0.0021)	3e-4 ($\pm 2e-4$)	0.0422 (± 0.0020)
W_6	0.0511 (± 0.0022)	0.0487 (± 0.0022)	3e-4 ($\pm 2e-4$)	0.0446 (± 0.0021)
W_7	0.0602 (± 0.0024)	0.0497 (± 0.0022)	1e-4 ($\pm 1e-4$)	0.0449 (± 0.0021)
W_8	0.0612 (± 0.0024)	0.0474 (± 0.0021)	3e-4 ($\pm 2e-4$)	0.0471 (± 0.0021)
W_9	0.0619 (± 0.0024)	0.0475 (± 0.0021)	3e-4 ($\pm 2e-4$)	0.0401 (± 0.0020)

Table A.1: Simulated type I error and respective Monte Carlo errors for the smooth tests computed on regions W_3 - W_9 and setting the local significance level to 0.05.

where $\hat{b}_{C_4}$ is the corrected background density for the combined region C_4 in (2.2.4.1); l_4 , u_4 corresponds to the lower and upper bounds of the wavelength range of region W_4 and $x \in [14.6, 15.15]$ Å. The corrected background density for region W_5 can be derived in a similar manner.

A.4 Additional tables

Tables A.1 and A.2 report the probability of Type I error for our smooth tests and adequate post-selection and multiple hypothesis testing adjustments. Specifically, Table A.1 reports the type I errors inclusive only of the post-selection adjustments. Specifically, this corresponds to the probabilities of false discoveries we would expect our testing procedures to have if we were to conduct an analysis on each of the regions considered individually, that is, without simultaneously testing for spectral lines in the remaining six regions. Conversely, in Table A.2, we report the probability of a false discovery across the entire spectrum, that is, when accounting for the fact that seven different tests are being conducted simultaneously.

Tables A.3 and A.4 report the upper limits obtained using smooth tests for the spectral lines of interest. Similarly to the upper limits obtained by means of the LRT, Table A.3 and Table A.4 provide the local and Sidak adjusted upper limits respectively described in detail in Section 2.4.3.

Regions of interest (W_r)	Bonferroni	K	Naive	LRT
W_3	0.0118 (± 0.0023)	0.0082 (± 0.0009)	0 (± 0)	0.0070 (± 0.0008)
W_4	0.0130 (± 0.0031)	0.0072 (± 0.0008)	0 (± 0)	0.0057 (± 0.0008)
W_5	0.0102 (± 0.0030)	0.0063 (± 0.0008)	0 (± 0)	0.0063 (± 0.0008)
W_6	0.0114 (± 0.0024)	0.0065 (± 0.0008)	0 (± 0)	0.0058 (± 0.0008)
W_7	0.0133 (± 0.0033)	0.0062 (± 0.0008)	0 (± 0)	0.0060 (± 0.0008)
W_8	0.0139 (± 0.0034)	0.0058 (± 0.0008)	0 (± 0)	0.0070 (± 0.0008)
W_9	0.0142 (± 0.0029)	0.0078 (± 0.0009)	0 (± 0)	0.0056 (± 0.0007)

Table A.2: Simulated type I error and respective Monte Carlo errors for the smooth tests computed on regions W_3 - W_9 and adjusted for multiple hypothesis testing using Sidak's correction with global significance $\alpha = 0.05$. In this case, the local significance is $\alpha_r = 1 - (1 - \alpha)^{\frac{1}{R}} = 0.0073$, for $r = 1, \dots, R$.

Local Regions (W_r)	50% Upper limits via smooth tests			90% Upper limits via smooth tests		
	Bonferroni	K	Naive	Bonferroni	K	Naive
W_3	45.26	45.99	61.32	70.08	71.54	85.41
W_4	28.90	31.37	38.53	44.71	48.17	53.60
W_5	35.32	36.74	47.57	55.11	56.99	65.47
W_6	32.76	34.32	44.07	51.09	53.82	60.84
W_7	26.85	30.07	36.70	42.06	45.82	50.12
W_8	26.82	29.87	37.56	41.76	45.53	50.75
W_9	58.50	66.30	79.56	91.65	100.6	110.0

Table A.3: 50% and 90% upper limits obtained using smooth tests with different post-selection adjustments. The 50% upper limits are calculated by the proportion of the expected lines η to achieve the power 0.5 (solid line in Figure 2.4) times the sample sizes n_r . Similarly, the 90% upper limits are calculated by the proportion of the expected lines η to achieve the power 0.9 times the sample sizes n .

Sidak adjusted Regions (W_r)	50% upper limits via smooth tests			90% upper limits via smooth tests		
	Bonferroni	K	Naive	Bonferroni	K	Naive
W_3	53.29	63.88	67.16	77.38	89.06	89.06
W_4	34.09	42.48	42.73	49.40	59.77	57.55
W_5	41.45	50.40	52.28	58.88	69.24	70.18
W_6	39.00	47.19	48.75	56.16	66.69	65.52
W_7	32.76	41.35	41.17	46.54	56.74	53.70
W_8	33.20	39.88	42.20	46.55	54.38	55.68
W_9	71.37	91.26	89.31	101.4	125.2	118.2

Table A.4: Sidak's corrected 50% and 90% upper limits obtained using smooth tests with different post-selection adjustments.

Appendix B

Supplementary material for Chapter 3

B.1 Proof of proposition 4

Proof. To show that K is a unitary operator, we need to demonstrate that it is surjective and preserves the inner product. It is surjective because for any function in $L^2(G_\beta)$, if $\phi \perp \text{span}(\mathbf{1}, l_{\gamma,\beta})$,

$$K\phi = \phi - \frac{l_{\gamma,\beta} - \mathbf{1}}{\mathbf{1} - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta}} \langle l_{\gamma,\beta} - \mathbf{1}, \phi \rangle_{G_\beta} = \phi.$$

Otherwise if $\phi = c_1 \mathbf{1} + c_2 l_{\gamma,\beta}$, we have

$$\begin{aligned} K(c_1 l_{\gamma,\beta} + c_2 \mathbf{1}) &= c_1 l_{\gamma,\beta} + c_2 \mathbf{1} - \frac{l_{\gamma,\beta} - \mathbf{1}}{\mathbf{1} - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{F_\gamma}} \langle l_{\gamma,\beta} - \mathbf{1}, c_1 l_{\gamma,\beta} + c_2 \mathbf{1} \rangle_{F_\gamma} \\ &= c_1 l_{\gamma,\beta} + c_2 \mathbf{1} - c_1 (l_{\gamma,\beta} - \mathbf{1}) + c_2 (l_{\gamma,\beta} - \mathbf{1}) \\ &= c_1 \mathbf{1} + c_2 l_{\gamma,\beta}. \end{aligned} \tag{B.B.1.1}$$

Notice here, by letting $c_1 = 1, c_2 = 0$ or $c_1 = 0, c_2 = 1$ in equation (B.B.1.1), we obtain

$$Kl_{\gamma,\beta} = \mathbf{1}, \quad K\mathbf{1} = l_{\gamma,\beta}.$$

The operator K preserves the inner product because for any functions $\phi_1, \phi_2 \in L^2(G_\beta)$

$$\begin{aligned} & \langle K\phi_1, K\phi_2 \rangle_{G_\beta} \\ &= \langle \phi_1, \phi_2 \rangle_{G_\beta} - \frac{2 \langle \mathbf{1} - l_{\gamma,\beta}, \phi_1 \rangle_{G_\beta} \langle \mathbf{1} - l_{\gamma,\beta}, \phi_2 \rangle_{G_\beta}}{1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta}} \\ & \quad + \frac{\langle \mathbf{1} - l_{\gamma,\beta}, \phi_1 \rangle_{G_\beta} \langle \mathbf{1} - l_{\gamma,\beta}, \phi_2 \rangle_{G_\beta} \langle \mathbf{1} - l_{\gamma,\beta}, \mathbf{1} - l_{\gamma,\beta} \rangle_{G_\beta}}{(1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta})^2} \\ &= \langle \phi_1, \phi_2 \rangle_{G_\beta} + \frac{\langle \mathbf{1} - l_{\gamma,\beta}, \phi_1 \rangle_{G_\beta} \langle \mathbf{1} - l_{\gamma,\beta}, \phi_2 \rangle_{G_\beta}}{(1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta})^2} \left(-2 + 2 \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta} + \langle \mathbf{1} - l_{\gamma,\beta}, \mathbf{1} - l_{\gamma,\beta} \rangle_{G_\beta} \right) \\ &= \langle \phi_1, \phi_2 \rangle_{G_\beta}. \end{aligned}$$

K is self-adjoint because

$$\begin{aligned} \langle K\phi_1, \phi_2 \rangle_{G_\beta} &= \left\langle \phi_1 - \frac{l_{\gamma,\beta} - \mathbf{1}}{1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta}} \langle l_{\gamma,\beta} - \mathbf{1}, \phi_1 \rangle_{G_\beta}, \phi_2 \right\rangle_{G_\beta} \\ &= \langle \phi_1, \phi_2 \rangle_{G_\beta} - \frac{\langle l_{\gamma,\beta} - \mathbf{1}, \phi_1 \rangle_{G_\beta} \langle l_{\gamma,\beta} - \mathbf{1}, \phi_2 \rangle_{G_\beta}}{1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta}} \\ &= \left\langle \phi_1, \phi_2 - \frac{l_{\gamma,\beta} - \mathbf{1}}{1 - \langle l_{\gamma,\beta}, \mathbf{1} \rangle_{G_\beta}} \langle l_{\gamma,\beta} - \mathbf{1}, \phi_2 \rangle_{G_\beta} \right\rangle_{G_\beta} \\ &= \langle \phi_1, K\phi_2 \rangle_{G_\beta}. \end{aligned}$$

The unitary and self-adjoint properties of K imply $K^2 = I$. ■

B.2 Required orthogonality conditions of $\tilde{c}_{\lambda_k}$

Proof. It can be verified that

$$\langle \tilde{c}_{\lambda_2}, \mathbf{1} \rangle_{G_\beta} = \left\langle U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_2}, \mathbf{1} \right\rangle_{G_\beta} = \left\langle c_{\lambda_2}, U_{b_{\beta_1} c_{\lambda_1}} \mathbf{1} \right\rangle_{G_\beta} = \langle c_{\lambda_2}, \mathbf{1} \rangle_{G_\beta} = 0,$$

$$\langle \tilde{c}_{\lambda_2}, b_{\beta_1} \rangle_{G_\beta} = \left\langle U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_2}, b_{\beta_1} \right\rangle_{G_\beta} = \left\langle c_{\lambda_2}, U_{b_{\beta_1} c_{\lambda_1}} b_{\beta_1} \right\rangle_{G_\beta} = \langle c_{\lambda_2}, c_{\lambda_1} \rangle_{G_\beta} = 0;$$

and

$$\langle \tilde{c}_{\lambda_3}, \mathbf{1} \rangle_{G_\beta} = \left\langle U_{b_{\beta_2} \tilde{c}_{\lambda_2}} U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_3}, \mathbf{1} \right\rangle_{G_\beta} = \left\langle c_{\lambda_3}, U_{b_{\beta_1} c_{\lambda_1}} U_{b_{\beta_2} \tilde{c}_{\lambda_2}} \mathbf{1} \right\rangle_{G_\beta} = \langle c_{\lambda_3}, \mathbf{1} \rangle_{G_\beta} = 0,$$

$$\langle \tilde{c}_{\lambda_3}, b_{\beta_1} \rangle_{G_\beta} = \left\langle U_{b_{\beta_2} \tilde{c}_{\lambda_2}} U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_3}, b_{\beta_1} \right\rangle_{G_\beta} = \left\langle c_{\lambda_3}, U_{b_{\beta_1} c_{\lambda_1}} U_{b_{\beta_2} \tilde{c}_{\lambda_2}} b_{\beta_1} \right\rangle_{G_\beta} = \langle c_{\lambda_3}, c_{\lambda_1} \rangle_{G_\beta} = 0,$$

$$\langle \tilde{c}_{\lambda_3}, b_{\beta_2} \rangle_{G_\beta} = \left\langle U_{b_{\beta_2} \tilde{c}_{\lambda_2}} U_{b_{\beta_1} c_{\lambda_1}} c_{\lambda_3}, b_{\beta_2} \right\rangle_{G_\beta} = \left\langle c_{\lambda_3}, U_{b_{\beta_1} c_{\lambda_1}} U_{b_{\beta_2} \tilde{c}_{\lambda_2}} b_{\beta_2} \right\rangle_{G_\beta} = \left\langle c_{\lambda_3}, U_{b_{\beta_1} c_{\lambda_1}}^2 c_{\lambda_1} \right\rangle_{G_\beta} = 0;$$

and this can proceed up to $\tilde{c}_{\lambda_p}$ by induction. ■

Appendix C

Supplementary Material for Chapter 4

C.1 The generalized χ^2 distribution of the angular power spectrum

Define the index set $\mathcal{I}_\ell = \{(\ell, m) \in \mathbb{N}^+ \times \mathbb{Z}; -\ell \leq m \leq \ell\}$ and define

$$\mathcal{I} = \bigcup_{\ell=1}^{\ell_{max}} \mathcal{I}_\ell.$$

From Equations (4.4.2.3)-(4.4.2.5) the vector of elements $\hat{a}_{f_b, \ell m}$ is a linear combination of the complex Gaussian distributed random variable $C(t; f)$ with density defined as in Equation (4.4.2.2). To ease notation, we write $\hat{a}_{\ell m}$ to denote $\hat{a}_{f_b, \ell m}$, ignoring the influence of f_b . Therefore, $\{\hat{a}_{\ell m}\}_{\mathcal{I}}$ is distributed as a multivariate complex Gaussian random variable with mean $\{a_{\ell m}\}_{\mathcal{I}}$ and covariance matrix $\mathbf{\Gamma}^{-1}$. That is,

$$\{\hat{a}_{\ell m}\}_{\mathcal{I}} \sim \mathcal{CN}(\{a_{\ell m}\}_{\mathcal{I}}, \mathbf{\Gamma}^{-1}),$$

where $\mathbf{\Gamma}^{-1} = \{\Gamma_{\ell m, \ell' m'}^{-1}\}_{(\ell, m), (\ell', m') \in \mathcal{I}}$. Define

$$\tilde{\chi}_\ell = \sum_{m=-\ell}^{\ell} |\hat{a}_{\ell m}|^2.$$

The goal is to show that $\tilde{\chi}_\ell$ has a generalized χ^2 distribution.

Let $\mathbf{X}$ be the random vector whose components are the real and imaginary parts of $\{\hat{a}_{\ell m}\}_{\mathcal{I}}$. Because the density of $\{\hat{a}_{\ell m}\}_{\mathcal{I}}$ is proportional to

$$\begin{aligned} & \exp\{-(\hat{a}_{\ell' m'} - a_{\ell' m'})^* \Gamma_{\ell' m', \ell m} (\hat{a}_{\ell m} - a_{\ell m})\} \\ &= \exp\{-\frac{1}{2}(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Omega}^{-1}(\mathbf{X} - \boldsymbol{\mu})\}, \end{aligned}$$

where

$$\mathbf{X} = \begin{pmatrix} \text{Re}(\{\hat{a}_{\ell m}\}_{\mathcal{I}}) \\ \text{Im}(\{\hat{a}_{\ell m}\}_{\mathcal{I}}) \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \text{Re}(\{a_{\ell m}\}_{\mathcal{I}}) \\ \text{Im}(\{a_{\ell m}\}_{\mathcal{I}}) \end{pmatrix}$$

and

$$\boldsymbol{\Omega}^{-1} = 2 \begin{pmatrix} \text{Re}(\mathbf{\Gamma}) & -\text{Im}(\mathbf{\Gamma}) \\ \text{Im}(\mathbf{\Gamma}) & \text{Re}(\mathbf{\Gamma}) \end{pmatrix},$$

it follows that $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Omega})$, i.e., a real multivariate normal with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Omega}$. Because $\mathbf{\Gamma} = \text{Re}(\mathbf{\Gamma}) + i \text{Im}(\mathbf{\Gamma})$ and $\mathbf{\Gamma}^{-1} = \text{Re}(\mathbf{\Gamma}^{-1}) + i \text{Im}(\mathbf{\Gamma}^{-1})$ are inverse of each other, we have

$$\boldsymbol{\Omega} = \frac{1}{2} \begin{pmatrix} \text{Re}(\mathbf{\Gamma}^{-1}) & -\text{Im}(\mathbf{\Gamma}^{-1}) \\ \text{Im}(\mathbf{\Gamma}^{-1}) & \text{Re}(\mathbf{\Gamma}^{-1}) \end{pmatrix}.$$

Subvectors of $\mathbf{X}$ also follow a (real) multivariate Gaussian distribution. In particular,

$$\mathbf{X}_\ell = \begin{pmatrix} \text{Re}(\{\hat{a}_{\ell m}\}_{\mathcal{I}_\ell}) \\ \text{Im}(\{\hat{a}_{\ell m}\}_{\mathcal{I}_\ell}) \end{pmatrix} \sim \mathcal{N}(\boldsymbol{\mu}_\ell, \boldsymbol{\Omega}_\ell),$$

with mean

$$\boldsymbol{\mu}_\ell = \begin{pmatrix} \text{Re}(\{a_{\ell m}\}_{\mathcal{I}_\ell}) \\ \text{Im}(\{a_{\ell m}\}_{\mathcal{I}_\ell}) \end{pmatrix},$$

and covariance matrix

$$\boldsymbol{\Omega}_\ell = \frac{1}{2} \begin{pmatrix} \text{Re}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} & -\text{Im}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} \\ \text{Im}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} & \text{Re}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} \end{pmatrix}.$$

Note that $\tilde{\chi}_\ell = \|\mathbf{X}_\ell\|^2$. Since $\boldsymbol{\Gamma}$ is a positive definite matrix, $(\boldsymbol{\Gamma}^{-1})_{\ell,\ell}$ is also positive definite. Therefore, if, for any non-zero vector $(\mathbf{x}^T, \mathbf{y}^T)^T \in \mathbb{R}^{2l+1} \times \mathbb{R}^{2l+1}$, $\mathbf{z} = \mathbf{x} + i\mathbf{y}$ and let $\mathbf{z}^\dagger$ be its conjugate transpose, then

$$\begin{aligned} & \frac{1}{2}(\mathbf{x}^T, \mathbf{y}^T) \begin{pmatrix} \text{Re}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} & -\text{Im}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} \\ \text{Im}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} & \text{Re}(\boldsymbol{\Gamma}^{-1})_{\ell,\ell} \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \\ &= \frac{1}{2}\mathbf{z}^\dagger(\boldsymbol{\Gamma}^{-1})_{\ell,\ell}\mathbf{z} > 0, \end{aligned} \tag{C.C.1.1}$$

which implies that $\boldsymbol{\Omega}_\ell$ is also positive definite.

Consider the eigendecomposition of $\boldsymbol{\Omega}_\ell$,

$$\boldsymbol{\Omega}_\ell = \mathbf{C}_\ell \boldsymbol{\Lambda}_\ell \mathbf{C}_\ell^T = \sum_{i=1}^{r_\ell} \lambda_{\ell,i} \mathbf{C}_{\ell,i} \mathbf{C}_{\ell,i}^T,$$

where $\mathbf{C}_\ell = [\mathbf{C}_{\ell,1}, \dots, \mathbf{C}_{\ell,r_\ell}]$ is an orthogonal matrix, and $\boldsymbol{\Lambda}_\ell = \text{diag}(\lambda_{\ell,1} \mathbf{I}_{k_{\ell,1}}, \dots, \lambda_{\ell,r_\ell} \mathbf{I}_{k_{\ell,r_\ell}})$ is a diagonal matrix. Here, r_ℓ represents the number of distinct eigenvalues of the matrix $\boldsymbol{\Omega}_\ell$, and $\{\lambda_{\ell,1}, \dots, \lambda_{\ell,r_\ell}\}$ is the set of these distinct eigenvalues. The values $k_{\ell,1}, \dots, k_{\ell,r_\ell}$ represent the algebraic multiplicities of the corresponding eigenvalues $\lambda_{\ell,1}, \dots, \lambda_{\ell,r_\ell}$ for the matrix $\boldsymbol{\Omega}_\ell$.

Set

$$\mathbf{U}_\ell = \mathbf{\Lambda}_\ell^{-1/2} \mathbf{C}_\ell^T (\mathbf{X}_\ell - \boldsymbol{\mu}_\ell) = \begin{pmatrix} \mathbf{U}_{\ell,1} \\ \mathbf{U}_{\ell,2} \\ \vdots \\ \mathbf{U}_{\ell,r} \end{pmatrix},$$

with $\mathbf{U}_{\ell,i} \in \mathbb{R}^{k_{\ell,i}}$ and

$$\boldsymbol{\mu}'_\ell = \mathbf{\Lambda}_\ell^{-1/2} \mathbf{C}_\ell^T \boldsymbol{\mu}_\ell = \begin{pmatrix} \boldsymbol{\mu}'_{\ell,1} \\ \boldsymbol{\mu}'_{\ell,2} \\ \vdots \\ \boldsymbol{\mu}'_{\ell,r} \end{pmatrix},$$

where $\boldsymbol{\mu}'_{\ell,i} = \lambda_{\ell,i}^{-1/2} \mathbf{C}_{\ell,i}^T \boldsymbol{\mu}_\ell \in \mathbb{R}^{k_{\ell,i}}$, for any $1 \leq i \leq r_\ell$. It is easy to verify that

$$\mathbf{U}_\ell \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{4\ell+2}) \text{ and } \mathbf{X}_\ell = \mathbf{C}_\ell \mathbf{\Lambda}_\ell^{1/2} (\mathbf{U}_\ell + \boldsymbol{\mu}'_\ell).$$

Therefore,

$$\begin{aligned} \|\mathbf{X}_\ell\|^2 &= \|\mathbf{C}_\ell \mathbf{\Lambda}_\ell^{1/2} (\mathbf{U}_\ell + \boldsymbol{\mu}'_\ell)\|^2 \\ &= \|\mathbf{\Lambda}_\ell^{1/2} (\mathbf{U}_\ell + \boldsymbol{\mu}'_\ell)\|^2 = \sum_{i=1}^{r_\ell} \lambda_{\ell,i} \|\mathbf{U}_{\ell,i} + \boldsymbol{\mu}'_{\ell,i}\|^2, \end{aligned}$$

and thus, due to the correspondence between $\tilde{\chi}_\ell$ and $\|\mathbf{X}_\ell\|^2$, it follows that $\tilde{\chi}_\ell$ and $\sum_{i=1}^{r_\ell} \lambda_{\ell,i} \|\mathbf{U}_{\ell,i} + \boldsymbol{\mu}'_{\ell,i}\|^2$ have the same distribution. Furthermore, $\|\mathbf{U}_{\ell,i} + \boldsymbol{\mu}'_{\ell,i}\|^2$ follows a noncentral chi-squared distribution, denoted by $\chi^2(\theta_{\ell,i}, k_{\ell,i})$, with noncentrality parameter

$$\theta_{\ell,i} = \|\boldsymbol{\mu}'_{\ell,i}\|^2 = \lambda_{\ell,i}^{-1} \boldsymbol{\mu}_\ell^T \mathbf{C}_{\ell,i} \mathbf{C}_{\ell,i}^T \boldsymbol{\mu}_\ell$$

and $k_{\ell,i}$ degrees of freedom.

It follows that the random variable $\tilde{\chi}_\ell$ can be represented as a sum of independent noncentral chi-squared random variables

$$\sum_{i=1}^{r_\ell} \lambda_{\ell,i} \cdot \chi^2(\theta_{\ell,i}, k_{\ell,i}). \quad (\text{C.C.1.2})$$

In general, the density of Equation (C.C.1.2) does not have a closed-form expression since it involves modified Bessel functions of the first kind [Kotz et al., 2000]. Because integrating the modified Bessel functions in the convolution lacks a simple closed-form solution, we cannot derive the density of the generalized chi-squared distribution unless $r_\ell = 1$, that is, when no convolution and integration are needed. This situation occurs only when $\mathbf{\Omega}_\ell = c\mathbf{I}_{4\ell+2}$ for some constant $c > 0$, which is equivalent to $(\mathbf{\Gamma}^{-1})_{\ell,\ell} = c'\mathbf{I}_{2\ell+1}$ for some constant $c' > 0$.

C.2 The covariance of the angular power spectrum

The goal of this section is to prove the following result about the covariance of the angular power spectrum.

Proposition 1. *At a given frequency bin f_b , and time segment s ,*

$$\begin{aligned} & \text{Cov}(\hat{A}_{f_b,s,\ell}, \hat{A}_{f_b,s,\ell'}) \\ &= \sum_{m,m'} \frac{|(\mathbf{\Gamma}_{f_b,s}^{-1})_{\ell m, \ell' m'}|^2 + 2 \text{Re}[a_{f_b,\ell m}^* (\mathbf{\Gamma}_{f_b,s}^{-1})_{\ell m, \ell' m'} a_{f_b,\ell' m'}]}{(1+2\ell)(1+2\ell')}. \end{aligned} \quad (\text{C.C.2.1})$$

The proof of Proposition 1 is deferred until two preliminary lemmas have been established.

Lemma 2. *If the random variables U and V satisfy*

$$\begin{pmatrix} U \\ V \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_U \\ \mu_V \end{pmatrix}, \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{pmatrix} \right),$$

then

$$\text{Cov}(U^2, V^2) = 2\sigma_{12}(2\mu_U\mu_V + \sigma_{12}). \quad (\text{C.C.2.2})$$

Proof. Notice that $\text{Cov}(U^2, V^2) = \langle U^2 V^2 \rangle - \langle U^2 \rangle \langle V^2 \rangle$ and

$$\langle U^2 \rangle = \mu_U^2 + \sigma_{11}, \text{ and } \langle V^2 \rangle = \mu_V^2 + \sigma_{22}. \quad (\text{C.C.2.3})$$

All that is left is to calculate $\langle U^2 V^2 \rangle$. The moment generating function of (U, V) , denoted as $M_{U,V}(a, b)$, is

$$M_{U,V}(a, b) = \langle e^{aU+bV} \rangle = \exp \left(\mu_U a + \mu_V b + \frac{1}{2} a^2 \sigma_{11} + ab \sigma_{12} + \frac{1}{2} b^2 \sigma_{22} \right)$$

and direct calculation yields

$$\begin{aligned} \langle U^2 V^2 \rangle &= \frac{\partial^4}{\partial a^2 \partial b^2} M_{U,V}(0, 0) \\ &= \mu_U^2 \mu_V^2 + \mu_V^2 \sigma_{11} + 4\mu_U \mu_V \sigma_{12} + 2\sigma_{12}^2 + \mu_U^2 \sigma_{22} + \sigma_{11} \sigma_{22}. \end{aligned} \quad (\text{C.C.2.4})$$

Combining Equations (C.C.2.3) and (C.C.2.4) yields the result. ■

Consider a random variable $\mathbf{Z}$ that follows a multivariate complex Gaussian distribution with mean vector $\boldsymbol{\mu} \in \mathbb{C}^n$ and covariance matrix $\boldsymbol{\Xi}$, which is Hermitian and non-negative definite, that is,

$$\mathbf{Z} = [Z_1, Z_2, \dots, Z_n]^T \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Xi}).$$

The real and imaginary parts of $\mathbf{Z}$ satisfy [Cf. Andersen, 1995, for details]

$$\begin{pmatrix} \text{Re}(\mathbf{Z}) \\ \text{Im}(\mathbf{Z}) \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \text{Re}(\boldsymbol{\mu}) \\ \text{Im}(\boldsymbol{\mu}) \end{pmatrix}, \frac{1}{2} \begin{pmatrix} \text{Re}(\boldsymbol{\Xi}) & -\text{Im}(\boldsymbol{\Xi}) \\ \text{Im}(\boldsymbol{\Xi}) & \text{Re}(\boldsymbol{\Xi}) \end{pmatrix} \right). \quad (\text{C.C.2.5})$$

Lemma 3. If $\mathbf{Z} = [Z_1, Z_2, \dots, Z_n]^T \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Xi})$, then, for any indices $i, j = 1, \dots, n$,

$$\text{Cov}(|Z_i|^2, |Z_j|^2) = |\Xi_{ij}|^2 + 2 \text{Re}(\mu_{Z_i}^* \Xi_{ij} \mu_{Z_j}), \quad (\text{C.C.2.6})$$

where μ_{Z_i} denotes the i -th element of $\boldsymbol{\mu}$ and Ξ_{ij} denotes the (i, j) -th element of the covariance matrix $\boldsymbol{\Xi}$.

Proof. Without loss of generality consider Z_1 and Z_2 . Let $Z_1 = X_1 + iY_1$ and $Z_2 = X_2 + iY_2$, where X_1, X_2, Y_1 , and Y_2 are real-valued Gaussian random variables. From Equation (C.C.2.5), if $j, k \in \{1, 2\}$, then $(X_j, Y_k)^T$ follows a bivariate Gaussian distribution. Let the means of Z_j, X_j , and Y_j be denoted by μ_{Z_j}, μ_{X_j} , and μ_{Y_j} , respectively. Then $\mu_{Z_j} = \mu_{X_j} + i\mu_{Y_j}$. If σ_{X_j, Y_k} represents the covariance between random variables X_j and Y_k , then, using Lemma 2, obtain

$$\begin{aligned} & \text{Cov}(|Z_1|^2, |Z_2|^2) \\ &= \text{Cov}(X_1^2, X_2^2) + \text{Cov}(X_1^2, Y_2^2) + \text{Cov}(Y_1^2, X_2^2) + \text{Cov}(Y_1^2, Y_2^2) \\ &= 2\sigma_{X_1, X_2}(2\mu_{X_1}\mu_{X_2} + \sigma_{X_1, X_2}) + 2\sigma_{X_1, Y_2}(2\mu_{X_1}\mu_{Y_2} + \sigma_{X_1, Y_2}) \\ & \quad + 2\sigma_{Y_1, X_2}(2\mu_{Y_1}\mu_{X_2} + \sigma_{Y_1, X_2}) + 2\sigma_{Y_1, Y_2}(2\mu_{Y_1}\mu_{Y_2} + \sigma_{Y_1, Y_2}). \end{aligned} \quad (\text{C.C.2.7})$$

If $\boldsymbol{\Xi}'$ is the covariance matrix between Z_1 and Z_2 , that is,

$$\boldsymbol{\Xi}' = \begin{pmatrix} \Xi_{11} & \Xi_{12} \\ \Xi_{21} & \Xi_{22} \end{pmatrix},$$

then

$$\text{Re}(\Xi') = \begin{pmatrix} \Xi_{11} & \text{Re}(\Xi_{12}) \\ \text{Re}(\Xi_{21}) & \Xi_{22} \end{pmatrix},$$

and

$$\text{Im}(\Xi') = \begin{pmatrix} 0 & -\text{Im}(\Xi_{12}) \\ \text{Im}(\Xi_{21}) & 0 \end{pmatrix}.$$

From Equation (C.C.2.5), the distribution of (X_1, X_2, Y_1, Y_2) is

$$\begin{pmatrix} X_1 \\ X_2 \\ Y_1 \\ Y_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_{X_1} \\ \mu_{X_2} \\ \mu_{Y_1} \\ \mu_{Y_2} \end{pmatrix}, \mathbf{\Omega} \right), \quad (\text{C.C.2.8})$$

where

$$\mathbf{\Omega} = \frac{1}{2} \begin{pmatrix} \Xi_{11} & \text{Re}(\Xi_{12}) & 0 & -\text{Im}(\Xi_{12}) \\ \text{Re}(\Xi_{21}) & \Xi_{22} & -\text{Im}(\Xi_{21}) & 0 \\ 0 & \text{Im}(\Xi_{12}) & \Xi_{11} & \text{Re}(\Xi_{12}) \\ \text{Im}(\Xi_{21}) & 0 & \text{Re}(\Xi_{21}) & \Xi_{22} \end{pmatrix}. \quad (\text{C.C.2.9})$$

Furthermore, since the covariance matrix Ξ is Hermitian, $\text{Re}(\Xi) = \text{Re}(\Xi)^T$, $\text{Im}(\Xi) = -\text{Im}(\Xi)^T$. Thus,

$$\begin{aligned} \sigma_{X_1, X_2} &= \sigma_{Y_1, Y_2} = \frac{1}{2} \text{Re}(\Xi_{12}) = \frac{1}{2} \text{Re}(\Xi_{21}), \\ \sigma_{X_1, Y_2} &= -\frac{1}{2} \text{Im}(\Xi_{12}) = \frac{1}{2} \text{Im}(\Xi_{21}), \quad \text{and} \\ \sigma_{Y_1, X_2} &= \frac{1}{2} \text{Im}(\Xi_{12}) = -\frac{1}{2} \text{Im}(\Xi_{21}). \end{aligned} \quad (\text{C.C.2.10})$$

By combining Equations (C.C.2.7) and (C.C.2.10) obtain

$$\begin{aligned}
& \text{Cov}(|Z_1|^2, |Z_2|^2) \\
&= (\text{Re}(\Xi_{21}))^2 + (\text{Im}(\Xi_{21}))^2 + 2(\mu_{X_1}\mu_{Y_2} - \mu_{X_2}\mu_{Y_1}) \text{Im}(\Xi_{21}) \\
&\quad + 2(\mu_{X_1}\mu_{X_2} + \mu_{Y_1}\mu_{Y_2}) \text{Re}(\Xi_{21}) \\
&= |\Xi_{21}|^2 + 2 \text{Re}(\Xi_{21}) \text{Re}(\mu_{Z_1}^* \mu_{Z_2}) + 2 \text{Im}(\Xi_{21}) \text{Im}(\mu_{Z_1}^* \mu_{Z_2}) \\
&= |\Xi_{12}|^2 + 2 \text{Re}(\mu_{Z_1}^* \Xi_{12} \mu_{Z_2}).
\end{aligned} \tag{C.C.2.11}$$

The only remaining case to consider is when the indices coincide. Specifically,

$$\text{Cov}(|Z_1|^2, |Z_1|^2) = \text{Var}(|Z_1|^2) = \text{Var}(X_1^2) + \text{Var}(Y_1^2).$$

The last equality follows since, by Equations (C.C.2.8) and (C.C.2.9), the random variables X_1 and Y_1 are independent. Also, note that from Equation (C.C.2.9), $\text{Var}(X_1) = \text{Var}(Y_1) = \frac{1}{2}\Xi_{11}$. Using standard results about the moments of Gaussian distributions, direct calculation yields

$$\begin{aligned}
\text{Var}(X_1^2) &= \langle X_1^4 \rangle - \langle X_1^2 \rangle^2 \\
&= \frac{3}{4}\Xi_{11}^2 + 3\mu_{X_1}^2 \Xi_{11} + \mu_{X_1}^4 - \left(\frac{1}{2}\Xi_{11} + \mu_{X_1}^2 \right)^2 \\
&= \frac{1}{2}\Xi_{11}^2 + 2\mu_{X_1}^2 \Xi_{11}
\end{aligned}$$

and, similarly,

$$\text{Var}(Y_1^2) = \frac{1}{2}\Xi_{11}^2 + 2\mu_{Y_1}^2 \Xi_{11}.$$

It follows that

$$\begin{aligned}
\text{Var}(|Z_1|^2) &= \Xi_{11}^2 + 2|\mu_{Z_1}|^2 \Xi_{11} \\
&= \Xi_{11}^2 + 2 \text{Re}(\mu_{Z_1}^* \Xi_{11} \mu_{Z_1}).
\end{aligned} \tag{C.C.2.12}$$



We are now in position to prove Proposition 1.

Proof of Proposition 1. Since

$$\{\hat{a}_{f_b,s,\ell m}\}_{\mathcal{I}} \sim \mathcal{CN}(\{a_{f_b,\ell m}\}_{\mathcal{I}}, \mathbf{\Gamma}_{f_b,s}^{-1}),$$

where the index set $\mathcal{I} = \{(\ell, m) \in \mathbb{Z}^2; 1 \leq \ell \leq \ell_{\max}, -\ell \leq m \leq \ell\}$, and $\mathbf{\Gamma}_{f_b,s}^{-1} = \{(\mathbf{\Gamma}_{f_b,s}^{-1})_{\ell m, \ell' m'}\}_{(\ell, m), (\ell', m') \in \mathcal{I}}$, applying Lemma 3 yields

$$\begin{aligned} & \text{Cov}(|\hat{a}_{f_b,s,\ell m}|^2, |\hat{a}_{f_b,s,\ell' m'}|^2) \\ &= |(\mathbf{\Gamma}_{f_b,s}^{-1})_{\ell m, \ell' m'}|^2 + 2 \text{Re}[a_{f_b,\ell m}^* (\mathbf{\Gamma}_{f_b,s}^{-1})_{\ell m, \ell' m'} a_{f_b,\ell' m'}]. \end{aligned}$$

Recalling the definition of $\hat{A}_{f_b,s,\ell}$ in Equation (4.4.2.6) obtain

$$\begin{aligned} & \text{Cov}(\hat{A}_{f_b,s,\ell}, \hat{A}_{f_b,s,\ell'}) \\ &= \frac{1}{(1+2\ell)(1+2\ell')} \sum_{m,m'} \text{Cov}(|\hat{a}_{f_b,s,\ell m}|^2, |\hat{a}_{f_b,s,\ell' m'}|^2), \end{aligned}$$

and Equation (C.C.2.1) follows. ■

C.3 Properties of the plug-in estimator of the covariance of angular power spectrum based on Equation (4.4.2.7)

Recall that at a given frequency band f_b , the unknown covariance function between $\hat{A}_{f_b,s,\ell}$ and $\hat{A}_{f_b,s,\ell'}$ is

$$(\Sigma_{f_b,s})_{\ell, \ell'} = \text{Cov}(\hat{A}_{f_b,s,\ell}, \hat{A}_{f_b,s,\ell'}).$$

Assume that $\{\hat{a}_{f_b,s,\ell m}\}_{\mathcal{I}}$, $s \geq 1$, is a sequence of independent complex Gaussian random variables, with distribution $\mathcal{CN}(\{a_{f_b,\ell m}\}_{\mathcal{I}}, \mathbf{\Gamma}_{f_b,s}^{-1})$, where $\mathbf{\Gamma}_{f_b,s}$ and $\mathbf{\Gamma}_{f_b,s}^{-1}$ are deterministic

and inverses of each other. The index set $\mathcal{I}$ is defined similarly as in Appendix C.1.

As described in Section 4.3.1, the elements of the plug-in estimator of $(\Sigma_{f_b,s})_{\ell,\ell'}$ are given by

$$\sum_{m,m'} \frac{|(\Gamma_{f_b,s}^{-1})_{\ell m,\ell' m'}|^2 + 2 \operatorname{Re} \left[\hat{a}_{f_b,\ell m}^* (\Gamma_{f_b,s}^{-1})_{\ell m,\ell' m'} \hat{a}_{f_b,\ell' m'} \right]}{(1+2\ell)(1+2\ell')}.$$

Recall from Equation (4.4.3.8), $\hat{a}_{f_b,\ell m}$ depends on S .

Let $\{X_n\}_{n=1}^\infty$ be a sequence of random variables defined on the same probability space, and let X be another random variable on this space. Say that X_n converges in probability to X as $n \rightarrow \infty$, denoted

$$X_n \xrightarrow{\mathbb{P}} X.$$

Let $\|\cdot\|_{op}$ denote the operator norm, which is equal to the largest singular value of the matrix.

Theorem 4. *The following statements hold.*

1. Both $\Sigma_{f_b,s}$ and $\hat{\Sigma}_{f_b,s}$ are positive definite matrices.
2. If there exists $\zeta > 0$ such that, for all $s \geq 1$ and f_b ,

$$\|\Gamma_{f_b,s}^{-1}\|_{op} \leq \zeta, \tag{C.C.3.1}$$

then $\hat{a}_{f_b,\ell m} \xrightarrow{\mathbb{P}} a_{f_b,\ell m}$, as $S \rightarrow \infty$.

3. If $\hat{a}_{f_b,\ell m} \xrightarrow{\mathbb{P}} a_{f_b,\ell m}$, then $\hat{\Sigma}_{f_b,s}$ is a consistent estimator of $\Sigma_{f_b,s}$, for any s .
4. If condition (C.C.3.1) holds, then, as $S \rightarrow \infty$,

$$\max_{f_b} \max_{1 \leq s \leq S} \|\hat{\Sigma}_{f_b,s}^{-1} - \Sigma_{f_b,s}^{-1}\|_{op} \xrightarrow{\mathbb{P}} 0.$$

Proof. Proof of statement 1: This proof will be the same for all frequency bins f_b and segments s considered. Therefore, for notational simplicity, let us denote $\Sigma = \Sigma_{f_b, s}$, $\mathbf{\Gamma} = \mathbf{\Gamma}_{f_b, s}$ and $a_{\ell m} = a_{f_b, \ell m}$.

Let $d_1, \dots, d_{\ell_{\max}}$ be complex numbers such that $|d_1|^2 + \dots + |d_{\ell_{\max}}|^2 > 0$. Let $\mathbf{d} = \{d_\ell / (2\ell + 1)\}_{(\ell, m) \in \mathcal{I}}$. A direct calculation shows that

$$\begin{aligned}
\sum_{\ell, \ell'} d_\ell^* (\Sigma)_{\ell, \ell'} d_{\ell'} &= \sum_{\ell m, \ell' m'} \frac{d_\ell^*}{1 + 2\ell} (\mathbf{\Gamma}^{-1})_{\ell' m', \ell m} (\mathbf{\Gamma}^{-1})_{\ell m, \ell' m'} \frac{d_{\ell'}}{1 + 2\ell'} \\
&\quad + 2 \sum_{\ell m, \ell' m'} \frac{d_\ell^*}{1 + 2\ell} \operatorname{Re}[a_{\ell m}^* (\mathbf{\Gamma}^{-1})_{\ell m, \ell' m'} a_{\ell' m'}] \frac{d_{\ell'}}{1 + 2\ell'} \\
&= \operatorname{tr} \left((\mathbf{\Gamma}^{-1} \operatorname{diag}(\mathbf{d}))^* \mathbf{\Gamma}^{-1} \operatorname{diag}(\mathbf{d}) \right) \\
&\quad + 2 \operatorname{Re} \left(\sum_{\ell m, \ell' m'} \left(\frac{d_{\ell'}}{1 + 2\ell'} a_{\ell m} \right)^* (\mathbf{\Gamma}^{-1})_{\ell m, \ell' m'} \left(a_{\ell' m'} \frac{d_\ell}{1 + 2\ell} \right) \right) \\
&\geq \operatorname{tr} \left((\mathbf{\Gamma}^{-1} \operatorname{diag}(\mathbf{d}))^* \mathbf{\Gamma}^{-1} \operatorname{diag}(\mathbf{d}) \right) > 0.
\end{aligned} \tag{C.C.3.2}$$

Thus, Σ is positive definite. The proof for $\hat{\Sigma}_{f_b, s}$ is similar.

Proof of statement 2: Recall that, as described in Section 4.3, $\mathbf{\Gamma}_{f_b} = \sum_{s=1}^S \mathbf{\Gamma}_{f_b, s}$. Hence, the smallest eigenvalue of $\mathbf{\Gamma}_{f_b}$ is bounded below by S/ζ . As $S \rightarrow \infty$, this lower bound goes to infinity; thus,

$$\|\mathbf{\Gamma}_{f_b}^{-1}\|_{op} \rightarrow 0.$$

Since

$$\{\hat{a}_{f_b, \ell m}\}_{\mathcal{I}} \sim \mathcal{CN}(\{a_{f_b, \ell m}\}_{\mathcal{I}}, \mathbf{\Gamma}_{f_b}^{-1}),$$

$\operatorname{Cov}(\{\hat{a}_{f_b, \ell m}\}_{\mathcal{I}}) = \mathbf{\Gamma}_{f_b}^{-1} \rightarrow 0$ as $S \rightarrow \infty$, which implies that

$$\hat{a}_{f_b, \ell m} \xrightarrow{\mathbb{P}} a_{f_b, \ell m} \quad \text{as } S \rightarrow \infty.$$

Proof of statement 3: Define the continuous function $g_{\ell,\ell'}$ as

$$g_{\ell,\ell'}(\{\hat{a}_{f,\ell m}\}_{\mathcal{I}}, \mathbf{\Omega}) = \sum_{m,m'} \frac{|\Omega_{\ell m,\ell' m'}|^2 + 2\text{Re}[\hat{a}_{f,\ell m}^* \Omega_{\ell m,\ell' m'} \hat{a}_{f,\ell' m'}]}{(1+2\ell)(1+2\ell')}.$$

By the continuous mapping theorem [Cf. [Van der Vaart, 2000](#), Ch. 2], we obtain that as $\hat{a}_{f_b,\ell m} \xrightarrow{\mathbb{P}} a_{f_b,\ell m}$

$$g_{\ell,\ell'}(\{\hat{a}_{f_b,\ell m}\}_{\mathcal{I}}, \Gamma_{f_b,s}^{-1}) \xrightarrow{\mathbb{P}} g_{\ell,\ell'}(\{a_{f_b,\ell m}\}_{\mathcal{I}}, \Gamma_{f_b,s}^{-1}).$$

Hence, $\hat{\Sigma}_{f_b,s}$ is a consistent estimator of $\Sigma_{f_b,s}$, as $\hat{a}_{f_b,\ell m} \xrightarrow{\mathbb{P}} a_{f_b,\ell m}$.

Proof of statement 4: Equation (C.C.3.2) implies that there exists $\beta > 0$ such that

$$\inf_{s \geq 1, f_b} \lambda_{\min}(\Sigma_{f_b,s}) \geq \beta, \text{ and } \inf_{s \geq 1, f_b} \lambda_{\min}(\hat{\Sigma}_{f_b,s}) \geq \beta,$$

where $\lambda_{\min}(\Sigma_{f_b,s})$ denotes the smallest eigenvalue of $\Sigma_{f_b,s}$. Since

$$\hat{\Sigma}_{f_b,s}^{-1} - \Sigma_{f_b,s}^{-1} = -\hat{\Sigma}_{f_b,s}^{-1}(\hat{\Sigma}_{f_b,s} - \Sigma_{f_b,s})\Sigma_{f_b,s}^{-1},$$

we have

$$\begin{aligned} \|\hat{\Sigma}_{f_b,s}^{-1} - \Sigma_{f_b,s}^{-1}\|_{op} &\leq \|\hat{\Sigma}_{f_b,s}^{-1}\|_{op} \|\hat{\Sigma}_{f_b,s} - \Sigma_{f_b,s}\|_{op} \|\Sigma_{f_b,s}^{-1}\|_{op} \\ &\leq \frac{1}{\beta^2} \|\hat{\Sigma}_{f_b,s} - \Sigma_{f_b,s}\|_{op}. \end{aligned}$$

Note that there exists a positive constant C such that

$$\begin{aligned}
& |(\hat{\Sigma}_{f_b, s})_{\ell, \ell'} - (\Sigma_{f_b, s})_{\ell, \ell'}| \\
& \leq \sum_{m, m'} \frac{2 \left| (a_{f_b, \ell m}^* a_{f_b, \ell' m'} - \hat{a}_{f_b, \ell m}^* \hat{a}_{f_b, \ell' m'}) (\Gamma_{f_b, s}^{-1})_{\ell m, \ell' m'} \right|}{(1 + 2\ell)(1 + 2\ell')} \\
& \leq C \cdot \|\Gamma_{f_b, s}^{-1}\|_{op} \sum_{m, m'} |a_{f_b, \ell m}^* a_{f_b, \ell' m'} - \hat{a}_{f_b, \ell m}^* \hat{a}_{f_b, \ell' m'}| \\
& \leq C \cdot \zeta \sum_{m, m'} |a_{f_b, \ell m}^* a_{f_b, \ell' m'} - \hat{a}_{f_b, \ell m}^* \hat{a}_{f_b, \ell' m'}|.
\end{aligned}$$

Since for any f_b and any $|m| \leq \ell \leq \ell_{max}$

$$\hat{a}_{f_b, \ell m} \xrightarrow{\mathbb{P}} a_{f_b, \ell m} \quad \text{as } S \rightarrow \infty,$$

there exists a positive constant C' such that

$$\begin{aligned}
& \max_{f_b} \max_{1 \leq s \leq S} \|\hat{\Sigma}_{f_b, s}^{-1} - \Sigma_{f_b, s}^{-1}\|_{op} \\
& \leq \frac{\zeta \cdot C'}{\beta^2} \max_{f_b} \sum_{m, m'} |a_{f_b, \ell m}^* a_{f_b, \ell' m'} - \hat{a}_{f_b, \ell m}^* \hat{a}_{f_b, \ell' m'}| \xrightarrow{\mathbb{P}} 0
\end{aligned}$$

as $S \rightarrow \infty$.

■

C.4 Properties of the operator in Equation (4.4.3.15)

Consider the operator $U_{\mathbf{a}, \mathbf{b}}$ on $\mathbb{R}^N$ introduced by Khmaladze [2013] and defined as

$$U_{\mathbf{a}, \mathbf{b}} = \mathbf{I} - \frac{\langle \mathbf{a} - \mathbf{b}, \cdot \rangle}{1 - \langle \mathbf{a}, \mathbf{b} \rangle} (\mathbf{a} - \mathbf{b}), \quad (\text{C.C.4.1})$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{R}^N$ satisfy

$$\|\mathbf{a}\| = \|\mathbf{b}\| = 1, \quad \langle \mathbf{a}, \mathbf{b} \rangle \neq 1.$$

The operator $U_{\mathbf{a}, \mathbf{b}}$ maps $\mathbf{a}$ to $\mathbf{b}$, maps $\mathbf{b}$ to $\mathbf{a}$, and keep the vectors that are orthogonal to both $\mathbf{a}$ and $\mathbf{b}$ unchanged, i.e.,

$$\begin{aligned} U_{\mathbf{a}, \mathbf{b}} \mathbf{a} &= \mathbf{b}, \quad U_{\mathbf{a}, \mathbf{b}} \mathbf{b} = \mathbf{a}, \\ \text{and } U_{\mathbf{a}, \mathbf{b}} \mathbf{x} &= \mathbf{x}, \quad \text{for } \mathbf{x} \perp \mathbf{a}, \mathbf{b}. \end{aligned} \tag{C.C.4.2}$$

Such an operator $U_{\mathbf{a}, \mathbf{b}}$ is unitary – that is, it is surjective and preserves the inner product.

The surjectivity can be easily shown from Equation (C.C.4.2). Moreover, for any vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^N$:

$$\begin{aligned} & \langle U_{\mathbf{a}, \mathbf{b}} \mathbf{x}, U_{\mathbf{a}, \mathbf{b}} \mathbf{y} \rangle \\ &= \left\langle \mathbf{x} - \frac{\langle \mathbf{a} - \mathbf{b}, \mathbf{x} \rangle}{1 - \langle \mathbf{a}, \mathbf{b} \rangle} (\mathbf{a} - \mathbf{b}), \mathbf{y} - \frac{\langle \mathbf{a} - \mathbf{b}, \mathbf{y} \rangle}{1 - \langle \mathbf{a}, \mathbf{b} \rangle} (\mathbf{a} - \mathbf{b}) \right\rangle \\ &= \langle \mathbf{x}, \mathbf{y} \rangle - \frac{\langle \mathbf{a} - \mathbf{b}, \mathbf{x} \rangle \langle \mathbf{a} - \mathbf{b}, \mathbf{y} \rangle}{(1 - \langle \mathbf{a}, \mathbf{b} \rangle)^2} (\|\mathbf{a} - \mathbf{b}\|^2 - 2 - 2\langle \mathbf{a}, \mathbf{b} \rangle) \\ &= \langle \mathbf{x}, \mathbf{y} \rangle, \end{aligned}$$

therefore, $U_{\mathbf{a}, \mathbf{b}}$ preserves the inner product. Moreover, $U_{\mathbf{a}, \mathbf{b}}$ is a self-adjoint operator because it satisfies the condition

$$\begin{aligned} \langle U_{\mathbf{a}, \mathbf{b}} \mathbf{x}, \mathbf{y} \rangle &= \langle \mathbf{x}, \mathbf{y} \rangle - \frac{\langle \mathbf{a} - \mathbf{b}, \mathbf{x} \rangle}{1 - \langle \mathbf{a}, \mathbf{b} \rangle} \langle \mathbf{a} - \mathbf{b}, \mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{y} - \frac{\langle \mathbf{a} - \mathbf{b}, \mathbf{y} \rangle}{1 - \langle \mathbf{a}, \mathbf{b} \rangle} (\mathbf{a} - \mathbf{b}) \rangle \\ &= \langle \mathbf{x}, U_{\mathbf{a}, \mathbf{b}} \mathbf{y} \rangle. \end{aligned} \tag{C.C.4.3}$$

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